6. B.Sc. MATHEMATICS

Duration – Three years
Eligibility – HSC (10+2) from any recognized Board in Science with Mathematics.
Medium – English only

Sl. No.	Course Code	Title of the Course	CIA Max.	ESE Max.	TOT Max.	C Max.
		FIRST YEAR				
		I Semester				
1.	**11311A / B**	Part-I: Tamil Paper-I / Communication Skills-I	25	75	100	4
2.	**11312**	Part-II : English-I	25	75	100	4
3.	**11313**	Classical Algebra	25	75	100	4
4.	**11314**	Calculus	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		II Semester				
5.	**11321 A / B**	Part-I: Tamil Paper-II / Communication Skills-II	25	75	100	4
6.	**11322**	Part-II : English-II	25	75	100	4
7.	**11323**	Analytical Geometry and Vector Calculus	25	75	100	4
8.	**11324**	Sequences and Series	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		SECOND YEAR				
		III Semester				
9.	**11331 A /B**	Part-I: Tamil Paper-III / Human Skills Development - I	25	75	100	4
10.	**11332**	Part-II : English-III	25	75	100	4
11.	**11333**	Differential Equations and its Applications	25	75	100	4
12.	**11334**	Mechanics	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		IV Semester				
13.	**11341 A/ B**	Part-I: Tamil Paper-IV / Human Skills Development - II	25	75	100	4
14.	**11342**	Part-II : English-IV	25	75	100	4
15.	**11343**	Analysis	25	75	100	4
16	**11344**	Statistics	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		V Semester				
17.	**11351**	Modern Algebra	25	75	100	4
18.	**11352**	Operations Research	25	75	100	4
19.	**11353**	Numerical Analysis	25	75	100	4
20.	**11354**	Transform Technics	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		VI Semester				
21.	**11361**	Discrete Mathematics	25	75	100	4
22.	**11362**	Fuzzy Algebra	25	75	100	4
23.	**11363**	Complex Analysis	25	75	100	4
24.	**11364**	Combinatorics	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		Grand Total	**600**	**1800**	**2400**	**96**

CIA: Continuous Internal Assessment, **ESE:** End Semester Examination,
TOT: Total, **C:** Credit Points **Max:** Maximum

FIRST SEMESTER

Course Code	Title of the Course
11311A	Part-I: Tamil Paper - I

கூறு 1

1. **கண்ணதாசன் - ஸ்ரீ கிருஷ்ண கானம்**
   1. புல்லாங்குழல் கொடுத்த
   2. குருவாயூருக்கு வாருங்கள்

கூறு 2

1. கோகுலத்து பசுக்கள்
2. கோகுலத்தில் ஒரு நாள் ராதை
3. ஆயர்பாடி மாளிகையில்

கூறு 3

பட்டுக்கோட்டை கல்யாண சுந்தரம்

1. நெஞ்சில் குடியிருக்கும்
2. செய்யும் தொழிலே தெய்வம்

கூறு 4

1. **பாரதியார்**

கண்ணன் என் விளையாட்டுப்பிள்ளை
பாரத மாதா திருப்பள்ளி எழுச்சி

கூறு 5

1. **பாரதிதாசன்** - உலகப்பன் பாட்டு (5)
2. **நாமக்கல் கவிஞர்** - நோயற்ற வாழ்வு 7 பாட்டு
3. **பெ.தூரன்** - நிலா பிஞ்சு

கூறு 6

1. **வல்லிக் கண்ணன்** - வெறும் புகழ்
2. **கு.ப.இராஜகோபாலன்** - எதற்காக?
3. **மீரா** - பதினைந்து

கூறு 7

1. **சிற்பி** - சர்ப்ப யாகம்1
2. **ஞானக்கூத்தன்** - தோழர் மோசிகீரனார்

கூறு 8

1. **அப்துல் ரகுமான்** - கண்ணும் எழுதேம்
2. **சண்முக சுப்பையா** - வயிறு

கூறு 9

1. **சிலப்பதிகாரம்** - வழக்குரை காதை
2. **கம்பராமாயணம்** - அயோத்தியா காண்டம்

கூறு 10

1. **சீறாப்புராணம்** - ஈத்தங்குலை வரவழைத்த படலம் (1)

கூறு 11

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

1. இன்னவாயில்
2. கொழுந்துறும்
3. பஞ்(ச) அரங்கில்

கூறு 12

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

4. எண்ணுளே
5. ஒண்தலங்கள்
6. இரவியேந்த கஞ்சக்

கூறு 13

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

7. கன்னியாயதாயும்
8. ஏந்தி ஓங்கு உளத்து
9. ஆவ தேமுனர்
10. கொல்லும் வேலொடும்

கூறு 14

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

11. என்ற வாசகம்
12. அம்பினால்
13. வேண்டும் ஓர் வினை

Course Code	Title of the Course
11311B	Part-I: Communication Skills - I

Unit-1: Communication - Meaning - Types- Importance.

Unit-2: Barriers to Effective Communication - Principles - Principles of Effective Communication.

Unit-3: Oral Communication - Meaning - Importance - Forms of Oral Communication.

Unit-4: Intonation - Meaning - Function - Types Preparation of Speech- Steps Involved.

Unit-5: Principles of Effective Oral Communication.

Unit-6: Written Communication - Meaning - Steps - Importance - Advantages - Use of words and Phrases.

Unit-7: Sentence - Meaning - Sentence formation - Characteristics of an Effective Sentence.

Unit-8: Paragraph Writing - Essay Writing - Steps Involved - Outline-Layout - Contents -Drafting- Correction - Final Draft.

Unit-9: Application for Employment and Curriculum Vitae - Steps involved.

Unit-10: Non-Verbal Communication - Meaning - Types - Body Language - Postures -Gestures - Facial Expressions - Eye Contact.

Unit-11: Report Writing - Report - Types of Reports - Format of a Report.

Unit-12: Essentials of a Good Report - Preparation of Report - Procedure Involved.

Unit-13: Meetings - Purpose of the Meeting - Procedure.

Unit-14: Group Discussion - Quality of Content - Participation - Logical Presentation - Behavioural Skills.

References:

1. Krishna Mohan & Meera Banerjee, Developing Communication Skills, 2005.
2. Geetha Nagaraj, Write to Communicate, 2004.
3. Wren & Martin, English Grammar and Composition, 2002.
4. Dale Carnegie, How to Win Friends and Influence People, 1981.
5. Dale R Jordan, Language Skills and Use.
6. Gartside L. Bahld, Nagammiah and McComas, Satterwhite, Modern Business Correspondence.
7. Rajendra Pal and Kortahalli J S, Essentials of Business Communication.
8. Wallace, Michael J, Study Skills in English.
9. Editors of Readers Digest, Super Word Power.

Course Code	Title of the Course
11312	**PART-II : ENGLISH PAPER - I**

Prose

Unit – I	Water-the Elixir of life	- C.V. Raman
Unit – II	Mrs. Packletide's Tiger	- SAKI
Unit – III	A Deed of Bravery	- Jim Carbett
Unit – IV	The Cat	- Catharine M.Willson
Unit – V	On Letter Writing	- Alpha of the Plough
Unit – VI	Our Ancestors	- Carl Sagan
Unit – VII	Our Civilization	- C.E.Foad
Unit – VIII	A Hero on Probation	- B.R. Nanda
Unit – IX	Dangers of Drug Abuse	- Hardin B. Fones
Unit – X	Food	- J.B.S. Haldane

Grammar

Unit – XI - Articles-Gerunds-Participles-Infinitives-Modals-Proposition – Tenses.

Unit – XII - Direct and Indirect Speech-Transformation of sentences- Active and passive voice.

Composition

Unit – XIII - Letter writing - Precis writing - Developing hints. Unit – XIV - Dialogue writing - Paragraph writing.

References:

1. Sebastian D K, *Prose for the Young Reader*, Macmillan.
2. *Active English Grammar*, Ed. by the Board of Editors, Macmillan.
3. Modern English – *A Book of Grammar Usage and Composition* by N.Krishnaswamy, Macmillan Publishers.

Course Code	Title of the Course
11313	PART-III : CLASSICAL ALGEBRA

UNIT- I

Introduction, Binomial theorem for rational index, other forms of Binomial Expansion, Summation of series using binomial theorem.

UNIT -II

Theory of equations - Introduction and basic theorems, Relation between the roots and coefficients, Symmetric functions of the roots.

UNIT -III

Sum of the powers of the roots of the equation, Transformation of equation by given quantity, formation of equations whose roots are diminished by h, formation of equations whose roots are equal in magnitude and opposite in sign.

UNIT -IV

Multiple Roots – Nature and position of roots – Descarte's rule of Signs, Rolle's theorem – Sturm's functions – Problems.

UNIT -V

Removal of terms – Introduction, Reciprocal roots, Newton's and Horner's methods.

UNIT -VI

Finding number and position of the real roots – Finding the nature and position of the roots (Cardans&Ferrar's method not included).

UNIT -VII

Inequalities – Arithmetic and geometric means, Weierstrass inequality.

UNIT -VIII

Determinants - Definition, Expansion of determinants, Properties of determinants

UNIT -IX

Matrices- Operation on matrices, adjoint of a square matrix- Problems.

UNIT -X

Singular and Non singular matirices-Inverse of a non-singular matrix.

UNIT -XI

System of equations - Cramer's rule, problems using Cramer's rule-Problems.

UNIT -XII

Rank of a matrix, Consistency of equations-related problems.

UNIT -XIII

Eigen values, Eigen vectors- Some problems

UNIT -XIV

Cayley Hamilton theorem – Statement, Verification of Cayley Hamilton theorem, finding inverse using Cayley Hamilton theorem.

REFERENCES:

1. Arumugam & Issac, Sequences and Series, New Gamma Publishing House, 2002 Edition.
2. Arumugam & Issac, Set Theory & Number System and Theory of Equations.
3. Venkataraman & Manorama, Algebra, National Publishing House, Chennai.
4. T.K.Manickavasagam Pillai & Others, Algebra Vol.I & Vol.II S.Viswanathan (Printers & Publishers) Pvt. Ltd, 1985-Revised Edition.

Course Code	Title of the Course
11314	PART-III : CALCULUS

UNIT- I

Differentiation – Introduction, Parametric differentiation, Logrithmic differentiation, differentiation of implicit functions.

UNIT –II

Successive differentiation – Introduction, n^{th} derivative of some standard functions, problems using higher order derivatives.

UNIT –III

Partial differentiation – Homogeneous functions, Euler's theorem, verification of Euler's theorem, Maxima and minima of functions of one variable and two variables.

UNIT –IV

Polar Coordinates – Radius of curvature in polar coordinates, p-r equation of a curve – Asymptotes – Method of finding asymptotes – problems

UNIT –V

Tangents and normal angle of intersection, curvature, Envelopes and Evolutes, working method to find envelope and involutes

UNIT –VI

Integration – Substitution methods, $1/(x_2 - a_2)$, $1/(x_2 + a_2)$, $1/(a_2 - x_2)$, $1/(x_2 - a_2)½$, $(x_2 - a_2)½$, $(x_2 + a_2)½$, $(a_2 - x_2)½$.

UNIT –VII

Definite Integrals and their properties –problems – Integration by parts — Reduction formulae – Bernoulli's formula.

UNIT -VIII

Double and triple integrals and their properties – Jacobian – Change of order of integration.

UNIT -IX

Beta and Gamma functions – properties – problems

UNIT -X

Differential equations – Solution of differential equations, variable separable methods.

UNIT -XI

Homogeneous equations in x and y-Methods and problems, First order linear equations.

UNIT -XII

Linear equations of order 2 with constant and variable coefficients, Variation of parameters.

UNIT -XIII

Laplace transform, Inverse Laplace transform, Solving differential equations using Laplace transforms.

UNIT -XIV

Partial differential equations – Forming differential equations by eliminating arbitrary constants and variables, First order partial order equations. Some standard forms – Charpit's method, Clairaut's form, Lagrange's multiplier method and problems.

REFERENCES:

1. Arumugam & Issac, Calculus, New Gamma Publishing House, 2005.
2. Arumugam & Issac, Differential Equations and Applications, New Gamma Publishing House, 2003.
3. A.K.Sharma, Text book of Differential Calculus, Discovery publishing house, New Delhi.
4. S.Narayanan & T.K. Manickavasagam Pillai, Differential Equations and its applications, S.Viswanathan (Printers & Publishers) Pvt. Ltd, 2009, Chennai.
5. Calculus and Fourier series by Dr. M.K.Venkataraman and Mrs. Manorama Sridhar, The National Publishing Company, Chennai.

SECOND SEMESTER

Course Code	Title of the Course
11321A	Part-I: Tamil Paper - II

கூறு 1

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

14. சொல் தவிர்ந்த
15. அன்னை
16. அஞ்சுவார்
17. சொல்லக் கேட்டனள்
18. மற்செய்கை
19. மண்கனியப்
20. அழுது ஆர்ந்த

கூறு 2

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

21. பொய் பொதுளும்
22. இன்பு அருந்தி
23. வழுதாயின இன்பு
24. மறம் ஏவினர்

கூறு 3

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

25. மண்ணோர்கள்
26. பொய்யா விதியோய்
27. விடியா இருள்
28. அழுவார் எவரும்

கூறு 4

சிறுகதை - நீலபத்மநாபனின் ‘‘வான வீதியில்’’

கூறு 5

உரைநடை - கம்பன் புறத்திணை - தி.சொக்கலிங்கம்

இலக்கணம் - எழுத்தும் சொல்லும்

கூறு 6

1. முதலெழுத்துகள், சார்பெழுத்துகள்
2. மொழி முதலெழுத்துகள் , மொழி இறுதி எழுத்துகள்

கூறு 7

1. ஒற்றெழுத்து மிகலும் மிகாமையும்,
2. ஆகு பெயர் , அன்மொழித் தொகை.
3. வினா-விடை வகைகள்

கூறு 8

1. தமிழ்ச் சொல்லமைப்பின் சிறப்பு – பெயர் , வினை, இடை, உரி வடிவங்கள் ,
2. பிற மொழிச் சொற்களைத் தமிழில் ஆளும் முறைகள்

கூறு 9

3. அல் வழி, வேற்றுமைப் புணர்ச்சிகள்
4. திணை, பால் , எண் , இட இயைபு.

தமிழ் இலக்கிய வரலாறு

கூறு 10

1. **இக்கால இலக்கிய வகைகள்**
   அ) மரபுக் கவிதை

   ஆ) புதுக் கவிதையின் தோற்றமும் வளர்ச்சியும்

கூறு 11

1. **உரை நடை இலக்கியங்கள் - தோற்றமும் வளர்ச்சியும்**

   அ) கட்டுரை ஆ) சிறுகதை

   இ) புதினம் ஈ) நாடகம்

கூறு 12

1. **இக்கால இலக்கியக் களங்கள்**
   திரைப்படம் , தொலைக்காட்சி, வானொலி, இதழ்கள் தமிழுக்கு ஆற்றி வரும் பணிகள்

கூறு 13

1. **தமிழும் சமயமும் :**
   அ) சைவம் ஆ)வைணவம் இ)சமணம் ஈ)பௌத்தம் உ)இசுலாம்

   ஊ) கிறித்துவம்

கூறு 14

1. **பிற்காலக் காப்பியங்கள் :**

   அ) கம்பராமாயணம் **ஆ)** பெரியபராணம்

2. **இணையம் - பற்றிய செய்திகள்**

Course Code	Title of the Course
11321B	Part-I: Communication Skills - II

Unit-1: Code and Content of Communication Skills.

Unit-2: Stimulus and Response of Communication Skills.

Unit-3: Effective Speaking Guidelines.

Unit-4: Pronunciation Etiquette of Communication Skills.

Unit-5: Phonetics in Communication Skills.

Unit-6: A Self-Assessment of Communicating Soft Skills.

Unit-7: Language Skills - Ability - Skill Selected Need - Learner Centre activities.

Unit-8: Listening Skills - Importance - Types of Listening - Interview Skills.

Unit-9: Conversation Skills - Modes.

Unit-10: Presentation Skills - Preparing - Planning - Presentation.

Unit-11: Written Communication - Structure of Effective Sentences - Paragraph.

Unit-12: Technical Writing - Creative Writing - Editing and Publishing.

Unit-13: Corporate Communication Skills - Internal - Effective business writing -Letters, Proposals, Resume.

Unit-14: Corporal Communication Skills - External - Press release - Newsletters- Interviewing skills.

References:

1. Dutt. Kiranmai & Geeta Rajjevan. Basic Communication Skills. Rev.ed. Foundation Books Pvt.Ltd. Cambridge House, New Delhi 2006.
2. Bill R. Swetmon. Communication Skills for the 21st Century. Chennai: Eswar Press. First South Asian Edition 2006.
3. Glass. Lillian. Talk to Win. New York: Perigee Books,1987.
4. Pease. Alan. Signals: How to Use Body Language for Power, Success and Love, New York: Bantam Books, 1981.
5. Walters. Lilly. Secrets of Successful Speakers. New York: McGraw-Hill, Inc., 1993.
6. Mandal. S.K. How to Succeed in Group Discussions & Personal Interviews. Mumbai: JAICO Publishing House.
7. Rogoff. Leonard and Ballenger. Grady. Office Guide to Business Letters, Memos & Reports. New York: Macmillan, 1994.
8. Krishna Mohan & Meera Banerjee, Developing Communication Skills, 2005.
9. Geetha Nagaraj, Write to Communicate, 2004.
10. Wren & Martin, English Grammar and Composition, 2002.
    Rajendra Pal and Kortahalli J S, Essentials of Business Communication.

Course Code	Title of the Course
11322	**PART-II : ENGLISH PAPER - II**

Poetry

Unit – I	Sonnet	- William Shakespeare
Unit – II	Lines Composed upon Westminster Bridge	-William Wordsworth
Unit – III	Grecian Urn	- John Keats (1795-1827)
Unit – IV	Andrea Del Sarto	- Robert Browning (1812-1889)
Unit – V	The Road Not Taken	- Robert Frost (1874-1963)
Unit – VI	Strange Meeting	- Wilfred Owen (1813-1918)
Unit – VII	Gitanjali	- Rabindranath Tagore (1861-1946)
Unit – VIII	The Coromandel Fishers	- Sarojini Naidu
Unit – IX	The Express	- Stephen Spender
Unit – X	**Shakespeare :** The Merchant of Venice	

Language Use:

Unit – XI	Essay writing
Unit – XII	Note Making
Unit – XIII	Report writing
Unit – XIV	Comprehension

References:

1. *The Golden Quill*, P.K. Seshadri, Macmillan.
2. *The Merchant of Venice*, Shakespeare. (Any overseas edition).
3. *Active English Grammar*, Ed. by the Board of Editors, Macmillan.
4. Modern English – *A Book of Grammar Usage and Composition* by N.Krishnaswamy, Macmillan Publishers.

Course Code	Title of the Course
11323	PART-III : ANALYTICAL GEOMETRY AND VECTOR CALCULUS

UNIT -I

Two dimension analytical geometry – Pair of straight lines, angle between pair of lines-Problems.

UNIT -II

Circle, System of circles, Radical axis– co axal system of circles.

UNIT -III

Polar coordinates-Equation of line in polar co-ordinates - Pole and polar conics.

UNIT -IV

Three dimension analytical geometry – Introduction, Direction ratios and direction coefficients – angle between the lines

UNIT -V

Plane – Plane equation – Angle between two planes – Length of the perpendicular – Distance between two planes

UNIT –VI

Straight lines- Equation of a straight line in various forms – problems – Image of a point, Image of a line about a plane.

UNIT -VII

Plane and straight lines – Coplanar lines-Problems.

UNIT -VIII

Cone – Definition – Equation of the Cone in various forms – Equation of a right circular Cone-problems

UNIT -IX

Cylinder – Definition – Equation of a right circular cylinder – simple problems.

UNIT -X

Skew lines – Shortest distance between two skew lines – Problems

UNIT -XI

Sphere – Equation of a sphere, Tangent plane – Problems

UNIT -XII

Equation of a circle on a sphere – Intersection of two spheres.

UNIT -XIII

Vector Calculus – Vector Differentiation– Vector Algebra – Differentiation of vectors - Gradient – Divergence and Curl – Solenoidal – irrotational – Harmonic Vector.

UNIT -XIV

Line and Surface Integrals – Line Integrals – Surface Integrals - Theorems of GREEN, GAUSS and STOKE'S(Statements only) problems.

REFERENCES:

1. Arumugam & Issac, Analytical Geometry 3D and Vector Calculus.
2. Analytical Geometry 3D and Vector Calculus by Dr. M.K.Venkataraman and Mrs. Manorama Sridhar, National Publishing Company, Chennai, 2001..
3. T.K. Manickavasagam Pillai & T.Natarajan, A text book of Analytical Geometry Part II-3D, S.Viswanathan(Printers & Publishers) Pvt. Ltd., 2001.
4. S.Narayanan & T.K. Manickavasagam Pillai, Vector Algebra & Analysis, S.Viswanathan(Printers & Publishers) Pvt. Ltd. 1995.

Course Code	Title of the Course
11324	**PART-III : SEQUENCES AND SERIES**

UNIT -I

Sequences – bounded sequences – Monotonic sequences.

UNIT -II

Convergent sequences – Cauchy's general principle of convergence - Cauchy's first theorem on Limits.

UNIT -III

Divergent and Oscillating sequences – Some problems.

UNIT -IV

The algebra of limits- Limit superior and Limit inferior.

UNIT -V

Behaviour of monotonic sequences – Some Theorems on limits

UNIT -VI

Subsequences – limit points –Some problems.

UNIT -VII

Cauchy sequences – The upper and lower limits of a sequence.

UNIT -VIII

Series of positive terms –infinite series – Some related problems.

UNIT -IX

Comparison test – Kummer's test – Root test – Simple problems.

UNIT -X

Cauchy's condensation Test, Cauchy's root test and their simple problems – Integral test – Problems.

UNIT -XI

Series of arbitrary terms – Alternating series – Problems.

UNIT -XII

Absolute convergence – Tests for convergence of series of arbitrary terms

UNIT -XIII

Rearrangement (Derangement) of Series – Multiplication of series.

UNIT -XIV

General summation of series including successive difference and recurring series.

REFERENCES:

1. Sequences and Series by Dr. S.Arumugam and Prof. A.ThangapandiIssac, New Gamma Publishing House, Palayamkottai, December 2015.
2. M.K.Singal & Asha Rani Singal, A first course in Real Analysis, R. Chand & Co. 1999.
3. Dr.S.Arumugam, Sequences & Series, New Gamma Publishers, 1999.

THIRD SEMESTER

Course Code	Title of the Course
11331A	Part-I: Tamil Paper - III

கூறு 1: பத்துப்பாட்டு – முல்லைப்பாட்டு

கூறு 2: எட்டுத்தொகை – ஐங்குறுநூறு

கூறு 3: கபிலர் - குறிஞ்சித்திணை

கூறு 4: மஞ்ஞைப்பத்து – முதல் மூன்று பாடல்கள்

கூறு 5: குறுந்தொகை – பரணர் பாடல்கள் பா. எண். 19, 24, 36, 128, 399

கூறு 6: நற்றிணை – பெருங்குன்றூர்கிழார் - பா. எண். 5
பெருவழுதியார் - பா. எண். 55
பெருங்கௌசிகனார் - பா. எண். 139

கூறு 7: நற்றிணை – கருவூர்க்கோசிகனார் - பா. எண். 214
உலோச்சனார் - பா. எண் 249

கூறு 8: அகநானூறு – சேந்தம்பூதனார் பாடல்கள் பா.எண். 84, 207

கூறு 9: புறநானூறு – மறோக்கத்து நப்பசலையார் பாடல்கள்
பா. எண். 37, 39, 126, 226, 280

கூறு 10: பதினெண் கீழ்க்கணக்கு – திருக்குறள் - வாழ்க்கைத் துணை நலம் (6), அறிவுடைமை (43), பிரிவாற்றாமை (116)

கூறு 11: நான்மணிக்கடிகை – எள்ளற்க (3), பறைபடவாழா (4),

கூறு 12: நான்மணிக்கடிகை - மண்ணயறிப (5),கள்ளிவயிற்றில் (6), கல்லிற்பிறக்கும்(7)

கூறு 13: நாடகம் - இராசராசசோழன் - அரு. இராமநாதன்

கூறு 14: நாவல் - சுவடுகள் - இரா. பாலசுப்பிரமணியன், சத்யா வெளியீடு, மதுரை.

Course Code	Title of the Course
11331B	Part-I: Human Skills Development - I

Unit-1: Human Skills - Developing skills - Types.

Unit-2: Mind-Levels of functions - Habits - Meaning - Types - Merits of good habits - Interpersonal Relationship - Features - Interpersonal Behaviour.

Unit-3: Thinking ahead - Significance of thinking ahead.

Unit-4: Developing Personality - Meaning - Need - Factors influencing personality, Ways of developing personality - Building positive personality.

Unit-5: Self-concept - Self-esteem - Meaning-Importance - Self-efficacy - Self-acceptance -Meaning-Importance - Etiquette - Meaning - Etiquettes in using mobile, telephones - Dais Etiquette.

Unit-6: Goal - setting Skills - Meaning - Types - Importance.

Unit-7: Decision-making skills - Meaning - Types - Steps in decision-making

Unit-8: Negotiating Skills - Styles - Structure - Creating negotiation - Competitive Negotiation.

Unit-9: Attitudes - Meaning - Types - Importance - Developing positive attitudes.

Unit-10: Coping with Change - Meaning - Characteristics - Importance of change Resistance to change - Dealing with change.

Unit-11: Leadership - Meaning - Importance - Characteristics - Styles.

Unit-12: Human Relations Skill - Need - Canons of good human relations.

Unit-13: Counselling - Meaning - Importance - Forms - Conflicts - Meaning - Types - Causes - Effects - Managements of conflicts

Unit-14: Stress-Meaning - Types - Causes - Effects - Managing the stress - Anger - Meaning - Causes - Consequences - Anger Management.

References:

1. Les Giblin, Skill with People, 1995.
2. Shiv Khera, You Can Win, 2002.
3. Christian H Godefroy, Mind Power.
4. Dale Carnegie, How to Enjoy Your Life and Your Job, 1985.
5. Natalie H Rogers, How to Speak without Fear, 1982.
6. Dale Carnegie, How to Develop Self-Confidence and Influence People by Public Speaking.

Course Code	Title of the Course
11332	**PART-II : ENGLISH PAPER - III**

Short Stories

Unit – I	A Hero	- R.K. Narayanan
Unit – II	The Diamond Necklace	- Guy de Maupassant
Unit – III	The Verger	- Somerset Maugham
Unit – IV	The Postmaster	- Rabindranath Tagore

One Act Plays

Unit – V	The Proposal	- Anton Chekhou
Unit – VI	The Boy Comes Home	- A.A. Milne
Unit – VII	The Silver Idol	- James R. Waugh
Unit – VIII	Progress	- St. John Ervine
Unit – IX	The Pie and the Tart	- Huge Chesterman
Unit – X	Reunion	- W.st. Joh Tayleur
Unit – XI	A kind of Justice	- Margaret Wood
Unit – XII	The Refugee	- Asif Currimbhoy

Grammar

Unit – XIII Parts of speech-Noun- Pronoun- Adjective Degrees of Comparison- Verb- Adverb

Composition

Unit – XIV Agenda- Minutes- Notice- Descriptive Writing

REFERENCES:

1. *Aroma*, Ed. by the Board of Editors, Publishers- New Century Book House, Chennai.
2. *Six Short Stories*, Ed. by the Board of Editors, Harrows Publications, Chennai.
2. *One Act Plays*, Ed. by the Board of Editors, Harrows Publications, Chennai.
3. Modern English – *A Book of Grammar Usage and Composition* by N.Krishnaswamy, Macmillan Publishers.
4. *English for Communication*, Ed. by the Board of Editors, Harrows Publications, Chennai.

Course Code	Title of the Course
11333	PART-III : DIFFERENTIAL EQUATIONS AND ITS APPLICATIONS

UNIT - I

Exact Differential Equations – Conditions for equation to be exact –Working rule for solving it – problems

UNIT - II

Equations of the first order but of higher degree – Equations solvable for p, x, y, clairaut's form-Equations that do not contain (i) x explicitly (ii) y explicitly.

UNIT - III

Equations homogenous in x and y - Linear Equation with constant coefficients- Problems.

UNIT- IV

Linear equations with variable coefficients – Equations reducible to the linear equations.

UNIT - VI

Simultaneous Differential Equations – First order and first degree – Simultaneous linear Differential Equations.

UNIT - VII

Linear equations of the second order – Complete Solution given a nown integral-Problems.

UNIT - VIII

Reduction to Normal form – Change of the independent variable-Problems.

UNIT - IX

Variation of parameters – Total Differential Equations – Problems.

UNIT - X

Necessary and Sufficient condition of integrability of Pdx + Qdy + Rdz = 0 -Rules - Problems.

UNIT - XI

Partial Differential Equations of the First oder – classifications of integrals

UNIT - XII

Derivations of Partial Differential Equations – Special methods – Standard forms – Charpit's method.

UNIT - XIII

Flow of water from an Orifice – Falling bodies and other rate problems – Brachistochrone Problem

UNIT- XIV

Tautochronous property of the Cycloid – Trajectories-Problems.

REFERENCE BOOKS:

1. Differential Equations and its Applications by S.Narayanan & T.K.ManickavachagomPillay, S.Viswanathan (Printers& Publishers) Pvt. Ltd., 2015.
2. Differential Equations and its Applications by Dr. S.Arumugam and Mr. A.Thangapandi Issac, New Gamma Publishing House, Palayamkottai, Edition, 2014.

Course Code	Title of the Course
11334	PART-III : MECHANICS

UNIT -I

Forces acting at a point – Resultant and Components – Definition – Simple cases of finding the resultant – Parallelogram law of forces – Analytical Expression for the resultant of two forces acting at a point

UNIT -II

Triangle of forces – Perpendicular Triangle of forces – Converse of Triangle of forces.

UNIT- III

The polygon of forces – Lami's Theorem – An Extended form of the parallelogram law of forces

UNIT -IV

Resolution of a force – Theorems on resolved parts – Resultant of any number of coplanar forces – Condition of equilibrium.

UNIT -V

Forces acting on a rigid body: Parallel forces – Resultant of two like and unlike parallel forces – Moment of a force – Varigon's theorem

UNIT -VI

Couples– Equivalence of two couples- couples in parallel planes-Resultant of a couple and a plane.

UNIT -VII

Three forces acting on as rigid body –Three coplanar forces, conditions of equilibrium- two trigonometrical theorems and simple problems.

UNIT -VIII

Friction- Statical, dynamical and limiting friction-Laws of friction –Coefficient of friction – Angle of friction – Cone of friction – Problems.

UNIT -IX

Uniform string under the action of gravity – Equation of the common catenary – axis, vertex, directrix, span and sag – Tension at any point – Important formulae – Geometrical properties of the Common Catenary

UNIT -X

Projectile – Definition – fundamental principles – path of the projectile – Characteristics of the motion of a projectile – Range on an inclined plane – greatest distance maximum range

UNIT -XI

Impulsive force – Impulse – Impact of two bodies – Loss of Kinetic energy in Impact – Collision of elastic bodies – Fundamental laws of Impact – Newton's experimental law – Impact of a smooth sphere on a fixed smooth plane,

UNIT -XII

Direct Impact of two smooth spheres – Loss of kinetic energy due to direct impact – Oblique impact of two smooth spheres – Loss of kinetic energy due to oblique impact.

UNIT -XIII

Motion under the action of Central forces – S.H.M – Equation of motion – Velocity and acceleration – Equation of motion in Polar Coordinates – Note on

equiangular spiral – Motion under a central force

UNIT -XIV

Central Orbits - Differential Equation of Central Orbits – Perpendicular from the pole on the tangent. Formulae in Polar Coordinates – Pedal Equation of the central orbit – Pedal equation of some of the well known curves – Velocities in a central orbit – Two folded problems.

REFERENCE BOOKS:

1. Dr. M.K. Venkataraman, Statics, Agasthiar Publications, 17th Edition, 2014.
2. Dr. M.K. Venkataraman, Dynamics, Agasthiar Publications, 13th Edition, 2009.
3. P.Duraipandian, Laxmi Duraipandian & Muthamizh Jayapragasam, Mechanics, S.Chand & Co. Pvt. Ltd, 2014.

FOURTH SEMESTER

Course Code	Title of the Course
11341A	Part-I: Tamil Paper - IV

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கூறு 1: செய்யுள் உறுப்புகள் - யாப்பு - எழுத்து, அசை, சீர்,

கூறு 2: செய்யுள் உறுப்புகள் - யாப்பு - தளை, அடி, தொடை

கூறு 3: வெண்பா, ஆசிரியப்பா, கலிப்பா, வஞ்சிப்பா,

கூறு 4: புதிய யாப்பு வடிவங்கள் - சிந்து, கண்ணி, கீர்த்தனை

கூறு 5: புதுக்கவிதையில் குறியீடு – படிமம்.

கூறு 6: அகப்பொருள் - புறப்பொருள் - ஐந்திணை விளக்கம்

கூறு 7: அகப்பொருள் துறைகள் - வரைவு கடாதல், அறத்தொடு நிற்றல், உடன்போக்கு

கூறு 8: புறப்பொருள் துறைகள் - வஞ்சினக்காஞ்சி, கையறுநிலை, செவியறிவுறூஉ

கூறு 9: அணி இலக்கணம் - உவமை, உருவகம், வேற்றுமை, பிறிது மொழிதல், தற்குறிப்பேற்றம், சிலேடை, பின்வருநிலை.

கூறு 10: நிறுத்தல் குறிகள்.

கூறு 11: தொல்காப்பியம் - சங்கஇலக்கியம் - எட்டுத்தொகை, பத்துப்பாட்டு,

கூறு 12: பதினெண்கீழ்க்கணக்கு.

கூறு 13: ஐம்பெருங்காப்பியங்கள் - பிற்காலக் காப்பியங்கள் - கம்பராமாயணம் - பெரியபுராணம்.

கூறு 14: இக்காலக் காப்பியங்கள் - பாரதியின் பாஞ்சாலி சபதம் - பாரதிதாசனின் பாண்டியன் பரிசு - கண்ணதாசனின் இயேசு காவியம் , சிற்பியின் - மௌன மயக்கங்கள்.

Course Code	Title of the Course
11341B	Part-I: Human Skills Development - II

Unit-1: Guidance & Counselling - Role of Counsellor - Importance and Techniques of counselling.

Unit-2: Managerial skill - Need - Importance.

Unit-3: Human relational skills - Communication - Attention.

Unit-4: Conceptual skills - Meaning - Importance.

Unit-5: Technical skills - Techniques - Practices - Tools - Procedures.

Unit-6: Presentation skills - Planning - Preparation - Delivery.

Unit-7: Organization skills - Meaning - Nature - Importance - Types.

Unit-8: Multi-Tasking skills Responsibilities - Causes.

Unit-9: Leader - Qualities of a good leader.

Unit-10: Understanding Skills - Human systems: Individual, Group, organization, and their major interactions.

Unit-11: Understanding Skills - Human systems: Community and Society, and their major interactions.

Unit-12: Problem solving skills - Handling - Facing - Importance.

Unit-13: Cooperative Learning Skills.

Unit-14: Making Social Responsibilities - Causes.

References:

1. Les Giblin, Skill with People, 1995.
2. Shiv Khera, You Can Win, 2002.
3. Christian H Godefroy, Mind Power.
4. Dale Carnegie, How to Enjoy Your Life and Your Job, 1985.
5. Natalie H Rogers, How to Speak without Fear, 1982.
6. Dale Carnegie, How to Develop Self-Confidence and Influence People by Public Speaking.

Course Code	Title of the Course
11342	PART-II : ENGLISH PAPER - IV

Short Stories

Unit – I Lalajee - Jim Corbelt

Unit – II A Day's Wait - Hemmingway

Unit – III Two old Men - Leo Tolstoy

Unit –IV Little Girls wiser than - Men Tolstoy

Unit – V Boy who Wanted more Cheese - William Elliot Griffir

Drama

Unit – VI Pygmalion - G.B. Shaw

Fiction

Unit – VII Swami and Friends - R.K. Narayanan

Tales from Shakespeare

Unit – VIII - The Merchant of

Venice

Unit – IX - Romeo and Juliet

Unit – X - The Winter's Tale

Biographies

Unit – XI - Martin-Luther king - R.N. Roy

Unit – XII - Nehru - A.J. Toynbee

Grammar

Unit – XIII - Concord- Phrases and Clauses-Question Tag

Composition

Unit – XIV - Expansion of Proverbs
- Group Discussion
- Conversation (Apologizing, Requesting, Thanking)

REFERENCES:

1. *Sizzlers*, by the Board of Editors, Publishers- :Manimekala Publishing House, Madurai.
2. *Pygmalion* – G.B. Shaw
3. *Swami and Friends* – R.K. Narayan
4. *Tales from Shakespeare* Ed. by the Board of Editors, Harrows Publications, Chennai.
5. Modern English – *A Book of Grammar Usage and Composition* by N.Krishnaswamy, Macmillan Publishers.

Course Code	Title of the Course
11343	**PART-III : ANALYSIS**

UNIT -I

Introduction – Sets and functions – Countable and Uncountable sets – Inequalities of Holder and Minkowski.

UNIT -II

Metric spaces: Definition and examples – Limits in metric spaces – Continuous functions on metric spaces.

UNIT -III

Functions continuous at a point in the real line – Reformulation -Bounded sets in Metric space- Problems.

UNIT -IV

Subspace – Interior of a set – Open sets- Closed sets – Closure – limit point – Dense sets -Problems

UNIT -V

Complete Metric spaces: Introduction- Completeness - Baire's Category theorem.

UNIT -VI

Continuity – Homeomorphism – Uniform continuity.

UNIT- VII

Differentiability of a function –Derivability & Continuity –Algebra of derivatives – Inverse Function Theorem – Daurboux's Theorem on derivatives.

UNIT -VIII

Rolle's Theorem –Mean Value Theorems on derivatives- Taylor's Theorem with remainder- Power series expansion .

UNIT - IX

Riemann integration –definition – Daurboux's theorem –conditions for integrability – Integrability of continuous & monotonic functions.

UNIT - X

Integral functions –Properties of Integrable functions - Continuity & derivability of integral functions –The First Mean Value Theorem and the Fundamental theorem of Calculus.

UNIT -XI

Contraction mapping - Definitions and Examples - Contraction mapping theorem- Applications.

UNIT - XII

Connectedness: Introduction - Connectedness definition and examples- Connected subsets of R- Connectedness and Continuity.

UNIT - XIII

Compactness: Introduction- Compact metric spaces – Continuous functions on compact metric spaces – Continuity of the inverse function – Uniform continuity.

UNIT - XIV

Sequence of functions and Series of functions- Pointwise convergent-Cauchy criterion for uniform convergence.

REFERENCE BOOKS:

1. Arumugam & Issac, Modern Analysis, New Gamma Publishing House, Palayamkottai, 2010.
2. Richard R. Goldberg, Methods of Real Analysis, Oxford & IBH Publishing Company, New Delhi.
3. D.Somasundaram & B.Choudhary, A first course in Mathematical Analysis, Narosa Publishing House, Chennai.
4. M.K,Singhal & Asha Rani Singhal , A First Course in Real Analysis, R.Chand & Co., June 1997 Edition.
5. Shanthi Narayan, A Course of Mathematical Analysis, S. Chand & Co., 1995

Course Code	Title of the Course
11344	PART-III : STATISTICS

UNIT- I

Central Tendencies – Introduction – Arithmetic Mean – Partition Values – Median-Mode

UNIT- II

Geometric Mean and Harmonic Mean – Related problems

UNIT- III

Measures of Dispersion-Problems.

UNIT- IV

Moments – Skewness and Kurtosis

UNIT-V

Curve fitting – Goodness of fit-Problems

UNIT-VII

Principle of least squares-Problems.

UNIT- VIII

Correlation – Rank correlation Regression –Related problems

UNIT-IX

Correlation Coefficient for a Bivariate Frequency Distribution.

UNIT-X

Interpolation – Finite Differences – Newton's Formula – Lagrange's Formula- Problems

UNIT-XI

Attributes – Consistency of Data – Independence and Association of Data -Problems

UNIT-XII

Index Numbers – Consumer Price Index Numbers – Problems.

UNIT –XIII

Analysis of Time series – Time series– Components of a Time series

UNIT- XIV

Measurement of Trends-Related problems.

REFERENCE BOOKS:

1. Arumugam & Issac, Statistics, New Gamma Publishing House, 2007.
2. S.P.Gupta, Statistical Methods, Sultan Chand & Sons, 37th Edition, 2008.
3. Statistics by Dr. S. Arumugam and Mr. A.ThangapandiIssac, New Gamma Publishing House, Palayamkottai, June 2015.

SEMESTER – V

Course Code	Title of the Course
11351	PART-III : MODERN ALGEBRA

UNIT- I

Set theory- Sets and mappings- concept of a set – Set inclusion- union, intersection of sets- Difference of sets- Complement of a set- Symmetric difference of two sets – Cartesian product of sets

UNIT- II

Relations – Equivalence relations – Partial order relations – Functions - Binary operations

UNIT- III

Group theory: Definition and examples – Properties – Permutation groups, Examples and problems

UNIT- IV

Subgroups – Cyclic groups – Order of an element-Problems.

UNIT- V

Cosets – Lagrange's theorem – Index of a subgroup- Euler's theorem- Fermat's theorem-Problems

UNIT-VI

Normal subgroups and Quotient groups - Homomorphism – Fundamental theorem of homomorphism

UNIT -VII

Isomorphism-Cayley's theorem- Automorphism-Problems.

UNIT -VIII

Ring theory: Definition and examples – Properties of rings– Isomorphism – Types of rings.

UNIT -IX

Integral domains – Fields – Characteristic of a ring – Subrings.

UNIT- X

Quotient Ring– Maximal and prime ideals –Ring homomorphism-Fundamental theorem of Ring homomorphism

UNIT- XI

Quotient field – Euclidean ring – Properties – Polynomial rings- Gauss lemma-Eisenstein' criterion.

UNIT -XII

Vector spaces: Definition and examples – Properties of vector space-Problems.

UNIT- XIII

Subspaces – Linear independence – Span of a set – Basis and dimension – Rank and nullity of a linear transformation

UNIT -XIV

Inner product spaces: Definition and examples– Orthogonality – Orthogonal complement.

REFERENCES:

1. Arumugam & Issac, Modern Algebra, Scitech Publications(India) Pvt. Ltd. , 2008.
2. A.R. Vasistha, Modern Algebra, Krishna Prakashan Mandir, Meerut, 1994-95.
3. T.K.Manickavasagam Pillai, T.Nagarajan & K.S.Ganapathy, Algebra Vol.I, S.Viswanathan (Printers & Publishers) Pvt. Ltd., 2012.

Course Code	Title of the Course
11352	PART-III : OPERATIONS RESEARCH

UNIT -I

Introduction – Origin and Development of O.R – Nature and features of O.R. – Scientific Methods in O.R. – Modelling in O.R. – Advantages and Limitations of Models – General solution methods of O.R. models – Applications of Operations Research

UNIT -II

Linear Programming problem – Mathematical formulation of the problem – Illustration on Mathematical formulation of linear programming problems – Graphical solution method – Some exceptional cases.

UNIT- III

General linear programming problem – Canonical and Standard forms of L.P.P – Simplex method.

UNIT- IV

Linear programming using artificial variables- Big M method – Two Phase method- Problems

UNIT -V

Duality in linear programming – General primal and dual pair – Formulating a Dual problem – Primal – Dual pair in matrix form – Duality Theorems – Complementary Slackness Theorem.

UNIT -VI

Integer Programming – Cutting plane technique, Dual simplex method.

UNIT -VII

Introduction – L.P. formulation of T.P. – Existence of solution in T.P. – The Transportation table – Loops in T.P. – Solution of a Transportation problem – Finding an initial basic – feasible solution (NWCM – LCM – VAM).

UNIT -VIII

Degeneracy in TP – Transportation Algorithm (MODI Method) – Unbalanced T.P – Maximization T.P.

UNIT- IX

Assignment problem – Introduction – Mathematical formulation of the problem – Test for optimality by using Hungarian method – Maximization case in Assignment problem

UNIT- X

Sequencing problem – Introduction – problem of sequencing – Basic terms used in Sequencing– n jobs to be operated on two machines – problems - n jobs to be operated on K machines–problems–Two jobs to be operated on K machines (Graphical method)–problems.

UNIT -XI

Game Theory – Two person Zero – Sum Games – Basic terms – Maximin – Minimax Principle.

UNIT -XII

Games without saddle points – Mixed strategies – Graphical solution of $2\times n$ and $m \times 2$ games

UNIT -XIII

Dominance Property – General solution of m×n rectangular games-Problems.

UNIT -XIV

Network Scheduling by PERT / CPM – Network Basic components – Drawing network – Critical path Analysis – PERT Analysis – Distinction between PERT and CPM.

REFERENCES:

1. R.S.Arumugam, Operations Research, New Gamma Publications, 2006.
2. V.Sundaresan, K.S.Ganapathy & K.Ganesan, Resource Management Techniques(Operations Research), A.R.Publications.
3. Kanti Swarup, P.K. Gupta & Man Mohan, Sultan Chand & Sons, 13th Edition, 2007.

Course Code	Title of the Course
11353	**PART-III : NUMERICAL ANALYSIS**

UNIT -I

Algebraic & Transcendental and polynomial equations: Bisection method, Iteration method, Method of false position, Newton-Raphson method.

UNIT -II

System of linear equations: Matrix inversion method, Cramer's rule, Guass elimination method, Guass-Jordan elimination method, Triangularisation method.

UNIT- III

Solutions to Linear Systems –Jacobi & Gauss Siedal iterative methods – Theory & problems.

UNIT -IV

Interpolation: Graphic method- Finite differences – Forward and Backward differences –Central differences- Fundamental theorem of finite differences.

UNIT- V

Interpolating Polynomials using finite differences- Other difference operators.

UNIT -VI

Lagrange and Newton interpolations-Applications.

UNIT -VII

Divided differences and their properties – Application of Newton's General Interpolating formula.

UNIT -VIII

Central differences Interpolation formulae - Guass formulae, Stirlings formula, Bessel's formula, Everett's formula, Hermite's formula.

UNIT -IX

Numerical differentiation - Methods based on interpolation-Problems.

UNIT -X

Numerical differentiation - Methods based on finite differences-Problems.

UNIT -XI

Numerical integration, Trapezodial rule, Simpson's 1/3 rule, Simpson's 3/8 rule, Weddle's rule, Cote's method.

UNIT -XII

Numerical solutions of ordinary differential equations: Taylor's series method, Picard's method, Euler's method, Runge-Kutta method

UNIT -XIII

Numerical solutions of ordinary differential equations using Runge Kutta 2nd and 4th order methods (Derivation of the formula not needed) - Theory & problems

UNIT- XIV

Predictor-Corrector methods-Milne's Predictor Corrector Methods-Adam's Predictor Corrector Method

REFERENCES:

1. Arumugam, Issac & Somasundaram, Numerical Methods, Scitech Publications(India) Pvt. Ltd., 2nd Edition, 2010.
2. P.P.Gupta & G.S.Malik, Calculus of finite differences and Numerical Analysis, Krishnaprakasham Mandhir, Meerut.
3. Dr.M.K.Venkatraman, Numerical Methods in Science and Engineering.

Course Code	Title of the Course
11354	**PART-III : TRANSFORM TECHNIQUES**

UNIT- I

Laplace Transform – Definition – Laplace Transform of Standard functions – Elementary Theorems.

UNIT- II

Laplace Transform of periodic functions – problems.

UNIT- III

Inverse Laplace Transforms – Standard formulae – Basic Theorems –Problems.

UNIT- IV

Solving Ordinary Differential Equations with constant coefficients using Laplace Transform-Problems.

UNIT- V

Solving Ordinary Differential Equations variable coefficients -using Laplace Transform-Problems.

UNIT- VI

Solving Simultaneous linear equations using Laplace Transform-Problems.

UNIT- VII

Fourier Series – Definition – To find the Fourier coefficients of Periodic functions of period 2π.

UNIT- VIII

Even and odd functions in Fourier series – Half range Fourier series – problems.

UNIT- IX

Fourier Transforms – Complex form of Fourier Integral Formula – Fourier Integral theorem.

UNIT- X

Properties of Fourier Transform – Fourier sine and cosine Transforms – Properties.

UNIT- XI

Parsival's Identity In Fourier Transforms- Problems.

UNIT- XII

Z Transforms – Definition – Proprieties – Z Transforms of some basic functions – Problems.

UNIT- XIII

Inverse Z Transforms – Methods to find the inverse Z Transform – Use of Z Transforms.

UNIT- XIV

Transforms to solve finite Difference Equations – problems.

REFERENCES:

1. Calculus Volume III by S.Narayanan and T.K.ManicavachagomPillay, S.Viswanathan (Printers & Publishers) Pvt. Ltd., 2014.
2. Engineering Mathematics 3rd Edition by T.Veerarajan, Tata McGraw Hill Publishing Company Limited, New Delhi.

SEMESTER – VI

Course Code	Title of the Course
11361	PART-III : DISCRETE MATHEMATICS

UNIT -I

Logic introduction – Connectives – Atomic and compound statements – Truth table – Problems.

UNIT- II

Tautology – Tautological implications and equivalence of formulae – Replacement Process- Law of duality- Tautological implications.

UNIT -III

Normal forms – Principal normal forms-Problems.

UNIT- IV

Theory of inference- Rules of inference-Open statements –Problems

UNIT- V

Quantifiers – bound and free variables -Theory of inference for predicate calculus.

UNIT-VI

Relations – Representation of a relation – Operations on relations – Equivalence relation.

UNIT -VII

Lattices – Some properties of Lattices, New Lattices – Modular and Distributive Lattices -Boolean Algebra, Boolean Polynomials.

UNIT -VIII

Coding theory – Introduction – Hamming Distance – Encoding a message – Group codes – Procedure for Generating Group codes – Decoding and Error correction.

UNIT -IX

Definition of a Graph – finite & infinite graphs – incidence, degree isolated & pendent vertices – isomorphisms –sub graphs – walks , paths & circuits –Connected & disconnected graphs.

UNIT -X

Matrix representation of a graph – Incidence matrix –Circuit Matrix - Fundamental Circuit Matrix and rank of the circuit matrix – Cut set matrix – adjacency matrix.

UNIT- XI

Chromatic Number - Chromatic partitioning – Chromatic polynomial-Problems.

UNIT -XII

Trees –properties of trees –pendent vertices in a tree – distances & centres in a tree – Rooted & binary trees.

UNIT -XIII

Spanning trees –Fundamental circuits – Finding all spanning trees of a Graph – Spanning trees in a weighted graph.

UNIT -XIV

Cut sets – Properties of a Cut set – all Cut sets in a graph – Fundamental circuits & Cut sets –Connectivity & separability - Eulerian and Hamiltionian graphs –Problems.

REFERENCES:

1. Venkatraman, Sridharan and Chandrasekaran, Discrete Mathematics, National Publishing House, Chennai, 2003.
2. J.P. Trembley and R.P. Manohar, Discrete Mathematics Structures with applications to Computer Science, Mc.Graw Hill Interamericana,, 1975.
3. S.Arumugam & S.Ramachandran, Scitech Publications, Chennai, 2001.
4. V.K.Balakrishnan, Introductory Discrete Mathematics, Dover Publications, INC. Newyork.
5. A First course in Graph Theory by S.A. Choudum, Macmillan India Ltd. New Delhi, 1987.

Course Code	Title of the Course
11362	PART-III : FUZZY ALGEBRA

UNIT-I

Fuzzy sets – Basic types – Basic concepts - α - cuts – Additional prosperities of α - cuts – Extension principle for Fuzzy sets.

UNIT- II

Operations on Fuzzy sets – Types of operations – Fuzzy complements – Fuzzy Union and intersections.

UNIT- III

Combinations of operations – Fuzzy Arithmetic – Fuzzy numbers

UNIT- IV

Arithmetic operations on intervals – Arithmetic operations on Fuzzy numbers.

UNIT- V

Fuzzy relations – Binary fuzzy relations – Fuzzy equivalence and similarity relations – Fuzzy compatibility relations.

UNIT- VI

Fuzzy ordering relations – fuzzy morphisms.

UNIT- VII

Fuzzy measures-Belief and Plausibility measures- Probability measures- Problems.

UNIT- VIII

Possibility measures- Necessity measures- Relationship among classes of fuzzy measures.

UNIT- IX

Types of uncertainity- Measures of fuzziness-Problems.

UNIT- X

Classical measures of uncertainity-Hartley information-Shannon Entropy-Boltzmann Entropy.

UNIT- XI

Measures of Dissonance- Body of evidence-Consonant body of evidence-Problems.

UNIT- XII

Measures of confusion-entropy like measures-Problems.

UNIT- XIII

Measures of nonspecificity - U- uncertainity –Problems.

UNIT- XIV

Uncertain and Information- syntactic, semantic, pragmatic-Problems.

REFERENCES:

1. George J.Klir and Bo Yuan, Fuzzy Sets and Fuzzy Logic, Theory and Applications, Prentice Hall Inc., New Jersey. 1995.
2. George J.Klir and Tina A. Folger, Fuzzy sets, Uncertainity and Information, Prentice Hall of India, New Delhi, 2007.
3. H.J.Zimmermann, Fuzzy Set Theory and its Applications, Allied Publishers Limited, New Delhi, 1991.

Course Code	Title of the Course
11363	**PART-III : COMPLEX ANALYSIS**

UNIT- I

The geometric representation of a complex number – the spherical representation and stereographic projection.

UNIT- II

Definitions of complex analytic function-Complex differentiation. The Cauchy- Riemann equations.

UNIT-III

Orthogonal trajectories and harmonic functions- Harmonic and Conjucate harmonic – To find an analytic function f(z)=u+iv if a harmonic function u is given – Milne Thomson method– Problems.

UNIT- IV

Power series – radius of convergence – Abel's limit theorem-Examples.

UNIT- V

Generating functions - Fibonacci numbers - An application of power series.

UNIT- VI

Conformal mappings – Bilinear transformations – Fixed point of bilinear transformations – Cross ratio.

UNIT- VII

General bilinear transformations which transforms unit disk onto the unit disk; half plane Im(z)□ the unit disk.

UNIT onto -VIII

Complex integration – Cauchy's theorem for a rectangle and for a disk – The index of a point with respect to a closed curve – Cauchy's integral formula-Problems.

UNIT- IX

Higher derivatives in complex integration– Taylor's theorem – Problems.

UNIT- X

Zeros and poles– The local mapping theorem – The maximum principle – Schwarz's lemma – Morera's theorem – Cauchy's estimate – Liouville's theorem.

UNIT- XI

The minimum- maximum theorem, Fundamental theorem of algebra.

UNIT -XII

Series expansions – Taylor's Series , Laurent series – Laurent's theorem- Problems.

UNIT- XIII

Singularities – Cauchy's residue theorem – The argument principle – Rouche's theorem –Problems.

UNIT- XIV

Evaluation of definite integrals for unit circles, Poles lie in the upper half of the plane and real axis.

REFERENCES:

1. Arumugam, Issac & Somasundaram, Complex Analysis, Scitech Publications(India) Pvt. Ltd., 2004.
2. T.K.Manickavasagam Pillai & others, S.Viswanathan(Printers & Publishers) Pvt. Ltd., Chennai, 1997.
3. P.Duraipandian & others, Complex Analysis, Emarald Publishers, Chennai.
4. V.Karunakaran, Complex Analysis, Alpha Science International Ltd., Harrow, U.K, Second Edition, 2005.
5. P.P Gupta – Kedarnath & Ramnath, Complex Variables, Meerut -Delhi
6. J.N. Sharma, Functions of a Complex variable, Krishna Prakasan Media (P) Ltd, 13th Edition, 1996-97.

Course Code	Title of the Course
11364	**PART-III : COMBINATORICS**

UNIT- I

Basic Combinatorial Numbers – Stirling Numbers of the First kind– Stirling Numbers of the Second kind.

UNIT-II

Recurrence Formula for S_n^m – Recurrence formula for P_n^m.

UNIT-III

Patterns of Distributions-Problems.

UNIT-IV

Generating Functions – The Algebra of Formal Power Series – Generating functions for Permutations – Generating functions for Partitions.

UNIT-V

Inventory of Maps – Recurrence Relations.

UNIT-VI

Symmetric functions – The Monomial Symmetric functions K_λ– The complete Homogeneous Symmetric Functions h_λ.

UNIT-VII

The Elementary Symmetric Functions a_λ – The Power sum Symmetric Function s_λ.

UNIT-VIII

Multinomials- Basic concepts- Problems.

UNIT-IX

Inclusion and Exclusion Principle –Theorems and Problems.

UNIT-X

Permutations with Forbidden Positions – The Menage problem

UNIT-XI

Problem of Fibonacci – Polya Theory – Problems

UNIT-XII

Necklace problem and Burnside's Lemma – Cyclic Index of a Permutation Group.

UNIT-XIII

Polya's Theorems and their Immediate Applications – Related problems.

UNIT-XIV

Binary operations on Permutation Groups.

REFERENCES:

1. Combinatorics Theory and Applications by V.Krishnamurthy, Affliated East-West Press Private Limited, New Delhi, 1985.
2. A First Course in Combinatorial Mathematics by IanAnderson, Oxford Applied Mathematics and Computing Science Series, U.K., 1974
3. Combinatorics by V.K.Balakrishnan, Schuam Series, 1996

7. B.Sc (Information Technology)

Duration – Three years
Eligibility – A pass in HSC (or) 3 year Diploma
Medium – English

Lateral Entry to II year B.Sc.(IT):

3 year Diploma in Computer Science Engineering / EEE / ECE / IT / Electronics/ Robotics.

Course of Study & Scheme of Examinations

S.No	Course code Bsc(IT)	Bsc(IT) (LE)	Name of the Course	*CIA Marks* Max.	*ESE Marks* Max.	*Total Marks* Max.	*C* Max.
FIRST YEAR							
SEMESTER I							
1	12911A 12911B		**Part I:** Tamil Paper I / Communication Skills – I	25	75	100	4
2	12912		**Part II:** English Paper – I	25	75	100	4
3	12913		**Part III: Core Paper:** Principles of Information Technology	25	75	100	4
4	12914		**Part III: Core Paper:** Office Automation Lab	25	75	100	4
			TOTAL	**100**	**300**	**400**	**16**
SEMESTER II							
5	12921A 12921B		**Part I:** Tamil Paper II: / Communication Skills – II	25	75	100	4
6	12922		**Part II:** English Paper – II	25	75	100	4
7	12923		**Part III: Core Paper:** Programming in C and Data Structures	25	75	100	4
8	12924		**Part III: Core Paper:** C AND Data structure Lab	25	75	100	4
			TOTAL	**100**	**300**	**400**	**16**
SECOND YEAR							
SEMESTER III							
9	12931A 12931B	13131A 13131B	**Part I:** Tamil Paper III / Human Skills Development I	25	75	100	4
10	12932	13132	**Part II:** English Paper – III	25	75	100	4
11	12933	13133	**Part III: Core Paper:** Internet and Java Programming	25	75	100	4
12	12934	13134	**Part III: Core Paper:** Internet and Java Programming Lab	25	75	100	4
			TOTAL	**100**	**300**	**400**	**16**
SEMESTER IV							
13	12941A 12941B	13141A 13141B	**Part I:** Tamil Paper IV / Human Skills Development II	25	75	100	4
14	12942	13142	**Part II:** Paper II: English – IV	25	75	100	4
15	12943	13143	**Part III: Core Paper:** Open Source Software	25	75	100	4

16	12944	13144	**Part III: Core Paper:** Open Source Lab	25	75	100	4
			TOTAL	**100**	**300**	**400**	**16**
THIRD YEAR							
SEMESTER V							
17	12951	13151	**Part III: Core Paper:** Discrete Mathematics	25	75	100	4
18	12952	13152	**Part III: Core Paper:** Operating Systems	25	75	100	4
19	12953	13153	**Part III: Core Paper:** Relational Database Management System (RDBMS)	25	75	100	4
20	12954	13154	**Part III: Core Paper:** Relational Database Management System (RDBMS)Lab	25	75	100	4
			TOTAL	**100**	**300**	**400**	**16**
SEMESTER VI							
21	12961	13161	**Part III: Core Paper:** ● NET Programming	25	75	100	4
22	12962	13162	**Part III: Core Paper:** System Analysis and Design	25	75	100	4
23	12963	13163	**Part III: Core Paper:** Multimedia and its Applications	25	75	100	4
24	12964	13164	**Part III: Core Paper:** ● NET Programming Lab	25	75	100	4
			TOTAL	**100**	**300**	**400**	**16**
			GRAND TOTAL	**600**	**1800**	**2400**	**96**

CIA: Continuous Internal Assessment, **ESE:** End Semester Examination,
TOT: Total, **C:** Credit Points **Max:** Maximum

FIRST YEAR
SEMESTER I

Course Code	Title of the Course
12911A	PART I: TAMIL PAPER I

கூறு 1

2. **கண்ணதாசன் - ஸ்ரீ கிருஷ்ண கானம்**
   3. புல்லாங்குழல் கொடுத்த
   4. குருவாயூருக்குவாருங்கள்

கூறு 2

   4. கோகுலத்துபசுக்கள்
   5. கோகுலத்தில்ஒருநாள் ராதை
   6. ஆயர்பாடிமாளிகையில்

கூறு 3

பட்டுக்கோட்டைகல்யாணசுந்தரம்
   3. நெஞ்சில் குடியிருக்கும்
   4. செய்யும் தொழிலேதெய்வம்

கூறு 4

2. **பாரதியார்**
கண்ணன் என் விளையாட்டுப்பிள்ளை
பாரதமாதாதிருப்பள்ளிஎழுச்சி

கூறு 5

4. **பாரதிதாசன்** - உலகப்பன்பாட்டு (5)
5. **நாமக்கல் கவிஞர்** - நோயற்றவாழ்வு7 பாட்டு
6. **பெ.தூரன்** - நிலாபிஞ்சு

கூறு 6

4. **வல்லிக் கண்ணன்** - வெறும் புகழ்
5. **கு.ப.இராஜகோபாலன்** - எதற்காக?
6. **மீரா** - பதினைந்து

கூறு 7

3. **சிற்பி** - சர்ப்பயாகம்
4. **ஞானக்கூத்தன்** - தோழர் மோசிகீரனார்

கூறு 8

3. **அப்துல் ரகுமான்** - கண்ணும் எழுதேம்
4. **சண்முகசுப்பையா** - வயிறு

கூறு 9

3. **சிலப்பதிகாரம்** - வழக்குரைகாதை
4. **கம்பராமாயணம்** - அயோத்தியாகாண்டம்

கூறு 10

2. **சீறாப்புராணம்** - ஈத்தங்குலைவரவழைத்தபடலம் (1)

கூறு 11

தேம்பாவணி - காட்சிப்படலம்
பாடல்எண்(ஒவ்வொருபாடலின் முதல்வரி)
   29. இன்னவாயில்
   30. கொழுந்துறும்
   31. பஞ்(ச) அரங்கில்

கூறு 12

தேம்பாவணி - காட்சிப்படலம்
பாடல்எண் (ஒவ்வொருபாடலின் முதல்வரி)
   32. எண்ணுளே
   33. ஒண்தலங்கள்

34. இரவியேந்தகஞ்சக்

கூறு 13

தேம்பாவணி - காட்சிப்படலம்

பாடல்எண் (ஒவ்வொருபாடலின் முதல்வரி)

35. கன்னியாயதாயும்
36. ஏந்திஓங்குஉளத்து
37. ஆவ தேமுனர்
38. கொல்லும் வேலொடும்

கூறு 14

தேம்பாவணி - காட்சிப்படலம்

பாடல்எண் (ஒவ்வொருபாடலின் முதல்வரி)

39. என்றவாசகம்
40. அம்பினால்
41. வேண்டும் ஓர் வினை

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Course Code	Title of the Course
12911B	PART I: COMMUNCATION SKILLS – I

Unit - I Communication – Meaning – Types- Importance

Unit – II Barriers to Effective Communication – Principles – Principles of Effective Communication

Unit – III Oral Communication – Meaning – Importance- Forms of Oral Communication

Unit – IV Intonation –Meaning – Function- Types
Preparation of Speech- Steps Involved

Unit – V Principles of Effective Oral Communication

Unit – VI Written Communication – Meaning –Steps – Importance- Advantages
Use of words and Phrases

Unit – VII Sentence – Meaning –Sentence formation- Characteristics of an Effective Sentence

Unit – VIII Paragraph Writing –Essay Writing –Steps Involved –Outline-Layout –Contents-Drafting-Correction- Final Draft

Unit – IX Application for Employment and Curriculum Vitae –Steps involved

Unit – X Non –Verbal Communication – Meaning –Types –Body Language –Postures-Gestures – Facial Expressions –Eye Contact

Unit – XI Report Writing –Report –Types of Reports –Format of a Report

Unit – XII Essentials of a Good Report –Preparation of Report-Procedure Involved

Unit – XIII Meetings-Purpose of the Meeting – Procedure

Unit – XIV Group Discussion –Quality of Content-Participation –Logical Presentation –Behavioural Skills

References:

1. Krishna Mohan & Meera Banerjee, Developing Communication Skills, 2005.
2. Geetha Nagaraj, Write to Communicate, 2004.
3. Wren & Martin, English Grammar and Composition, 2002.
4. Dale Carnegie, How to Win Friends and Influence People, 1981.
5. Dale R Jordan, Language Skills and Use.
6. Gartside L. Bahld, Nagammiah and McComas, Satterwhite, Modern Business Correspondence.
7. Rajendra Pal and Kortahalli J S, Essentials of Business Communication.
8. Wallace, Michael J, Study Skills in English.
9. Editors of Readers Digest, Super Word Power.

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Course Code	Title of the Course
12912	**Part II: ENGLISH – Paper I**

Prose

Unit – I Water-the Elixir of life - C.V. Raman
Unit – II Mrs. Packletide's Tiger - SAKI
Unit – III A Deed of Bravery - Jim Carbett
Unit – IV The Cat - Catharine M.Willson
Unit – V On Letter Writing - Alpha of the Plough
Unit – VI Our Ancestors - Carl Sagan
Unit – VII Our Civilization - C.E.Foad
Unit – VIII A Hero on Probation - B.R. Nanda
Unit – IX Dangers of Drug Abuse - Hardin B. Fones
Unit – X Food - J.B.S. Haldane

Grammar

Unit – XI - Articles-Gerunds-Participles-Infinitives-Modals-Proposition –Tenses.
Unit – XII - Direct and Indirect Speech-Transformation of sentences- Active and passive voice.

Composition

Unit – XIII - Letter writing - Precis writing - Developing hints.
Unit – XIV - Dialogue writing - Paragraph writing.

References:

1. Sebastian D K, *Prose for the Young Reader*, Macmillan.
2. *Active English Grammar*, Ed. by the Board of Editors, Macmillan.
3. Modern English – *A Book of Grammar Usage and Composition* by N.Krishnaswamy, Macmillan Publishers.

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Course Code	Title of the Course
12913	PRINCIPLES OF INFORMATION TECHNOLOGY

Unit No.	Contents
	BLOCK 1 : AN OVERVIEW OF THE REVOLUTION IN COMPUTERS AND COMMUNICATIONS:
1	**From the analog to the digital age :** The " New Story" of computers and communications - The six Elements of a Computer & Communications System
2	**Communications:** Development in Computer Technology, Developments in Communications Technology
3	**Computer and Communications Technology Combined:** Connectivity and Interactivity - The Ethics of Information Technology.
	BLOCK 3 : NUMBER SYSTEM
4	**Introduction**: Binary, Octal, Decimal and Hexadecimal number systems – Conversion from one base to another base – Use of complements – binary arithmetic – Numeric and Character codes.
5	**Boolean algebra:** Fundamental concepts of Boolean Algebra – De Morgan's theorems – Simplification of expressions – Sum of products and products of sums
6	**Karnaugh map simplification** – Quine - McKluskey method – two level implementation of Combinational Circuits
	BLOCK 2 : SOFTWARE
7	**Application Software:** Kinds of Software - The five types of applications software - Word processing - Spreadsheets
8	**Other Software:** Database software, Presentation graphics software - Communications software **Programming Languages** - Object Oriented and Visual Programming - Internet Programming - HTML, XML, JAVA and ActiveX.
9	**Desktop accessories and personal information managers** - integrated software and suites - Groupware - Internet Web browsers - Specialised software - Ethics and Intellectual property rights.
	BLOCK 4 : COMMUNCATIONS
10	**Communications:** The practical uses of communications and connectivity - Telephone related communications services - Video/voice communication: Video conferencing and picture phones - online information services - The Internet - Shared resources : Workgroup computing, Electronic Data Interchange, and Intranets - Telecomputing and virtual offices -
11	**Using computer to communicate:** Analog and Digital Signals - modems and communication Software, ISDN lines, and Cable Modems
12	**Communications Channels:** Communications Networks - Local Networks - Factors affecting Data transmission - Cyberethics: Netiquette, Controversial material and censorship, and privacy issues.
	BLOCK 5 : STORAGE AND DATABASE
13	**Storage fundamentals** - Compression and Decompression - Criteria for Rating

	Secondary Storage Devices - Diskettes - Hard Disks - Optical Disks - Magnetic Tapes
14	**Organising Data in Secondary Storage:** Databases, Data Storage - Hierarchy and the concept of the key field - File Management: Basic concepts - File Management Systems .Data Management Systems - Types of Database Organization - Features of a DBMS.

Text Books

1. Stacey C Sawyer, Brain K Williams, Sarah E Hutchinson Using Information Technology – Brief Version A Practical Introduction to Computer and Communications Second Edition, The McGraw Hill Companies 2009.
2. Stacey C Sawyer, Brain K Williams, Sarah E Hutchinson Using Information Technology – Brief Version A Practical Introduction to Computer and Communications Third Edition, McGraw Hill Companies 2011

Book for Reference:

1. J Hames O'Brien – Introduction to Information systems.
2. Digital Computer Fundamentals, 6th Edition, Thomas C. Bartee, Tata McGraw Hill, 2008.
3. Digital Logic and Computer Design, M. Morris Mano, Pearson Education, 2008.

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Course Code	Title of the Course
12914	**OFFICE AUTOMATION LAB**

Unit No.	Contents
	BLOCK 1 : MS-WORD
1	**Working with Files** – Creating and opening documents, Saving documents, Renaming documents, working on multiple documents. Working with Text – Formatting, Moving, copying and pasting text
2	Styles – Apply a style, Apply from the Style dialog box, Create a new style from a model, Modify or rename a style, Delete style. Lists – Bulleted and numbered lists, Nested lists, Formatting lists.
3	**Table Manipulations.** Graphics – Adding clip Art, Add an image from a file, Editing graphics, Spelling and Grammar, AutoCorrect
4	**Page formatting**-Page margins, page size and orientation, Header and footers, page numbers, Mail Merge. Macros – Recording a macro, Running a macro- Web wizard – Using the Web Wizard, Creating & Saving web pages, Hyper links.Mail Merge.
	BLOCK 2 : MS-EXCEL
5	**Modifying a Worksheet** – Moving through cells, Adding worksheets, rows and columns Resizing rows and columns, Selecting cells, Moving and copying cells, Freezing panes - Macros – recording and running.
6	**Formatting cells** – Formatting toolbar, Dates and times, Auto formatting. Formula and Functions. Linking worksheets – Relative, absolute and mixed referencing
7	**Sorting and Filling** – Basic ascending and descending sorted, Complex sorts, Alternating text and numbers with Auto fill, Autofilling functions.
8	**Graphics** – Adding clip art, add an image from a file, Charts – Using chart Wizard, Copy a chart to Microsoft Word
	BLOCK 3 : MS-POWER POINT
9	**Create a Presentation from a template.** Working with Slides – Insert a new slide, Applying a design template, Changing slide layouts
10	**Slides:**Reordering slides, Hide slides, Create a Custom slide show. Adding Content – Resizing a text box, Text box properties, Delete a text box.
11	**Video and Audio effects**, Color Schemes & Backgrounds Adding clip art, Adding an image from a file, Save as a web page.
	BLOCK 4 : MS-ACCESS
12	**Using Access database wizard, pages and projects.** Creating Tables – Create a Table in design view, Primary key, Indexes, Field validation rules.
13	**Datasheet Records** – Adding, Editing, Deleting records, Adding and deleting columns Resizing rows and columns, Finding data in a table & replacing, Print a datasheet.Declaring Table Relationships. Sorting and Filtering – Sorting, Filter by selection, by form, saving & removing a filter. Queries – Create a query in design view, Query Wizard, Find duplicates query
	BLOCK 5 : FORMS IN MS ACCESS

14	**Forms** – Create a form using the wizard, Create a form in Design View. Form Controls. Sub forms-Create a form and sub form at once, Sub form wizard, Drag and drop method. Reports – Using the wizard, Create in Design View, Printing reports. Importing, Exporting, Linking.

REFERENCE BOOKS:

1. Laura Acklen et al, Microsoft Office 97 Professional Essentials, Prentice-Hall India (1998).

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SEMESTER II

Course Code	Title of the Course
12921A	PART I: TAMIL PAPER I

கூறு 1

தேம்பாவணி - காட்சிப்படலம்

பாடல்எண் (ஒவ்வொருபாடலின் முதல்வரி)

42. சொல் தவிர்ந்த
43. அன்னை
44. அஞ்சுவார்
45. சொல்லக் கேட்டனள்
46. மற்செய்கை
47. மண்கனியப்
48. அழுதுஆர்ந்த

கூறு 2

தேம்பாவணி - காட்சிப்படலம்

பாடல்எண் (ஒவ்வொருபாடலின் முதல்வரி)

49. பொய் பொதுளும்
50. இன்புஅருந்தி
51. வழுதாயின இன்பு
52. மறம் ஏவினர்

கூறு 3

தேம்பாவணி - காட்சிப்படலம்

பாடல்எண் (ஒவ்வொருபாடலின் முதல்வரி)

53. மண்ணோர்கள்
54. பொய்யாவிதியோய்
55. விடியா இருள்
56. அழுவார் எவரும்

கூறு 4

சிறுகதை - நீலபத்மநாபனின் ‘‘வானவீதியில்’’

கூறு 5

உரைநடை - கம்பன் புறத்திணை - தி.சொக்கலிங்கம்

இலக்கணம் - எழுத்தும் சொல்லும்

கூறு 6

3. முதலெழுத்துகள்,சார்பெழுத்துகள்
4. மொழிமுதலெழுத்துகள்,மொழி இறுதிஎழுத்துகள்

கூறு 7

4. ஒற்றெழுத்துமிகலும் மிகாமையும்,
5. ஆகு பெயர்,அன்மொழித் தொகை.
6. வினா-விடை வகைகள்

கூறு 8

5. தமிழ்ச் சொல்லமைப்பின் சிறப்பு–பெயர்,வினை, இடை,உரிவடிவங்கள்,
6. பிறமொழிச் சொற்களைத் தமிழில் ஆளும் முறைகள்

கூறு 9

7. அல் வழி,வேற்றுமைப் புணர்ச்சிகள்
8. திணை,பால்,எண், இட இயைபு.

தமிழ் இலக்கியவரலாறு

கூறு 10

2. இக்கால இலக்கிய வகைகள்

அ) மரபுக் கவிதை

ஆ) புதுக் கவிதையின் தோற்றமும் வளர்ச்சியும்

கூறு 11

1. உரைநடை இலக்கியங்கள் - தோற்றமும் வளர்ச்சியும்

அ) கட்டுரை ஆ) சிறுகதை

இ) புதினம் ஈ) நாடகம்

கூறு 12

2. இக்கால இலக்கியக் களங்கள்

திரைப்படம்,தொலைக்காட்சி,வானொலி, இதழ்கள் தமிழுக்குஆற்றிவரும் பணிகள்

கூறு 13

1. **தமிழும் சமயமும் :**

அ) சைவம் ஆ)வைணவம் இ)சமணம் ஈ)பௌத்தம் உ)இசுலாம்

ஊ) கிறித்துவம்

கூறு 14

1. பிற்காலக் காப்பியங்கள் :

அ) கம்பராமாயணம் **ஆ)** பெரியபராணம்

2. இணையம் - பற்றியசெய்திகள்

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Course Code	Title of the Course
12921B	PART I: COMMUNCATION SKILLS – II

Unit – I Code and Content of Communication Skills
Unit – II Stimulus and Response of Communication Skills
Unit – III Effective Speaking Guidelines
Unit – IV Pronunciation Etiquette of Communication Skills
Unit – V Phonetics in Communication Skills
Unit – VI A self Assessment of Communicating Soft Skills
Unit – VII Language Skills –Ability –Skill Selected Need- Learner Centre activities
Unit – VIII Listening Skills –Importance –Types of Listening- Interview Skills
Unit – IX Conversation Skills –Modes
Unit – X Presentation Skills - Preparing –Planning-Presentation
Unit – XI Written Communication –Structure of Effective Sentences –Paragraph
Unit – XII Technical Writing-Creative Writing- Editing and Publishing
Unit – XIII Corporate Communication Skills-Internal –Effective business writing –Letters, Proposals, Resume
Unit – XIV Corporal Communication Skills-External - Press release - Newsletters- Interviewing skills

References:

1. Dutt. Kiranmai & Geeta Rajjevan. Basic Communication Skills. Rev.ed. Foundation Books Pvt.Ltd. Cambridge House, New Delhi 2006.
2. Bill R. Swetmon. Communication Skills for the 21st Century. Chennai: Eswar Press. First South Asian Edition 2006.
3. Glass. Lillian. Talk to Win. New York: Perigee Books,1987.
4. Pease. Alan. Signals: How to Use Body Language for Power, Success and Love, New York: Bantam Books, 1981.
5. Walters. Lilly. Secrets of Successful Speakers. New York: McGraw-Hill, Inc., 1993.
6. Mandal. S.K. How to Succeed in Group Discussions & Personal Interviews. Mumbai: JAICO Publishing House.
7. Rogoff. Leonard and Ballenger. Grady. Office Guide to Business Letters, Memos & Reports. New York: Macmillan, 1994.
8. Krishna Mohan & Meera Banerjee, Developing Communication Skills, 2005.
9. Geetha Nagaraj, Write to Communicate, 2004.
10. Wren & Martin, English Grammar and Composition, 2002.
11. Rajendra Pal and Kortahalli J S, Essentials of Business Communication.

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Course Code	Title of the Course
12922	Part II: ENGLISH – PAPER II

Poetry

Unit – I	Sonnet	- William Shakespeare
Unit – II	Lines Composed upon Westminster Bridge	-William Wordsworth
Unit – III	Grecian Urn	- John Keats (1795-1827)
Unit – IV	Andrea Del Sarto	- Robert Browning (1812-1889)
Unit – V	The Road Not Taken	- Robert Frost (1874-1963)
Unit – VI	Strange Meeting	- Wilfred Owen (1813-1918)
Unit – VII	Gitanjali	- Rabindranath Tagore (1861-1946)
Unit – VIII	The Coromandel Fishers	- Sarojini Naidu
Unit – IX	The Express	- Stephen Spender
Unit – X	**Shakespeare :** The Merchant of Venice	

Language Use:

Unit – XI	Essay writing
Unit – XII	Note Making
Unit – XIII	Report writing
Unit – XIV	Comprehension

References:

1. *The Golden Quill*, P.K. Seshadri, Macmillan.
2. *The Merchant of Venice*, Shakespeare. (Any overseas edition).
3. *Active English Grammar*, Ed. by the Board of Editors, Macmillan.
4. Modern English – *A Book of Grammar Usage and Composition* by N.Krishnaswamy, Macmillan Publishers.

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Course Code	Title of the Course
12923	PROGRAMMING IN C AND DATA STRUCTURES

Unit no.	Description
	BLOCK 1 : INTRODUCTION TO C
1	**Program Development styles and Basics of C**. Introduction to C – Character set – Identifiers and keywords – Data types – constants – Variables – declarations – Declaring variables, - Rules for defining variables. Initializing variables - Type conversion. Operator and Expressions
2	**Data input, output** and Preliminaries – single character input and output – Entering input data – Writing output data – gets and puts functions –
3	**control statements:** Branching and looping – Nested control structures.
4	Switch – Break – Continue and Goto.
	BLOCK 2 : FUNCTIONS,ARRAYS AND POINTERS
5	**Function:** defining a function – Accessing a function – Passing arguments to a function – Recursion – Library function – Macros – C preprocessor – Program structure: Storage classes – Automatic variables – Global variables – Static variables– Multiple programming – Bitwise operation.
6	**Arrays** – Array initialization, Definition of Array, Characteristic of Array, One dimensional array, Two dimensional array, Multidimensional arrays, Character array and strings – string handling functions.
7	**Pointers** – Features of Pointers, Pointer declaration, Arithmetic operation with pointers, Pointers and Arrays, Pointers and two dimensional arrays, Array of Pointers, Pointers to Pointers, Pointers and strings.
	BLOCK 3 : STRUCTURE UNION AND FILES
8	Structures and Unions: defining a structure – Processing a structure – Structures and pointers– Passing structures to functions – Self referential structures – Bit fields – Unions – Enumerations.
9	Data file: Opening and Closing a data file – Creating a data file – Processing a data file – Unformatted data file – Command line parameters.
	BLOCK 4 : LINEAR DATA STRUCUTRE
10	**Introduction to Data Structure,** Stack, Stack related terms, operation on a stack, Representation of Stack, Implementation of a stack - Polish notation. Queues, Various Positions of Queue, Representation of Queues.
11	**List,** Merging lists, Linked list, Single linked list, Double Linked List, Header Linked list, Insertion and Deletion of linked list, Traversing a linked list. Representation of Linked list
	BLOCK 5 : NON-LINEAR DATA STRUCTURE
12	**Introduction** – Trees, Binary Trees, Types of Binary trees,
13	**Binary Tree Representation**, Traversing Binary Trees,
14	**Binary Search tree**, Insertion and Deletion operations, trees and their applications Hashing Techniques.

TEXT BOOKS:

1. Programming in ANSI C, Fifth Edition, E.Balagurusamy, Tata McGraw-Hill Publishing Company Ltd, 2011
2. Data Structures, Seymour Lipschutz, G.A.Vijayalakshmi Pai, Second Edition , Schaum's Outlines, Tata Mc-Graw Hill Private Ltd., 2006.

REFERENCE BOOKS :

1. Fundamentals of Data structures in C, Second edition, Ellis Horowitz and Sartaj Sahini, Universities press, 2007.
2. Programming and Data Structure, Pearson Edition, Ashok N Kamthane, 2007.

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Course Code	Title of the Course
12924	C AND DATA STRUCTURE LAB

Experiments based on C programming and Data Structures

Unit No.	Contents
	BLOCK 1 : C PROGRAM FUNDAMENTALS
1	Simple C Programs
2	Using if and switch constructs Programs
3	Looping statements Problems
	BLOCK 2 : FUNCTIONS,ARRAYS,STRINGS,FILEAND POINTERS
4	Functions and Recursive programs
5	Arrays ,Strings and Matrices Programs
6	File Handling Programs
7	Pointers and Arrays Programs programs
	BLOCK 3 – STRUCTURE , UNION AND FILES
8	Structure and union : Programs using structure and union
9	Files : Programs based on file handling
	BLOCK 4 : LINEAR DATA STRUCTURE PROGRAMS
10	Stacks, queues ,expression evaluation programs
11	Infix to postfix conversion
12	Linked List programs: List, Merging lists, Linked list, Single linked list, Double Linked List, Header Linked list, Insertion and Deletion of linked list, Traversing a linked list.
	BLOCK 5 : NON LINEAR DATA STRUCURE EXPERIMENTS
13	Tree Programs : Trees, Binary Trees, Types of Binary trees, Binary Tree Representation,
14	Traversing Binary Trees, Binary Search tree, Insertion and Deletion operations,

REFERENCE BOOKS:

1. Programming in ANSI C, Fifth Edition, E.Balagurusamy, Tata McGraw-Hill Publishing Company Ltd, 2011
2. Data Structures, Seymour Lipschutz, G.A.Vijayalakshmi Pai, Second Edition , Schaum's Outlines, Tata Mc-Graw Hill Private Ltd., 2006.
3. Fundamentals of Data structures in C, Second edition, Ellis Horowitz and Sartaj Sahini, Universities press, 2007.
4. Programming and Data Structure, Pearson Edition, Ashok N Kamthane, 2007.

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SECOND YEAR
SEMESTER III

Course Code	Title of the Course
12931A / 13131A	**PART I: TAMIL PAPER I**

கூறு 1 :பத்துப்பாட்டு–முல்லைப்பாட்டு

கூறு 2 :எட்டுத்தொகை–ஐங்குறுநூறு

கூறு 3 :கபிலர் - குறிஞ்சித்திணை

கூறு 4 :மஞ்ஞைப்பத்து–முதல் மூன்றுபாடல்கள்

கூறு 5 :குறுந்தொகை–பரணர் பாடல்கள் பா. எண். 19, 24, 36, 128, 399

கூறு 6 : நற்றிணை– பெருங்குன்றூர்கிழார் - பா. எண். 5

பெருவழுதியார் - பா. எண். 55

பெருங்கௌசிகனார் - பா. எண். 139

கூறு 7 :நற்றிணை– கருவூர்க்கோசிகனார் - பா. எண். 214

உலோச்சனார் - பா. எண் 249

கூறு 8 :அகநானூறு –சேந்தம்பூதனார் பாடல்கள் பா.எண். 84, 207

கூறு 9 :புறநானூறு –மறோக்கத்துநப்பசலையார் பாடல்கள்

பா. எண். 37, 39, 126, 226, 280

கூறு 10 :பதினெண் கீழ்க்கணக்கு–திருக்குறள் - வாழ்க்கைத்துணைநலம் (6),

அறிவுடைமை (43),பிரிவாற்றாமை (116)

கூறு 11: நான்மணிக்கடிகை–எள்ளற்க (3),பறைபடவாழா (4),

கூறு 12:நான்மணிக்கடிகை - மண்ணயறிப (5),கள்ளிவயிற்றில் (6),கல்லிற்பிறக்கும்(7)

கூறு 13: நாடகம் - இராசராசசோழன் - அரு. இராமநாதன்

கூறு 14: நாவல் - சுவடுகள் - இரா. பாலசுப்பிரமணியன்,சத்யாவெளியீடு,மதுரை

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Course Code	Title of the Course
12931B / 13131B	**PART I: HUMAN SKILL DEVELOPMENT – I**

Unit – I Human Skills –Developing skills-Types

Unit – II Mind-Levels of functions
Habits-Meaning-Types-Merits of good habits - Interpersonal Relationship-Features-Interpersonal Behaviour

Unit – III Thinking ahead- Significance of thinking ahead

Unit – IV Developing Personality-Meaning -Need- Factors influencing personality, Ways of developing personality -Building positive personality

Unit – V Self-concept - Self-esteem – Meaning - Importance - Self- efficacy - Self-acceptance-Meaning-Importance - Etiquette-Meaning-Etiquettes in using mobile, telephones-Dais Etiquette

Unit – VI Goal-setting Skills-Meaning-Types-Importance-

Unit – VII Decision-making skills-Meaning-Types-Steps in decision-making

Unit – VIII Negotiating Skills-Styles-Structure-Creating negotiation-Competitive Negotiation

Unit – IX Attitudes-Meaning-Types-Importance-Developing positive attitudes

Unit – X **C**oping with Change – Meaning – Characteristics - Importance of change Resistance to change-Dealing with change

Unit – XI Leadership-Meaning-Importance-Characteristics-Styles-

Unit – XII Human Relations Skill-Need-Canons of good human relations

Unit – XIII **C**ounselling-Meaning-Importance-Forms- Conflicts-Meaning-Types-Causes-Effects-Managements of conflicts

Unit – XIV Stress-Meaning-Types-Causes-Effects-Managing the stress - Anger-Meaning-Causes-Consequences-Anger Management

References:

7. Les Giblin, Skill with People, 1995.
8. Shiv Khera, You Can Win, 2002.
9. Christian H Godefroy, Mind Power.
10. Dale Carnegie, How to Enjoy Your Life and Your Job, 1985.
11. Natalie H Rogers, How to Speak without Fear, 1982.
12. Dale Carnegie, How to Develop Self-Confidence and Influence People by Public Speaking.

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Course Code	Title of the Course
12932 / 13132	Part II: ENGLISH - PAPER– III

Short Stories

Unit – I	A Hero	- R.K. Narayanan
Unit – II	The Diamond Necklace	- Guy de Maupassant
Unit – III	The Verger	- Somerset Maugham
Unit – IV	The Postmaster	- Rabindranath Tagore

One Act Plays

Unit – V	The Proposal	- Anton Chekhou
Unit – VI	The Boy Comes Home	- A.A. Milne
Unit – VII	The Silver Idol	- James R. Waugh
Unit – VIII	Progress	- St. John Ervine
Unit – IX	The Pie and the Tart	- Huge Chesterman
Unit – X	Reunion	- W.st. Joh Tayleur
Unit – XI	A kind of Justice	- Margaret Wood
Unit – XII	The Refugee	- Asif Currimbhoy

Grammar

Unit – VIII Parts of speech-Noun- Pronoun- Adjective
Degrees of Comparison- Verb- Adverb

Composition

Unit – XIV Agenda- Minutes- Notice- Descriptive Writing

References:

1. *Aroma*, Ed. by the Board of Editors, Publishers- New Century Book House, Chennai.
2. *Six Short Stories*, Ed. by the Board of Editors, Harrows Publications, Chennai.
2. *One Act Plays*, Ed. by the Board of Editors, Harrows Publications, Chennai.
3. Modern English – *A Book of Grammar Usage and Composition* by N.Krishnaswamy, Macmillan Publishers.
4. *English for Communication*, Ed. by the Board of Editors, Harrows Publications, Chennai.

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COURSE CODE	TITLE OF THE COURSE
12933 / 13133	INTERNET AND JAVA PROGRAMMING

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	**Basic Internet Concepts:** Connecting to the Internet – Domain Name System – E-mail
2	**The World Wide Web** – Internet Search Engines – Web Browsers – Chatting and conferencing on the Internet
3	**Online Chatting** –Messaging – Usenet Newsgroup – Internet Relay chat (IRC) – FTP – Telnet.
	BLOCK 2 : FUNDAMENTALS OF OBJECT-ORIENTED PROGRAMMING:
4	**Basic concepts** of OOP – Benefits – Applications
5	**Java Evolution:** Features – how java differs from C and C++ - java and internet- java support system – java environment -
6	**Overview of Java Language** –constants variables and data types- Operators and Expressions - Decision Making and Branching - Looping
	BLOCK 3 : CLASSES, OBJECTS AND METHODS
7	**class:** Defining a class –fields –methods –creating objects – accessing class members – constructors – methods overloading –static members –nesting of methods – Inheritance –overriding methods –final variables-classes –methods
8	**Arrays, Strings and Vectors** :One dimensional Arrays –creating of array – Two dimensional arrays- strings –vectors –Wrapper classes – Enumerated Types - Interfaces: Multiple Inheritance
9	**Packages:** Defining interface –Extending interfaces – Implementing Interfaces -Putting Classes Together
	BLOCK 4: MULTITHEREADING , EXCEPTION AND APPLETS
10	**Multithreaded Programming** – Creating Threads –Extending the thread class – Stopping and Blocking a thread –Life cycle of a thread –using thread methods –Thread Exceptions –Priority –Synchronization –Implementing the 'Runnable' Interface
11	**Managing Error and Exceptions:** Types of errors –Exceptions –Syntax of Exception Handling code – Multiple Catch statements –using finally statement – Throwing our own Exceptions – using exceptions for Debugging - Graphics Programming: The Graphics Class – Lines and Rectangles – Circles and Ellipses – Drawing Arcs – Drawing Polygons – Line Graphs – Using Control Loops in Applets – Drawing Bar Charts.
12	**Applet Programming:** How applets differ from Applications – preparing to write applets – Building Applet Code – Applet life cycle – creating an Executable Applet – Designing a Web Page – Applet Tag – Adding Applet to HTML file – Running the Applet – Passing parameters to Applets – Displaying Numerical values – Getting input from the user
	BLOCK 5 : MANAGING INPUT/OUTPUT FILES IN JAVA
13	**Introduction** – concept of streams –stream classes – byte stream classes – character Stream
14	**I/O classes**: –using stream –using the file class –Input / output Exceptions –creation of files – Reading / writing characters – reading writing bytes Random access files- Interactive input and output –Other stream classes

Text Books:

1. R. Krishnamoorthy and S. Prabhu, Internet and Java Programming, New Age International Publishers, 2004 (Unit I).
2. Programming with Java, 4e, E. Balagurusamy, Tata McGraw-Hill, 2010.

Reference books:

1. Deitel, Deitel and Nieto, Internet and World Wide Web – How to program, Pearson Education, 2000.
2. Naughton and H.Schildt, Java 2 - The complete reference, Tata McGraw-Hill, Fourth edition, 2006.
3. Elliotte Rusty Harold, Java Network Programming, O'Reilly Publishers, 2000.
4. B.Mohamal Ibrahim , Java : J2SE – A Practical Approach, Firewall media, 2006.
5. Cay S. Horstmann, Gary Cornell, Core Java, Volume I and II, 5th Edition, Pearson Education, 2003.
6. Topley, J2ME in A Nutshell, O'Reilly Publishers, 2002.
7. Hunt, Guide to J2EE Enterprise Java, Springer Publications, 2004.
8. Ed Roman, Enterprise Java Beans, Wiley Publishers, 1998.

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COURSE CODE	TITLE OF THE COURSE
12934 / 13134	INTERNET AND JAVA PROGRAMMING LAB

Experiments based on Internet Programming Theory

Unit No.	Contents
	BLOCK 1 : JAVA FUNDAMENTAL PROBLEMS
1	Simple Java Problems
2	class and objects
3	Conditional control using java
4	Looping using java
	BLOCK 2 : OOP CONCEPTS
5	Function overloading programs
6	Operator overloading programs
7	Inheritance programs, Packages
8	Polymorphism programs Message passing programs
	BLOCK 3 : THREAD & VIRTUAL FUNCTION
9	Threads
10	Virtual function programs
	BLOCK 4 : I/O AND EXCEPTION HANDLING
11	Exception handling programs
12	I/O manipulation programs,
	BLOCK 4 :NETWORK PROGRAMMING
13	Applet programs
14	Implementation of simple network programs using java

Reference books:

1. R. Krishnamoorthy and S. Prabhu, Internet and Java Programming, New Age International Publishers, 2004).
2. Programming with Java, 4e, E. Balagurusamy, Tata McGraw-Hill, 2010.
3. Deitel, Deitel and Nieto, Internet and World Wide Web – How to program, Pearson Education, 2000.
4. Naughton and H.Schildt, Java 2 - The complete reference, Tata McGraw-Hill, Fourth edition, 2006.
5. Elliotte Rusty Harold, Java Network Programming, O'Reilly Publishers, 2000.
6. B.Mohamal Ibrahim , Java : J2SE – A Practical Approach, Firewall media, 2006.

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SEMESTER IV

Course Code	Title of the Course
12941A / 13141A	PART I: TAMIL PAPER IV

கூறு 1 : செய்யுள் உறுப்புகள் - யாப்பு -எழுத்து,அசை,சீர்,

கூறு 2 :செய்யுள் உறுப்புகள் - யாப்பு -தளை,அடி,தொடை

கூறு 3 :வெண்பா,ஆசிரியப்பா,கலிப்பா,வஞ்சிப்பா,

கூறு 4 :புதியயாப்புவடிவங்கள் - சிந்து,கண்ணி,கீர்த்தனை

கூறு 5 :புதுக்கவிதையில் குறியீடு–படிமம்.

கூறு 6 :அகப்பொருள் - புறப்பொருள் - ஐந்திணைவிளக்கம்

கூறு 7 :அகப்பொருள் துறைகள் - வரைவுகடாதல்,அறத்தொடுநிற்றல், உடன்போக்கு

கூறு 8 :புறப்பொருள் துறைகள் - வஞ்சினக்காஞ்சி,கையறுநிலை,செவியறிவுறூஉ

கூறு 9 :அணி இலக்கணம் - உவமை,உருவகம்,வேற்றுமை,பிறிதுமொழிதல்,
தற்குறிப்பேற்றம்,சிலேடை,பின்வருநிலை

கூறு 10 :நிறுத்தல் குறிகள்

கூறு 11 : தொல்காப்பியம் - சங்கஇலக்கியம் - எட்டுத்தொகை,பத்துப்பாட்டு,

கூறு 12 :பதினெண்கீழ்க்கணக்கு ஐம்பெருங்காப்பியங்கள் - பிற்காலக் காப்பியங்கள் - கம்பராமாயணம் - பெரியபுராணம்

கூறு 13 :இக்காலக் காப்பியங்கள் - பாரதியின் பாஞ்சாலிசபதம் -
பாரதிதாசனின் பாண்டியன் பரிசு

கூறு 14. :கண்ணதாசனின் இயேசுகாவியம் ,சிற்பியின் -மௌனமயக்கங்கள்.

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Course Code	Title of the Course
12941B / 13141B	**PART I: HUMAN SKILL DEVELOPMENT – II**

Unit – I	Guidance & Counselling – Role of Counsellor - Importance and Techniques of counselling
Unit – II	Managerial skill- Need – Importance
Unit – III	Human relational skills-Communication-Attention
Unit – IV	Conceptual skills-Meaning-Importance
Unit – V	Technical skills-Techniques-Practices-Tools-Procedures
Unit – VI	Presentation skills-Planning-Preparation-Delivery
Unit – VII	Organization skills-Meaning-Nature-Importance-Types
Unit – VIII	Multi-Tasking skills Responsibilities-Causes
Unit – IX	Leader- Qualities of a good leader
Unit – X	Understanding Skills -Human systems: Individual, Group, organization, and their major interactions
Unit – XI	Understanding Skills -Human systems: Community and Society, and their major interactions
Unit – XII	Problem solving skills – Handling –Facing - Importance
Unit – XIII	Cooperative Learning Skills
Unit – XIV	Making Social Responsibilities-Causes

References:

7. Les Giblin, Skill with People, 1995.
8. Shiv Khera, You Can Win, 2002.
9. Christian H Godefroy, Mind Power.
10. Dale Carnegie, How to Enjoy Your Life and Your Job, 1985.
11. Natalie H Rogers, How to Speak without Fear, 1982.
12. Dale Carnegie, How to Develop Self-Confidence and Influence People by Public Speaking.

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Course Code	Title of the Course
12942 / 13142	Part II: ENGLISH –PAPER – IV

Short Stories

Unit – I Lalajee - Jim Corbelt
Unit – II A Day's Wait - Hemmingway
Unit – III Two old Men - Leo Tolstoy
Unit –IV Little Girls wiser than - Men Tolstoy
Unit – V Boy who Wanted more Cheese - William Elliot Griffir

Drama

Unit – VI Pygmalion - G.B. Shaw

Fiction

Unit – VII Swami and Friends - R.K. Narayanan

Tales from Shakespeare

Unit – VIII - The Merchant of Venice
Unit – IX - Romeo and Juliet
Unit – X - The Winter's Tale

Biographies

Unit – XI - Martin-Luther king - R.N. Roy
Unit – XII - Nehru - A.J. Toynbee

Grammar

Unit – XIII - Concord- Phrases and Clauses-Question Tag

Composition

Unit – XIV - Expansion of Proverbs
- Group Discussion
- Conversation (Apologizing, Requesting, Thanking)

References:

1. *Sizzlers*, by the Board of Editors, Publishers-:Manimekala Publishing House, Madurai.
2. *Pygmalion* – G.B. Shaw
3. *Swami and Friends* – R.K. Narayan
4. *Tales from Shakespeare* Ed. by the Board of Editors, Harrows Publications, Chennai.
5. Modern English – *A Book of Grammar Usage and Composition* by N.Krishnaswamy, Macmillan Publishers.

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SEMESTER IV

COURSE CODE	TITLE OF THE COURSE
12943 / 13143	OPEN SOURCE SOFTWARE

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	Introduction to Open sources – Need of Open Sources – Advantages of Open Sources–Application of Open Sources.
2	Open source operating systems: LINUX: Introduction– General Overview–Kernel Mode and user mode–Process–
3	Advanced Concepts–Scheduling – Personalities – Cloning – Signals – Development with Linux. .
	BLOCK 2 : OPEN SOURCE DATABASE
4	MySQL: Introduction Setting up account Starting, terminating and writing your own SQL programs
5	Record selection Technology– Working with strings – Date and Time
6	Sorting Query Results – Generating Summary – Working with metadata –Using sequences – MySQL and Web.
	BLOCK 3 :OPEN SOURCE PROGRAMMING LANGUAGE - PHP
7	PHP: Introduction – Programming in web environment – variables – constants–data types – operators –Statements
8	Functions– Arrays – OOP –String Manipulation and regular expression –File handling and data storage
9	PHP and SQL database – PHP and LDAP – PHP Connectivity –Sending and receiving E-mails –Debugging and error handling – Security – Templates.
	BLOCK 4 :OPEN SOURCE PROGRAMMING LANGUAGE - PYTHON
10	Syntax and Style – Python Objects – Numbers – Sequences – Strings –Lists and Tuples – Dictionaries –Conditionals and Loops .
11	Files – Input and Output –Errors and Exceptions – Functions – Modules –Classes and OOP – Execution Environment.
	BLOCK 5 :OPEN SOURCE PROGRAMMING LANGUAGE - PERL
12	Perl backgrounder – Perl overview– Perl parsing rules – Variables and Data – Statements
13	Control structures – Subroutines
14	Packages, and Modules-Working with Files –Data Manipulation.

Text Books:

1. Remy Card, Eric Dumas and Frank Mevel, “The Linux Kernel Book”, Wiley Publications, 2003
2. Steve Suchring, “MySQL Bible”, John Wiley, 2002

Books for Reference:

1. Rasmus Lerdorf and Levin Tatroe, “Programming PHP”, O’Reilly, 2002
2. Wesley J. Chun, “Core Phython Programming”, Prentice Hall, 2001
3. Martin C. Brown, “Perl: The Complete Reference”, 2nd Edition, Tata McGraw-Hill Publishing Company Limited, Indian Reprint 2009.
4. Steven Holzner, “PHP: The Complete Reference”, 2nd Edition, Tata McGraw-Hill Publishing Company Limited, Indian Reprint 2009.
5. Vikram Vaswani, “MYSQL: The Complete Reference”, 2nd Edition, Tata McGraw-Hill Publishing CompanyLimited, Indian Reprint 2009.

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COURSE CODE	TITLE OF THE COURSE
12944 / 13144	OPEN SOURCE LAB

Unit No.	Contents
	BLOCK 1 : INTRODUCTION TO LINUX
1	**Kernel** configuration, compilation and installation.
2	**Install various software** on Linux. Install and configure XAMP.Unix Commands and shell programming.
	BLOCK 2 : MYSQL
3	**Creating simple table** with constraints Insertion, Updation and Deletion of rows in MYSQL tables Searching of data by different criteria,Sorting of data
4	**Demonstration of joining tables** Usage of subqueries,aggregate functions Working with set operators Working with string, numeric and date functions
5	**Database connectivity** in PHP with MySQL Validating Input Formatting the Output.
	BLOCK 3 : PHP
6	**PHP Simple Programs** PHP program to perform the arithmetic operation. PHP program Adding numbers using function.
7	**PHP Web programs arrays and functions** Creating simple webpage using PHP Use of conditional statements in PHP Use of looping statements in PHP Creating different types of arrays Usage of array functions Creating user defined functions
8	**File manipulation using PHP** Creation of files,sessions and cookies Creating simple applications using PHP Creating simple table with constraints
	BLOCK 4 : PERL AND PYTHON PROGRMAMING
9	**PERL programs** : Simple programs using PERL
10	**Python Programming:** Use of conditional statements Use of looping statements
11	**Python Programming: Arrays** Creating different types of arrays Usage of array functions Creating user defined functions
12	**Python Programming: String**: String Handling
	BLOCK 5 : APPLICATION DEVELOPMENT
13	**Connect to a MYSQL** database with PHO, PERL and Python.
14	**Developing simple applications** using PHP and MYSQL

Books for Reference:

1. Remy Card, Eric Dumas and Frank Mevel, “The Linux Kernel Book”, Wiley Publications, 2003
2. Steve Suchring, “MySQL Bible”, John Wiley, 2002.
3. Rasmus Lerdorf and Levin Tatroe, “Programming PHP”, O’Reilly, 2002.
4. Wesley J. Chun, “Core Phython Programming”, Prentice Hall, 2001
5. Martin C. Brown, “Perl: The Complete Reference”, 2nd Edition, Tata McGraw-Hill Publishing Company Limited, Indian Reprint 2009.

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THIRD YEAR
SEMESTER V

COURSE CODE	TITLE OF THE COURSE
12951 / 13151	DISCRETE MATHAMATICS

Unit No.	Contents
	BLOCK 1 : MATHEMATICAL LOGIC
1	**Mathematical Logic:** Statements and Notation - connectives -normal forms – The theory of inference for the statement calculus -
2	**Predicate Calculus**: The predicate calculus - Inference theory and predicate calculus.
3	**Set theory:** Sets – Basic concepts – notation - inclusion and equality of sets - the power set
	BLOCK 2 : RELATIONS
4	**Relations and ordering** properties – relation matrix and graph of a relation
5	**Relations** Partition – equivalence and compatibility relations
6	**Composition and partial ordering**:Composition – partial ordering – partially ordered set
	BLOCK 3 : FUNCTIONS
7	**Functions** – definition – composition – inverse – binary and n-ary operations
8	**Other Functions** : Characteristic function – hashing function.
	BLOCK 4 : ALGEBRAIC STRUCTURES
9	**Algebraic Structures:** Algebraic Systems: Examples and General Properties
10	**Semigroups and Monoids:** Definitions and Examples - Homomorphism of Semigroups and Monoids - Subsemigroups and Submonoids
11	**Groups:** Definitions and Examples - Cosets and Lagrange's Theorem
12	**Normal Subgroups** – Algebraic Systems with two Binary Operations.
	BLOCK 5 : GRAPH AND FINITE PROBABILITY
13	**Graph theory:** Basic concepts – definition – paths - reach - ability and connectedness – matrix representation of graphs - trees.
14	**Finite Probability** – Probability Distributions – Conditional Probability Independence – Bayes' Theorem – Mathematical Expectation

TEXT BOOKS:

1. J.P. Tremblay and R. Manohar Discrete mathematical structures with applications to Computer Science TMH Publishing Company 2003.
2. Judith L. Gersting, Mathematical Structures for Computer Science, 5thEdition, W.H. Freeman and Company, 2003.

REFERENCE BOOKS:

1. Venkatraman M K, Sridharan N and Chandrasekaran N, Discrete Mathematics, The National Publishing Company, 2004.
2. Narsingh Deo, Graph Theory with Applications to Engineering and Computer Science PHI,2003.
3. Ramasamy, Discrete Mathematical Structures with application to combinatorics, Univeristies Press, 2006.

4. Bernand Kolman, Roberty C. Busby, Sharn Cutter Ross, Discrete Mathematical Structures, Pearson Education, 2006.
5. Richard Johnsonbaugh, Discrete Mathematics, Fifth Edition, Pearson Education. 2001.
6. Garry Haggard and others, Discrete Mathematics for Computer science, Thomson.

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COURSE CODE	TITLE OF THE COURSE
12952 / 13152	OPERATING SYSTEMS

Unit No.	ntents
	BLOCK 1 : INTRODUCTION
1	**Introduction:** Definition of Operating Systems – Computer System Organization
2	**Computer System Architecture** – Operating System Structure – Operating System Operations
3	**System Structures:** Operating System Services – System Calls – System Programs – Operating System Design and Implementation.
	BLOCK 2 : PROCESS CONCEPT
4	**Process Concept:** Process Scheduling – Operations on Processes – Inter Process Communication
5	**Process Scheduling:** Scheduling Concepts – Scheduling Criteria – Scheduling Algorithms – Multiple Processor Scheduling
	BLOCK 3 : SYNCHRONIZATION
6	**Synchronization:** The Critical Section Problem – Synchronization Hardware – Semaphores – Classic Problems of Synchronization – Monitors
7	**Deadlocks:** Deadlocks Characterization – Methods for Handling Deadlocks
8	**Deadlock** Prevention – Avoidance – Detection –Recovery from Deadlock.
	BLOCK 4 : MEMORY MANAGEMENT
9	**Memory Management Strategies:** Swapping – Contiguous Memory Allocation – Paging – Segmentation
	BLOCK 5 : FILE SYSTEM
10	**File Concept** – Access Methods – Directory
11	**Structure** – File System Mounting – File Sharing – Protection.
12	**Implementing File Systems:** File System Structure – File System Implementation
13	**Directory Implementation** – Allocation Methods – Free Space Management
14	**Secondary Storage Structure:** Overview of Mass Storage Structure – Disk Structure – Disk Attachment – Disk Scheduling – Disk Management.

TEXT BOOK:

1. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, “Operating System Principles”, 7th Edition, Wiley India Edition, 2008.

REFERENCE BOOKS:

1. Andrew S.Tanenbaum, Operating Systems Design and Implementation, 3rd edition, Prentice Hall, 2006.
2. Harvey M. Deitel, An Introduction to Operating Systems, 3rd Edition, Addison Wesley 2007.

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COURSE CODE	TITLE OF THE COURSE
12953 / 13153	RELATIONAL DATABASE MANAGEMENT SYSTEMS (RBDMS)

Unit No.	ntents
	BLOCK 1 INTRODUCTION
1	**Data base System Applications**, data base System VS file System – View of Data – Data Abstraction –Instances and Schemas – data Models – the ER Model
2	**odel :**Relational Model – Other Models – Database Languages – DDL – DML – database Access for applications Programs – data base Users and Administrator – Transaction Management – data base System Structure – Storage Manager – the Query Processor.
3	**History of Data base Systems -** Data base design and ER diagrams – Beyond ER Design Entities, Attributes and Entity sets – Relationships and Relationship sets – Additional features of ER Model – Concept Design with the ER Model – Conceptual Design for Large enterprises.
	BLOCK 2 : RELATIONAL MODEL
4	**Introduction**– Integrity Constraint Over relations – Enforcing Integrity constraints – Querying relational data – Logical data base Design – Introduction to Views – Destroying / altering Tables and Views.
5	**Relational Algebra** – Selection and projection set operations – renaming – Joins – Division – Examples of Algebra overviews –
6	**Relational calculus** – Tuple relational Calculus – Domain relational calculus – Expressive Power of Algebra and calculus.
	BLOCK 3 : SQL QUERY
7	**Form of Basic SQL Query** – Examples of Basic SQL Queries – Introduction to Nested Queries – Correlated Nested Queries Set – Comparison Operators – Aggregative Operators – NULL values – Comparison using Null values – Logical connectivity's – AND, OR and NOT – Impact on SQL Constructs – Outer Joins – Disallowing NULL values – Complex Integrity Constraints in SQL Triggers and Active Data bases. Schema refinement
8	**Normal forms :**Problems Caused by redundancy – Decompositions – Problem related to decomposition – reasoning about FDS – FIRST, SECOND, THIRD Normal forms – BCNF–
9	**Join:** Lossless join Decomposition – Dependency preserving Decomposition – Schema refinement in Data base Design – Multi valued Dependencies – FORTH Normal Form.
	BLOCK 4 TRANSACTION
10	**Introduction :**Transaction Concept- Transaction State- Implementation of Atomicity and Durability – Concurrent – Executions – Serializability- Recoverability – Implementation of Isolation – Testing for serializability
11	**Protocols :** Lock Based Protocols – Timestamp Based Protocols- Validation- Based Protocols – Multiple Granularity.
12	**Recovery and Atomicity** – Log – Based Recovery – Recovery with Concurrent Transactions – Buffer Management – Failure with loss of nonvolatile storage-Advance Recovery systems- Remote Backup systems
	BLOCK 5 STORAGE
13	**Data on External Storage** – File Organization and Indexing – Cluster Indexes, Primary and Secondary Indexes – Index data Structures – Hash Based Indexing – Tree base Indexing – Comparison of File Organizations – Indexes

14	**Performance Tuning-** Intuitions for tree Indexes – Indexed Sequential Access Methods (ISAM) – B+ Trees: A Dynamic Index Structure.

Text Books:

1. Raghurama Krishnan, Johannes Gehrke, Data base Management Systems, 3rd Edition, TATA McGrawHill.2003.
2. Silberschatz, Korth, Data base System Concepts, 6th Edition, Tata McGraw Hill, 2011.

Reference Books:

1. Relational Database Principles 2nd Edition, Colin Ritchie, 2004
2. Sharad Maheswari and Ruchin Jain, Database management systems Complete Practical Approach,Firewall media, 2006
3. Peter Rob & Carlos Coronel, Data base Systems design, Implementation, and Management, 7th Edition.
4. Elmasri Navrate , Fundamentals of Database Systems, Pearson Education.

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COURSE CODE	TITLE OF THE COURSE
12954 / 13154	**RELATIONAL DATABASE MANAGEMENT SYSTEMS (RBDMS) - LAB**

Use the concepts like data normalization, link between table by means of foreign keys and other relevant database concepts for the following applications. The implementation of each should have necessary input screen (forms) Menu-driven query processing and reports. Necessary validations should be made for each table;

Unit No.	Contents
	BLOCK 1 : TABLE MANIPULATION
1	**Table** creation, Renaming a Table, Copying another table, Dropping a Table
2	**Table Description:** Describing Table Definitions, Modifying Tables
	BLOCK 2 : SQL QUERIES AND SUB QUERIES
3	**SQL Queries:** Queries, Sub Queries, and aggregate functions
4	**DDL:** Experiments using database DDL SQL statements
5	**DML:** Experiment using database DML SQL statements
6	**DCL:** Experiment using database DCL SQL statements
	BLOCK 3 : INDEX AND VIEW
7	**Index :** Experiment using database index creation, Renaming a index, Copying another index, Dropping a index
8	**Views:** Create Views, Partition and locks
	BLOCK 4 : EXCEPTION HANDLING AND PL/SQL
9	**Exception Handling:** PL/SQL Procedure for application using exception handling
10	**Cursor:** PL/SQL Procedure for application using cursors
11	**Trigger:** PL/SQL Procedure for application using triggers
12	**Package:** PL/SQL Procedure for application using package
13	**Reports:** DBMS programs to prepare report using functions
	BLOCK 5 : APPLICATION DEVELOPMENT
14	**Design and Develop Application:** Library information system, Students mark sheet processing, Telephone directory maintenance, Gas booking and delivering, Electricity bill processing, Bank Transaction, Pay roll processing. Personal information system, Question database and conducting Quiz and Personal diary

Reference Books:

1. Raghurama Krishnan, Johannes Gehrke, Data base Management Systems, 3rd Edition, TATA McGrawHill.2003.
2. Silberschatz, Korth, Data base System Concepts, 6th Edition, Tata McGraw Hill, 2011.
3. Relational Database Principles 2nd Edition, Colin Ritchie, 2004
4. Sharad Maheswari and Ruchin Jain, Database management systems Complete Practical Approach,Firewall media, 2006
5. Peter Rob & Carlos Coronel, Data base Systems design, Implementation, and Management, 7th Edition.
6. Elmasri Navrate , Fundamentals of Database Systems, Pearson Education.

SEMESTER VI

COURSE CODE	TITLE OF THE COURSE
12961 / 13161	● NET PROGRAMMING

Unit No.	Contents
	BLOCK 1 – INTRODUCTION
1	**Overview of Microsoft .NET Framework** - The .NET Framework components-The Common Language Runtime (CLR) Environment- The .NET Framework class Library
2	**Getting Started with Visual Basic .net IDE :** Set up of work environment, start page, the menu system, toolbars, the new project dialog box, graphical designers, code designers, the object explorer, the toolbox, the solution explorer, the class view window, the properties window, the dynamic help window, the server explorer, the output window, the command window.
3	**Visual basic language concept :** variables, Constants, Data Types, Operators, Control Structures and loops - Arrays : single and multidimensional array, declaring, dynamic array.
	BLOCK 2 : INTRODUCTION TO WINDOWS COMMON CONTROLS
4	**Working with Form** - Properties : appearance, behaviour, layout, windows style etc,
5	**methods and events** - Differentiate procedure oriented, object oriented and event driven programming – Input box- Message box
6	**Working with Common Tool Box Controls:** Label , button, Textbox , NumericUpDown , Check Box, Radio Button , Group Box , control and all important methods and events.
	BLOCK 3 : ADDITIONAL CONTROLS AND MENUS OF WINDOWS
7	Working with other controls of toolbox: Date Time Picker, List Box, Combo box, Picture Box, Rich Text Box, Progress bar, Masked Text box, Link Label, Checked List box
8	Working with Menus: creating menu,inserting,deleting,assigning short cut keys, popup menu.
	BLOCK 4 : INBUILT FUNCTIONS, DIALOG BOX,MDI & EXCEPTION
9	**Inbuilt Functions :** Mathematical Functions-String manipulation - Dialog Boxes: OpenFileDialog, SaveFileDialog, FontDialog, ColorDialog, PrintDialog
10	**Sub Procedures and functions :** declaring, passing and returning arguments, exiting from it, pass by value and pass by ref
11	**Exception Handling and MDI:** Structured Error Handling (Try ....Catch ....finally), Unstructured Error Handling (On error go to line, goto 0, goto -1, resume next) - Multiple document interface (MDI) : MDI Parent form and child form.
	BLOCK 5 : DATABASE ACCESS USING ADO.NET
12	**ADO .NET Object Model:** Dataprovider - Dataset - ADO .NET Programming : Creating a Database Application
13	**Creating Connection to a Database using ADO.NET** , Populating Data in ADO.NET,
14	**Browsing Records**, Datagrid view, Editing, Saving, Adding and Deleting Records using bounded and unbounded.

Text Books:

1. Shelly, cashman, Quasney ‘ Microsoft Visual Basic .NET : Comprehensive Concepts And Techniques ‘ Cengage learning, 2012
2. StevenHolzner , Visual Basic .NET Programming Black Book , Dreamtech Press Publications, New Delhi

Book for Reference:

1. Julia Case Bradley and Anita C Millspaugh Programming in Visual Basic.NET McGraw Hill Higher Education (September 2002)

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COURSE CODE	TITLE OF THE COURSE
12962 / 13162	SYSTEM ANALYSIS AND DESIGN

Unit No.	Contents
	BLOCK 1 : SYSTEM CONCEPTS AND SYSTEM DEVELOPMENT LIFE CYCLE
1	**System Concepts** - Characteristics - Elements of a system - Types of Systems: Abstract, Physical, Open, Closed and Man-made Information system - Computer Based Information Systems: MIS, DSS, TPS and OAS
2	**System Development Life Cycle** - Problem Definition - Feasibility Study - Analysis - Design - Development - Implementation - Post Implementation and Maintenance
3	**System Analyst :** Interpersonal Skills - Technical Skill - Communication Skills - Role of Systems Analyst.
	BLOCK 2 : SYSTEM ANALYSIS
4	**System Analysis :** Bases for planning in System Analysis - Preliminary Investigation - Determining the User's information requirements, Case Scenario, Problem Definition and Project Initiation, Background Analysis
5	**Fact Finding Techniques :** Interview - Questionnaire - Record Review - Observation. Systems Analysis: Analysing Systems data - Feasibility Study: Technical, Economical and Operational - Steps in Feasibility Analysis, Feasibility Report, Oral Presentation
6	**Systems Costs & Benefits:** Categories of Cost - Benefits - Cost Benefit Analysis: Break Even, Present Value, Pay Back and Cash Flow. **Analysis Tools :** Data flow concept - Data Flow Diagram - Data Dictionary - Decision Table - Decision Tree - Structured English.
	BLOCK 3 : SYSTEM DESIGN
7	**System Design :** Process and stages of System Design : Logical and Physical Design. Design Methodologies: Structured design - Form Driven Methodology - Major Development Activities
8	**Input Output and Form Design:** Input Design : Capturing Data for input - Input Validation - Input Design of on-line systems. Output Design - Printed, Display and Audio.
9	**Forms Design :** Definition - Classification of Forms, Requirements of Forms Design - Types of Forms - Forms Control.
	BLOCK 4 : FILE AND DATABASE DESIGN:
10	**File concepts** - Types of Files - Methods of File Organization - Sequential - Direct - Indexed - Database Design: Database concept
11	**Types of Databases :** Hierarchical, Network and Relational.
12	**System Development**: Software Design - Top Down Approach - Flow Chart: System Flow Chart - Program Flow Chart - HIPO - IPO - VTOC - Warnier Orr Diagram - Structured Walkthrough - Quality Assurance - Levels of Assurance - System Testing - Special Systems Tests
	BLOCK 4 : SYSTEM EVALUATION,IMPLEMENTATION AND MAINTENANCE
13	**System Evaluation and Implementation** Training Personnel - Training Methods - Conversion: Conversion Methods - Parallel, Direct, Pilot and Phase-in. Conversion Plan - Site Preparation - Data and File Preparation - Post Implementation Review -
14	**System Maintenance :** Corrective - Adaptive - Hardware and Software Selection :

	Computer Industry - Software Industry - Procedure of Hardware and Software Selection: Major phases in Hardware and Software selection - Evaluation Process - Financial considerations.

Text Book

Elias M.Awad, Systems Analysis and Design, 1990, Galgotia Publication Pvt. Ltd.

Reference Book:

1. James A. Sen, Analysis and Design of Information System, 1985, McGraw Hill.

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COURSE CODE	TITLE OF THE COURSE
12963 / 13163	MULTIMEDIA AND ITS APPLICATIONS

Unit No.	Contents
	BLOCK 1 : FUNDAMENTAL CONCEPTS
1	**Fundamental concepts** in Text and Image: Multimedia and hypermedia, world wide web, overview of multimedia software tools.
2	**Graphics and image** data representation graphics/image data types, file formats
3	**Color in image and video:** color science, color models in images, color models in video.
	BLOCK 2 : FUNDAMENTAL CONCEPTS IN VIDEO AND DIGITAL AUDIO
4	Fundamental concepts in video and digital audio: Types of video signals, analog video, digital video
5	**Digitization of sound**, MIDI
6	**Quantization** and transmission of audio
	BLOCK 3 : MULTIMEDIA DATA COMPRESSION:
7	**Lossless compression algorithm:** Run-Length Coding, Variable Length Coding, Dictionary Based Coding, Arithmetic Coding
8	Lossless Image Compression, Lossy compression algorithm: Quantization.
	BLOCK 4 : BASIC VIDEO COMPRESSION TECHNIQUES
9	**Introduction** to video compression, video compression based on motion compensation
10	**search for motion vectors**, MPEG
11	**Basic Audio Compression** Techniques.
	BLOCK 5 : MULTIMEDIA NETWORKS:
12	**Multimedia Networks:** Basics of Multimedia Networks, Multimedia Network.
13	**Communications and Applications:** Quality of Multimedia Data Transmission, Multimedia over IP
14	**Multimedia over ATM Networks**, Transport of MPEG-4, Media-on-Demand(MOD).

Text Book:

1. Fundamentals of Multimedia by Ze-Nian Li and Mark S. Drew PHI/Pearson Education 2004

Reference Books:

1. Digital Multimedia, Nigel chapman and jenny chapman, Wiley-Dreamtech 2009
2. Multimedia and Communications Technology, Steve Heath, Elsevier (Focal Press) 1999

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COURSE CODE	TITLE OF THE COURSE
12964 / 13164	● NET PROGRAMMING LAB

Unit No.	Contents
	BLOCK 1
1	Building simple applications, Observe and draw visual .net IDE layout and hands on practice to create, save and open the project.
2	Working with intrinsic controls ,Control Arrays,Sub Procedures and functions
	BLOCK 2
3	Application with multiple forms
4	Application with dialogs
5	Application with Menus
6	Application using data controls
7	Application using Common Dialogs
	BLOCK 3
8	Drag and Drop Events,Inbuild functions, Mathematical and string functions
9	Database Management
10	Creating ActiveX Controls
11	Database object (DAO) and properties
12	Active Data Objects (ADO) ADO and OLE DB
	BLOCK 4
13	**Database : Bounded and Unbounded Mode :**Connecting to the database ,Retrieving a recordset Creating a query dynamically Using a parameterized query using action queries - Adding records Editing records closing the database connection
	BLOCK 5
14	**Simple Application development:** 1. Library information system 2. Students mark sheet processing 3. Telephone directory maintenance 4. Gas booking and delivering 5. Electricity bill processing 6. Bank Transaction 7. Pay roll processing 8. Personal information system 9. Question database and conducting Quiz 10. Personal diary

Text Books

1. Gary Cornwell Visual basic 6 , Tata McGraw Hill
2. Shelly, cashman, Quasney ‘ Microsoft Visual Basic .NET : Comprehensive Concepts And Techniques ‘Cengage learning, 2012
3. StevenHolzner , Visual Basic .NET Programming Black Book , Dreamtech Press Publications, New Delhi

8. B.Sc(Computer Science)

Duration – Three years
Eligibility – A pass in HSC (or) 3 year Diploma
Medium – English

Lateral Entry to II year B.Sc.(CS):
3 year Diploma in Computer Science Engineering, EEE / ECE / IT.

Course of Study & Scheme of Examinations

Sl. No.	Course Code	Title of the Course	CIA Max.	ESE Max.	TOT Max.	C Max.
		I SEMESTER				
1	13011 A / B	Part I: Tamil - Paper I /Communication Skills I	25	75	100	4
2	13012	Part II: English Paper I	25	75	100	4
3	13013	Programming in C	25	75	100	4
4	13014	Lab : Programming in C	25	75	100	4
			100	**300**	**400**	**16**
		II SEMESTER				
1	13021 A / B	Part I: Tamil Paper- II/Communication Skills- II	25	75	100	4
2	13022	Part II: English Paper- II	25	75	100	4
3	13023	Object Oriented Programming and C++	25	75	100	4
4	13024	Lab : Object Oriented Programming and C++	25	75	100	4
			100	**300**	**400**	**16**
		III SEMESTER				
1	13031 A / B	Part I: Paper : Tamil Paper- III/Human Skill Development- I	25	75	100	4
2	13032	Part II: English Paper- III	25	75	100	4
3	13033	Data Structures and Algorithms	25	75	100	4
4	13034	Lab : Data Structures and Algorithms	25	75	100	4
			100	**300**	**400**	**16**
		IV SEMESTER				
1	13041 A / B	Part I: Tamil Paper IV /Human Skill Development- II	25	75	100	4
2	13042	Part II: English Paper IV	25	75	100	4
3	13043	Java Programming	25	75	100	4
4	13044	Lab : Java Programming	25	75	100	4
			100	**300**	**400**	**16**
		V SEMESTER				
1	13051	Operating Systems	25	75	100	4
2	13052	Relational Database Management Systems	25	75	100	4
3	13053	Computer Architecture	25	75	100	4
4	13054	Lab : RDBMS	25	75	100	4

			100	**300**	**400**	**16**
VI SEMESTER						
1	13061	Computer Network	25	75	100	4
2	13062	Visual Basic Programming	25	75	100	4
3	13063	Software Engineering	25	75	100	4
4	13064	Lab : Visual Basic Programming	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		Grand Total	**600**	**1800**	**2400**	**96**

CIA: Continuous Internal Assessment, **ESE:** End Semester Examination,
TOT: Total, **C:** Credit Points **Max:** Maximum

SEMESTER I

Course Code	Title of the Course
13011A	Part I: Tamil - Paper I

PART- I TAMIL

பொதுத்தமிழ்

நோக்கம் : மொழி அறிவு, இலக்கண அறிவை வளர்த்தல்

கூறு 1

3. **கண்ணதாசன் - ஸ்ரீ கிருஷ்ண கானம்**

1. புல்லாங்குழல் கொடுத்த

2. குருவாயூருக்கு வாருங்கள்

கூறு 2

1. கோகுலத்து பசுக்கள்

2. கோகுலத்தில் ஒரு நாள் ராதை

3. ஆயர்பாடி மாளிகையில்

கூறு 3

பட்டுக்கோட்டை கல்யாண சுந்தரம்

1. நெஞ்சில் குடியிருக்கும்

2. செய்யும் தொழிலே தெய்வம்

கூறு 4

3. **பாரதியார்**

கண்ணன் என் விளையாட்டுப்பிள்ளை

பாரத மாதா திருப்பள்ளி எழுச்சி

கூறு 5

7. **பாரதிதாசன்** - உலகப்பன் பாட்டு (5)

8. **நாமக்கல் கவிஞர்** - நோயற்ற வாழ்வு 7 பாட்டு

9. **பெ.தூரன்** - நிலா பிஞ்சு

கூறு 6

7. **வல்லிக் கண்ணன்** - வெறும் புகழ்

8. **கு.ப.இராஜகோபாலன்** - எதற்காக?

9. **மீரா** - பதினைந்து

கூறு 7

5. **சிற்பி** - சர்ப்ப யாகம்

6. **ஞானக்கூத்தன்** - தோழர் மோசிகீரனார்

கூறு 8

5. **அப்துல் ரகுமான்** - கண்ணும் எழுதேம்

6. **சண்முக சுப்பையா** - வயிறு

கூறு 9

5. **சிலப்பதிகாரம்** - வழக்குரை காதை

6. **கம்பராமாயணம்** - அயோத்தியா காண்டம்

கூறு 10

3. **சீறாப்புராணம்** - ஈத்தங்குலை வரவழைத்த படலம் (1)

கூறு 11

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

1. இன்னவாயில்

57. கொழுந்துறும்

58. பஞ்(ச) அரங்கில்

கூறு 12

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

59. எண்ணுளே

60. ஒண்தலங்கள்

61. இரவியேந்த கஞ்சக்

கூறு 13

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

62. கன்னியாயதாயும்

63. ஏந்தி ஓங்கு உளத்து

64. ஆவ தேமுனர்

65. கொல்லும் வேலொடும்

கூறு 14

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

66. என்ற வாசகம்

67. அம்பினால்

68. வேண்டும் ஓர் வினை

Course Code	Title of the Course
13011B	**PART- I : COMMUNICATION SKILLS I**

Unit – 1 Communication – Meaning – Types- Importance

Unit – 2 Barriers to Effective Communication – Principles – Principles of Effective Communication

Unit – 3 Oral Communication – Meaning – Importance- Forms of Oral Communication

Unit – 4 Intonation –Meaning – Function- Types
Preparation of Speech- Steps Involved

Unit – 5 Principles of Effective Oral Communication

Unit – 6 Written Communication – Meaning –Steps – Importance- Advantages
Use of words and Phrases

Unit – 7 Sentence – Meaning –Sentence formation- Characteristics of an Effective Sentence

Unit – 8 Paragraph Writing –Essay Writing –Steps Involved –Outline-Layout –Contents-Drafting-Correction- Final Draft

Unit – 9 Application for Employment and Curriculum Vitae –Steps involved

Unit – 10 Non –Verbal Communication – Meaning –Types –Body Language –Postures-Gestures – Facial Expressions –Eye Contact

Unit – 11 Report Writing –Report –Types of Reports –Format of a Report

Unit – 12 Essentials of a Good Report –Preparation of Report-Procedure Involved

Unit – 13 Meetings-Purpose of the Meeting – Procedure

Unit – 14 Group Discussion –Quality of Content-Participation –Logical Presentation –Behavioral Skills

References:

1. Krishna Mohan & Meera Banerjee, Developing Communication Skills, 2005.
2. Geetha Nagaraj, Write to Communicate, 2004.
3. Wren & Martin, English Grammar and Composition, 2002.
4. Dale Carnegie, How to Win Friends and Influence People, 1981.
5. Dale R Jordan, Language Skills and Use.
6. Gartside L. Bahld, Nagammiah and McComas, Satterwhite, Modern Business Correspondence.
7. Rajendra Pal and Kortahalli J S, Essentials of Business Communication.
8. Wallace, Michael J, Study Skills in English.
9. Editors of Readers Digest, Super Word Power.

***

Course Code	Title of the Course
13012	Part- II: English Paper I

Prose

Unit – 1	Water-the Elixir of life	- C.V. Raman
Unit – 2	Mrs. Packletide's Tiger	- SAKI
Unit – 3	A Deed of Bravery	- Jim Carbett
Unit – 4	The Cat	- Catharine M.Willson
Unit – 5	On Letter Writing	- Alpha of the Plough
Unit – 6	Our Ancestors	- Carl Sagan
Unit – 7	Our Civilization	- C.E.Foad
Unit – 8	A Hero on Probation	- B.R. Nanda
Unit – 9	Dangers of Drug Abuse	- Hardin B. Fones
Unit – 10	Food	- J.B.S. Haldane

Grammar

Unit – 11 - Articles-Gerunds-Participles-Infinitives-Modals-Proposition –Tenses.

Unit – 12 - Direct and Indirect Speech-Transformation of sentences- Active and passive voice.

Composition

Unit – XIII - Letter writing - Precis writing - Developing hints.

Unit – XIV - Dialogue writing - Paragraph writing.

References:

1. Sebastian D K, Prose for the Young Reader, Macmillan.
2. Active English Grammar, Ed. by the Board of Editors, Macmillan.
3. Modern English – A Book of Grammar Usage and Composition by N.Krishnaswamy, Macmillan Publishers.

Course Code	Title of the Course
13013	Programming in C

Unit No.	Contents
	BLOCK 1: INTRODUCTION
1	**Introduction and Features**: History of C, Importance of C, Basic Structure of C program, character set, Tokens, keywords and identifiers
2	**Constants and variables and data types:** declaration of variables, defining symbolic constants, declaring a variable as a constant
3	**Operators and expressions:** arithmetic, relational, logical, assignment operators, arithmetic expression, Evaluation of expressions, precedence of arithmetic operators
	BLOCK 2 : I/O OPERATIONS AND DECISION MAKING
4	**Managing I/O operations:** reading and writing a character, formatted input, output
5	**Decision making and branching:** IF statement, If..else statement, nesting if else statement, else if ladder, switch statement, goto statement, while statement, do statement, for statement
6	**arrays:** one-dimensional arrays, declaration, initialization, two dimensional arrays, multi dimensional arrays, dynamic arrays
7	**Strings :** Declaration, Initialization of string variables, reading and writing strings, string handling functions
	BLOCK 3 : USER DEFINED FUNCTIONS
8	**Functions basics:** Elements of user defined functions, definitions, return values and their types, function calls, declaration, nesting of functions, recursion
9	**Structures and Unions:** Defining a structure, declaring a structure variable, accessing structure members, array of structures, array within structures, structures within structures, structures and functions
	BLOCK 4 : POINTERS
10	**pointers:** Basics, declaring, initialization of pointer variables, address of variable, accessing a variable through its pointer
11	**Pointer as Functions:** Chain of pointers, pointer increments and scale factors
12	**Strings with Pointer:** pointers and character strings, pointers and structures
	BLOCK 5 : FILES
13	**Introduction:** Introduction, Defining, opening and closing files, I/O operations on files
14	**Error Handling methods:** Error Handling during I/O operations, command line arguments

TEXT BOOK:

1. Programming with C, Schaum outline series, Gottfried, TataMcHill,2006
2. Programming with ANSI and Turbo C, Ashok N Kamthane, Pearson Education, 2006

REFERENCE BOOK:

1. C: The complete reference,H Schildt, TMH Edition, 2000
2. Kanetkar, Let Us C, BPB publications, 1999.

Course Code	Title of the Course
13014	Lab : Programming in C

Unit No.	Contents
	BLOCK 1: C PROGRAM FUNDAMENTALS
1	Simple C programs
2	Using IF and switch constructs programs
3	Looping related problems
	BLOCK 2 : FUNCTIONS, ARRAYS, STRINGS
4	Programs using functions
5	IF statement, If..else statement, nesting if else statement, else if ladder, switch statement, goto statement, while statement, do statement, for statement
6	One-dimensional arrays, two dimensional arrays, multi dimensional arrays
7	Initialization of string variables, reading and writing strings, string handling functions
	BLOCK 3 : STRUCTURE AND UNIONS
8	Programs using structures
9	Programs using unions
	BLOCK 4 : POINTERS
10	Initialization of pointer variables, address of variable, accessing a variable through its pointer
11	Pointer as Functions
12	Strings with Pointer: pointers and character strings, pointers and structures
	BLOCK 5 : FILES
13	Programs based on file handling
14	Error Handling methods: Error Handling during I/O operations, command line arguments

II SEMESTER

Course Code	Title of the Course
13021 *A*	Part I: Tamil Paper - II

கூறு 1

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

1. சொல் தவிர்ந்த
2. அன்னை
3. அஞ்சுவார்
4. சொல்லக் கேட்டனள்
5. மற்செய்கை
6. மண்கனியப்
7. அழுது ஆர்ந்த

கூறு 2

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

1. பொய் பொதுளும்
2. இன்பு அருந்தி
3. வழுதாயின இன்பு
4. மறம் ஏவினர்

கூறு 3

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

1. மண்ணோர்கள்
2. பொய்யா விதியோய்
3. விடியா இருள்
4. அழுவார் எவரும்

கூறு 4

சிறுகதை - நீலபத்மநாபனின் ‘‘வான வீதியில்’’

கூறு 5

உரைநடை - கம்பன் புறத்திணை - தி.சொக்கலிங்கம்

இலக்கணம் - எழுத்தும் சொல்லும்

கூறு 6

1. முதலெழுத்துகள், சார்பெழுத்துகள்
2. மொழி முதலெழுத்துகள் , மொழி இறுதி எழுத்துகள்

கூறு 7

1. ஒற்றெழுத்து மிகலும் மிகாமையும்,
2. ஆகு பெயர் , அன்மொழித் தொகை.
3. வினா-விடை வகைகள்

கூறு 8

1. தமிழ்ச் சொல்லமைப்பின் சிறப்பு – பெயர் , வினை, இடை, உரி வடிவங்கள் ,
2. பிற மொழிச் சொற்களைத் தமிழில் ஆளும் முறைகள்

கூறு 9

1. அல் வழி, வேற்றுமைப் புணர்ச்சிகள்

2. திணை, பால் , எண் , இட இயைபு.

தமிழ் இலக்கிய வரலாறு

கூறு 10

1. **இக்கால இலக்கிய வகைகள்**

அ) மரபுக் கவிதை

ஆ) புதுக் கவிதையின் தோற்றமும் வளர்ச்சியும்

கூறு 11

1. உரை நடை இலக்கியங்கள் - தோற்றமும் வளர்ச்சியும்

அ) கட்டுரை ஆ) சிறுகதை

இ) புதினம் ஈ) நாடகம்

கூறு 12

2. **இக்கால இலக்கியக் களங்கள்**

திரைப்படம் , தொலைக்காட்சி, வானொலி, இதழ்கள் தமிழுக்கு ஆற்றி வரும் பணிகள்

கூறு 13

2. **தமிழும் சமயமும் :**

அ) சைவம் ஆ)வைணவம் இ)சமணம் ஈ)பௌத்தம் உ)இசுலாம்

ஊ) கிறித்துவம்

கூறு 14

1. பிற்காலக் காப்பியங்கள் :

அ) கம்பராமாயணம் **ஆ)** பெரியபராணம்

2. இணையம் - பற்றிய செய்திகள்

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Course Code	Title of the Course
13021 B	**PART-I : COMMUNICATION SKILLS - II**

Unit – 1 Code and Content of Communication Skills

Unit – 2 Stimulus and Response of Communication Skills

Unit – 3 Effective Speaking Guidelines

Unit – 4 Pronunciation Etiquette of Communication Skills

Unit – 5 Phonetics in Communication Skills

Unit – 6 A self Assessment of Communicating Soft Skills

Unit – 7 Language Skills –Ability –Skill Selected Need- Learner Centre activities

Unit – 8 Listening Skills –Importance –Types of Listening- Interview Skills

Unit – 9 Conversation Skills –Modes

Unit – 10 Presentation Skills - Preparing –Planning-Presentation

Unit – 11 Written Communication –Structure of Effective Sentences –Paragraph

Unit – 12 Technical Writing-Creative Writing- Editing and Publishing

Unit – 13 Corporate Communication Skills-Internal –Effective business writing –Letters, Proposals, Resume

Unit – 14 Corporal Communication Skills-External - Press release - Newsletters- Interviewing skills

References:

1. Dutt. Kiranmai & Geeta Rajjevan. Basic Communication Skills. Rev.ed. Foundation Books Pvt.Ltd. Cambridge House, New Delhi 2006.
2. Bill R. Swetmon. Communication Skills for the 21st Century. Chennai: Eswar Press. First South Asian Edition 2006.
3. Glass. Lillian. Talk to Win. New York: Perigee Books,1987.
4. Pease. Alan. Signals: How to Use Body Language for Power, Success and Love, New York: Bantam Books, 1981.
5. Walters. Lilly. Secrets of Successful Speakers. New York: McGraw-Hill, Inc., 1993.
6. Mandal. S.K. How to Succeed in Group Discussions & Personal Interviews. Mumbai: JAICO Publishing House.
7. Rogoff. Leonard and Ballenger. Grady. Office Guide to Business Letters, Memos & Reports. New York: Macmillan, 1994.
8. Krishna Mohan & Meera Banerjee, Developing Communication Skills, 2005.
9. Geetha Nagaraj, Write to Communicate, 2004.
10. Wren & Martin, English Grammar and Composition, 2002.
11. Rajendra Pal and Kortahalli J S, Essentials of Business Communication.

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Course Code	Title of the Course
13022	Part II: English Paper- II

Poetry

Unit – 1	Sonnet	- William Shakespeare
Unit – 2	Lines Composed upon Westminster Bridge -William Wordsworth	
Unit – 3	Grecian Urn	- John Keats (1795-1827)
Unit – 4	Andrea Del Sarto	- Robert Browning (1812-1889)
Unit – 5	The Road Not Taken	- Robert Frost (1874-1963)
Unit – 6	Strange Meeting	- Wilfred Owen (1813-1918)
Unit – 7	Gitanjali	- Rabindranath Tagore (1861-1946)
Unit – 8	The Coromandel Fishers	- Sarojini Naidu
Unit – 9	The Express	- Stephen Spender
Unit – 10	**Shakespeare:** The Merchant of Venice	

Language Use:

Unit – 11	Essay writing
Unit – 12	Note Making
Unit – 13	Report writing
Unit – 14	Comprehension

References:

1. The Golden Quill, P.K. Seshadri, Macmillan.

2. The Merchant of Venice, Shakespeare. (Any overseas edition).

3. Active English Grammar, Ed. by the Board of Editors, Macmillan.

4. Modern English – A Book of Grammar Usage and Composition by N.Krishnaswamy, Macmillan Publishers.

Course Code	Title of the Course
13023	OBJECT ORIENTED PROGRAMMING and C++

Unit No.	Contents
	BLOCK 1: INTRODUCTION
1	**Introduction and Features**: Evolution of Object Oriented Language, Object oriented Paradigm, Basic concept of object-oriented programming- objects, classes, encapsulation and data abstraction, inheritance, polymorphism, dynamic binding, message passing
2	**Popular OOP languages.** Moving from C to C++ Introduction – Predefined console streams, hierarchy of console stream classes,
3	**I/O operations;** Unformatted I/O operations, formatted console I/O operations, manipulators, custom/user-defined manipulators.
	BLOCK 2 : CLASSES AND OBJECTS
4	**Classes and Objects:** Introduction, class specification, class objects, accessing class members, defining member functions, accessing member functions within a class, outside member functions as inline, private member function,
5	**Memory allocation for objects:** array of objects, function prototype, call by reference, return by reference, objects as function arguments, inline function, friend function, constant parameter and member function.
6	**Object Initialization:** Introduction - constructors, default constructor, parameterized constructors, multiple constructors in a class, dynamic initialization through constructors, copy constructor, dynamic constructor, destructor. Dynamic Objects: Introduction, pointers to objects, array of pointers to objects, this pointer.
	BLOCK 3 : INHERITANCE, POLYMORPHISM AND DATA CONVERSION
7	**Inheritance:** Introduction, derived class declaration, forms of inheritance, inheritance and member accessibility, multilevel inheritance, multiple inheritance, hierarchical inheritance, hybrid inheritance.
8	**Polymorphism:** Introduction, Function overloading, Operator overloading introduction, unary operator overloading, binary operator overloading, assignment operator overloading, overloading with friend functions.
9	**Data conversion:** conversion between basic data types, conversion between objects and basic types, conversion between objects of different classes. Virtual function: Introduction, need for virtual functions, pure virtual functions, abstract classes.
	BLOCK 4 : TEMPLATES AND FILES
10	**Generic Programming with Templates:** Introduction - class templates – class template with multiple arguments
11	**Function template:** function template with multiple arguments. Inheritance of class template.
12	**Streams with Files:** Introduction, hierarchy of file stream classes, opening and closing of files, file pointers and their manipulators, sequential access to a file, file input/output with stream class, random access to a file.
	BLOCK 5 : EXCEPTION HANDLING
13	**Exception Handling:** Introduction – Basics of exception handling, exception handling mechanism, throwing mechanism, catching mechanism. Exceptions in constructors and destructors
14	**Other Exception Handling methods:** Handling uncaught exceptions, exceptions in operator overloaded functions, exception in inheritance tree, exceptions in class templates, memory allocation failure exception.

TEXT BOOK:

1. E.Balagurusamy, Object oriented programming in C++, Third Edition, Tata McGraw Hill Publications, 2007.
2. Mastering C++, K.R Venugopal and Rajkumar, T.Ravishankar, Tata McGraw Hill Publishing Company Ltd., 2006.

REFERENCE BOOK:

1. Object Oriented Programming in C++, Fourth Edition, Robert Lafore, Galgotia Publications Pvt. Ltd., New Delhi. 2010.

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Course Code	Title of the Course
13024	LAB: OBJECT ORIENTED PROGRAMMING and C++

Unit No.	Contents
	BLOCK 1: INTRODUCTION
1	Writing simple C++ programs
2	Using if and switch constructs Programs
3	**Looping , Arrays ,Structure statements**: for, while, do-while, Strings and Matrices Programs Problems
	BLOCK 2 : OOPs CONCEPT
4	**Functions:** static function, friend function ,constructor , destructor and operator overloading and Recursive programs
5	**Inheritance :** Inheritance types
6	**Polymorphism :** polymorphism types, Virtual function
	BLOCK 3 : FILE AND POINTERS
7	**File:** File Handling C++ Programs, opening and closing a data file - creating a data file, processing a data file.
8	**Pointers :** Pointers and Pointers with Arrays Programs
9	**Virtual functions:** Pure virtual functions
	BLOCK 4 : TEMPLATES AND FILES
10	**Generic Programming with Templates:** Demonstrating class templates, class template with multiple arguments
11	**Function template:** Demonstrating function template with multiple arguments. Inheritance of class template.
12	**Streams with Files:** opening and closing of files, file pointers and their manipulators, sequential access to a file, file input/output with stream class, random access to a file.
	BLOCK 5 : EXCEPTION HANDLING
13	**Exception Handling:** programs using exception handling, Exceptions in constructors and destructors
14	**Other Exception Handling methods:** Handling uncaught exceptions, exceptions in operator overloaded functions, exception in inheritance tree, exceptions in class templates, memory allocation failure exception.

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III SEMESTER

Course Code	Title of the Course
13031 *A*	**Part I: Tamil Paper- III**

கூறு 1 : பத்துப்பாட்டு – முல்லைப்பாட்டு

கூறு 2 : எட்டுத்தொகை – ஐங்குறுநூறு

கூறு 3 : கபிலர் - குறிஞ்சித்திணை

கூறு 4 : மஞ்ஞைப்பத்து – முதல் மூன்று பாடல்கள்

கூறு 5 : குறுந்தொகை – பரணர் பாடல்கள் பா. எண். 19, 24, 36, 128, 399

கூறு 6 : நற்றிணை – பெருங்குன்றூர்கிழார் - பா. எண். 5

பெருவழுதியார் - பா. எண். 55

பெருங்கௌசிகனார் - பா. எண். 139

கூறு 7 : நற்றிணை – கருவூர்க்கோசிகனார் - பா. எண். 214

உலோச்சனார் - பா. எண் 249

கூறு 8 : அகநானூறு – சேந்தம்பூதனார் பாடல்கள் பா.எண். 84, 207

கூறு 9 : புறநானூறு – மறோக்கத்து நப்பசலையார் பாடல்கள்

பா. எண். 37, 39, 126, 226, 280

கூறு 10 : பதினெண் கீழ்க்கணக்கு – திருக்குறள் - வாழ்க்கைத் துணை நலம் (6), அறிவுடைமை (43), பிரிவாற்றாமை (116)

கூறு 11: நான்மணிக்கடிகை – எள்ளற்க (3), பறைபடவாழா (4),

கூறு 12: நான்மணிக்கடிகை - மண்ணயறிப (5),கள்ளிவயிற்றில் (6), கல்லிற்பிறக்கும்(7)

கூறு 13: நாடகம் - இராசராசசோழன் - அரு. இராமநாதன்

கூறு 14: நாவல் - சுவடுகள் - இரா. பாலசுப்பிரமணியன், சத்யா வெளியீடு, மதுரை

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Course Code	Title of the Course
13031B	**PART-I : HUMAN SKILLS DEVELOPMENT - I**

Unit – 1 Human Skills –Developing skills-Types

Unit – 2 Mind-Levels of functions
Habits-Meaning-Types-Merits of good habits - Interpersonal Relationship-Features-Interpersonal Behaviour

Unit – 3 Thinking ahead- Significance of thinking ahead

Unit – 4 Developing Personality-Meaning -Need- Factors influencing personality, Ways of developing personality -Building positive personality

Unit – 5 Self-concept-Self-esteem-Meaning-Importance - Self- efficacy-Self-acceptance-Meaning-Importance - Etiquette-Meaning-Etiquettes in using mobile, telephones-Dais Etiquette

Unit – 6 Goal-setting Skills-Meaning-Types-Importance-

Unit – 7 Decision-making skills-Meaning-Types-Steps in decision-making

Unit – 8 Negotiating Skills-Styles-Structure-Creating negotiation-Competitive Negotiation

Unit – 9 Attitudes-Meaning-Types-Importance-Developing positive attitudes

Unit – 10 coping with Change- Meaning – Characteristics – Importance of change Resistance to change-Dealing with change

Unit – 11 Leadership-Meaning-Importance-Characteristics-Styles-

Unit – 12 Human Relations Skill-Need-Canons of good human relations

Unit – 13 **C**ounselling-Meaning-Importance-Forms- Conflicts-Meaning-Types-Causes-Effects-Managements of conflicts

Unit – 14 Stress-Meaning-Types-Causes-Effects-Managing the stress - Anger-Meaning-Causes-Consequences-Anger Management

References:

1. Les Giblin, Skill with People, 1995.
2. Shiv Khera, You Can Win, 2002.
3. Christian H Godefroy, Mind Power.
4. Dale Carnegie, How to Enjoy Your Life and Your Job, 1985.
5. Natalie H Rogers, How to Speak without Fear, 1982.
6. Dale Carnegie, How to Develop Self-Confidence and Influence People by Public Speaking.

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Course Code	Title of the Course
13032	Part II: English Paper- III

Short Stories

Unit – 1	A Hero	- R.K. Narayanan
Unit – 2	The Diamond Necklace	- Guy de Maupassant
Unit – 3	The Verger	- Somerset Maugham
Unit – 4	The Postmaster	- Rabindranath Tagore

One Act Plays

Unit – 5	The Proposal	- Anton Chekhou
Unit – 6	The Boy Comes Home	- A.A. Milne
Unit – 7	The Silver Idol	- James R. Waugh
Unit – 8	Progress	- St. John Ervine
Unit – 9	The Pie and the Tart	- Huge Chesterman
Unit – 10	Reunion	- W.st. Joh Tayleur
Unit – 11	A kind of Justice	- Margaret Wood
Unit – 12	The Refugee	- Asif Currimbhoy

Grammar

Unit – 13 Parts of speech-Noun- Pronoun- Adjective Degrees of Comparison- Verb- Adverb

Composition

Unit – 14 Agenda- Minutes- Notice- Descriptive Writing

References:

1. Aroma, Ed. by the Board of Editors, Publishers- New Century Book House, Chennai.
2. Six Short Stories, Ed. by the Board of Editors, Harrows Publications, Chennai.
3. One Act Plays, Ed. by the Board of Editors, Harrows Publications, Chennai.
4. Modern English – A Book of Grammar Usage and Composition by N.Krishnaswamy, Macmillan Publishers.
5. English for Communication, Ed. by the Board of Editors, Harrows Publications,Chennai.

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Course Code	Title of the Course
13033	DATA STRUCTURES AND ALGORITHMS

Unit No	Contents
	BLOCK 1 : INTRODUCTION
1	**Introduction to Data Structure :** Types of Data Structure , Primitive data types
	Algorithms: –Time and space Complexity of algorithms
2	**Arrays:** Array initialization, Definition of Array, Characteristic of Array ,One-dimensional Array, Two-dimensional array and Multi dimensional array
	BLOCK 2 : LINEAR DATA STRUCTURE
3	**Stack :** Stack related terms, Operations on a stack,
4	**Representation of Stack:** Implementation of a stack – application of Stack. Expression Evaluation Polish notation.
5	**Queues:** Operations on queue Circular Queue, Representation of Queues, Application of Queues
6	**List:** Merging lists, Linked list, Single linked list, Double Linked List, Header Linked list
7	**Operation on Linked List :** Insertion and Deletion of linked list
8	**Traversal:** Traversing a linked list , Representation of linked list.
	BLOCK : 3 NON-LINEAR DATA STRUCTURE
9	**Trees:** Binary Trees, Types of Binary trees, Binary Tree Representation
10	**Binary Tree operations / Applications :** Traversing Binary Trees, Binary Search tree,
11	**Operations on Binary Tree:** Insertion and Deletion operations, Hashing Techniques.
	BLOCK 4 : SEARCHING TECHNIQUES
12	**Searching :** Introduction, Searching, Linear Search, Binary Search
	BLOCK 5 : SORTING TECHNIQUES
13	**Sorting:** Bubble sort, Insertion sort, Radix sort
14	**Other sorting Techniques:** Selection sort, Quick sort, Tree sort.

Text Books:

1. Fundamentals of Data structures , Second edition, Ellis Horowitz and Sartaj Sahini, Universities press, 2007.
2. Data Structures, Seymour Lipschutz, G.A.Vijayalakshmi Pai, Second Edition , Schaum's Outlines, Tata Mc-Graw Hill Private Ltd., 2006.

Reference Books:

1. Programming and Data Structure, Pearson Edition, Ashok N Kamthane, 2007.

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Course Code	Title of the Course
13034	Lab : Data Structures and Algorithms

Experiments based on c++ programming and Data Structures

Unit No.	Contents
	BLOCK 1 : SIMPLE C++ PROGRAMS
1	**Introduction** Simple C++ Programs
2	**Control Structures:** Using if and switch constructs Programs
3	**Looping , Arrays ,Structure statements**: for, while, do-while, Strings and Matrices Programs Problems
	BLOCK 2 : OOPs CONCEPTS
4	**Functions:** static function, friend function ,constructor , destructor and operator overloading and Recursive programs
5	**Inheritance and polymorphism:** Inheritance types and polymorphism types, Virtual function
6	**File:** File Handling C++ Programs, opening and closing a data file - creating a data file, processing a data file.
7	**Pointers :** Pointers and Pointers with Arrays Programs
	BLOCK 3: LINEAR DATA STRUCUTURE
8	**Stacks :** Stack Implementation, expression evaluation, Polish notation
9	**Queues:** Queue Implementation, Applications of Queue
10	**Linked List programs:** List, Merging lists, Linked list, Single linked list, Double Linked List, Header Linked list, Insertion and Deletion of linked list, Traversing a linked list.
	BLOCK 4 : NON LINEAR DATA STRUCTURE
11	**Tree Programs :** Trees, Binary Trees, Types of Binary trees, Binary Tree Representation, Traversing Binary Trees, Binary Search tree, Insertion and Deletion operations,
12	**Graphs:** Shortest Path Algorithms o Dijkstra's Algorithm o Graphs with Negative Edge costs o Acyclic Graphs o All Pairs Shortest Paths Algorithm **Minimum cost Spanning Trees** o Kruskal's Algorithm o Prims's Algorithm o Applications □ Breadth First Search
	BLOCK 5 : SEARCHING AND SORTING ALGORITHMS
13	**Searching Techniques:** Linear and Binary search Programs
14	**Sorting techniques**: Bubble sort, Quick sort, Insertion sort, Merge sort

Reference Books:

1. Data Structures, Seymour Lipschutz, G.A.Vijayalakshmi Pai, Second Edition , Schaum's Outlines, Tata Mc-Graw Hill Private Ltd., 2006.
2. Fundamentals of Data structures in C, Second edition, Ellis Horowitz and Sartaj Sahini, Universities press, 2007.
3. Programming and Data Structure, Pearson Edition, Ashok N Kamthane, 2007.

SEMESTER IV

Course Code	Title of the Course
13041 A	Part I: Tamil Paper - IV

கூறு 1 : செய்யுள் உறுப்புகள் - யாப்பு - எழுத்து, அசை, சீர்,

கூறு 2 : செய்யுள் உறுப்புகள் - யாப்பு - தளை, அடி, தொடை

கூறு 3 : வெண்பா, ஆசிரியப்பா, கலிப்பா, வஞ்சிப்பா,

கூறு 4 : புதிய யாப்பு வடிவங்கள் - சிந்து, கண்ணி, கீர்த்தனை

கூறு 5 : புதுக்கவிதையில் குறியீடு – படிமம்.

கூறு 6 : அகப்பொருள் - புறப்பொருள் - ஐந்திணை விளக்கம்

கூறு 7 : அகப்பொருள் துறைகள் - வரைவு கடாதல், அறத்தொடு நிற்றல், உடன்போக்கு

கூறு 8 : புறப்பொருள் துறைகள் - வஞ்சினக்காஞ்சி, கையறுநிலை, செவியறிவுறூஉ

கூறு 9 : அணி இலக்கணம் - உவமை, உருவகம், வேற்றுமை, பிறிது மொழிதல், தற்குறிப்பேற்றம், சிலேடை, பின்வருநிலை

கூறு 10 : நிறுத்தல் குறிகள்

கூறு 11 : தொல்காப்பியம் - சங்கஇலக்கியம் - எட்டுத்தொகை, பத்துப்பாட்டு,

கூறு 12 : பதினெண்கீழ்க்கணக்கு

கூறு 13 : ஐம்பெருங்காப்பியங்கள் - பிற்காலக் காப்பியங்கள் - கம்பராமாயணம் - பெரியபுராணம்

கூறு 13 : இக்காலக் காப்பியங்கள் - பாரதியின் பாஞ்சாலி சபதம் - பாரதிதாசனின் பாண்டியன் பரிசு

கூறு 14. : கண்ணதாசனின் இயேசு காவியம் , சிற்பியின் - மௌன மயக்கங்கள்.

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Course Code	Title of the Course
13041 B	**PART-I : HUMAN SKILLS DEVELOPMENT - II**

Unit – 1 Guidance & Counselling – Role of Counsellor - Importance and Techniques of counselling

Unit – 2 Managerial skill- Need – Importance

Unit – 3 Human relational skills-Communication-Attention

Unit – 4 Conceptual skills-Meaning-Importance

Unit – 5 Technical skills-Techniques-Practices-Tools-Procedures

Unit – 6 Presentation skills-Planning-Preparation-Delivery

Unit – 7 Organization skills-Meaning-Nature-Importance-Types

Unit – 8 Multi-Tasking skills Responsibilities-Causes

Unit – 9 Leader- Qualities of a good leader

Unit – 10 Understanding Skills -Human systems: Individual, Group, organization, and their major interactions

Unit – 11 Understanding Skills -Human systems: Community and Society, and their major interactions

Unit – 12 Problem solving skills – Handling –Facing - Importance

Unit – 13 Cooperative Learning Skills

Unit – 14 Making Social Responsibilities-Causes

References:

1. Les Giblin, Skill with People, 1995.
2. Shiv Khera, You Can Win, 2002.
3. Christian H Godefroy, Mind Power.
4. Dale Carnegie, How to Enjoy Your Life and Your Job, 1985.
5. Natalie H Rogers, How to Speak without Fear, 1982.
6. Dale Carnegie, How to Develop Self-Confidence and Influence People by Public Speaking.

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Course Code	Title of the Course
13042	Part II: English Paper - IV

Short Stories

Unit – 1 Lalajee - Jim Corbelt
Unit – 2 A Day's Wait - Hemmingway
Unit – 3 Two old Men - Leo Tolstoy
Unit – 4 Little Girls wiser than - Men Tolstoy
Unit – 5 Boy who wanted more Cheese - William Elliot Griffir

Drama

Unit – 6 Pygmalion - G.B. Shaw

Fiction

Unit – 7 Swami and Friends - R.K. Narayanan

Tales from Shakespeare

Unit – 8 - The Merchant of Venice
Unit – 9 - Romeo and Juliet
Unit – 10 - The Winter's Tale

Biographies

Unit – 11 - Martin-Luther king - R.N. Roy
Unit – 12 - Nehru - A.J. Toynbee

Grammar

Unit – 13 - Concord- Phrases and Clauses-Question Tag

Composition

Unit – 14 - Expansion of Proverbs
- Group Discussion
- Conversation (Apologizing, Requesting, Thanking)

References:

1. Sizzlers, by the Board of Editors, Publishers-:Manimekala Publishing House, Madurai.
2. Pygmalion – G.B. Shaw
3. Swami and Friends – R.K. Narayan
4. Tales from Shakespeare Ed. by the Board of Editors, Harrows Publications, Chennai.
5. Modern English – A Book of Grammar Usage and Composition by N.Krishnaswamy, Macmillan Publishers.

Course Code	Title of the Course
13043	JAVA PROGRAMMING

Unit No	ntents
	BLOCK 1 INTRODUCTION
1	**Java Evolution:** Java history, features, java and Internet, WWW, web browsers
2	**Overview :**simple java program, program structure, tokens, statements
3	**Writing Java programs:** JVM, constants, variables, data types, type casting
	BLOCK 2 : OPERATORS AND EXPRESSIONS
4	**Operators** : arithmetic, relational, logical, assignment, increment and decrement, conditional, bitwise, special operators
5	**Expressions :** arithmetic, Evaluation of expression, operator precedence and associatively
6	**Decision making and branching:** If, If..Else, nesting of If..Else, else..if, switch, ? Operators, while..do, for jump in loops
	BLOCK 3 : CLASSES,OBJECTS
7	**Defining a class:** adding variables, methods, creating objects, accessing members, constructors, method overloading, nesting of methods, inheritance, overriding methods, final classes
8	**Arrays, strings and vectors:** arrays, one dimensional arrays, two dimensional arrays, strings, vectors, wrapper classes
9	**Interfaces :** multiple inheritance, defining interfaces, extending interfaces, implementing interfaces, accessing interface variables
	BLOCK 4 PACKAGES AND MULTITHREADED PROGRAMMING
10	**API packages :** using system packages, naming conventions, creating packages, accessing packages, using a package, adding a class to a package
11	**Basics :** creating threads, extending the thread class, stopping and blocking a thread, life cycle of a thread, using thread methods, thread exceptions, synchronization, implementing the 'Runnable' interface
12	**Managing Errors :** types of errors, exception handling code, multiple catch statements, using finally statement
	BLOCK 5 APPLET AND GRAPHICS PROGRAMMING
13	**INTRODUCTION:** preparing to write applets, applet life cycle, applet tag, adding applet to a HTML file, running the applet
14	**The Graphics class**: lines and rectangles, circles and ellipses, drawing arcs, drawing polygons, line graphs

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Course Code	Title of the Course
13044	LAB : JAVA PROGRAMMING

Unit No	ntents
	BLOCK 1 JAVA FUNDAMENTALS
1	**Simple Java programs**
2	**Programs using classes and objects**
3	**Conditional statements using Java**
4	**Looping statements using Java**
	BLOCK 2 : OOP CONCEPTS
5	Operator overloading programs
6	Function overloading programs
7	Inheritance, packages
8	Polymorphism and message passing programs
	BLOCK 3 : VIRTUAL FUNCTION & THREADS
9	Threads
10	Virtual functions
	BLOCK 4 : I/O AND EXCEPTION HANDLING
11	Exception handling programs
12	I/O manipulation programs
	BLOCK 5 : APPLET AND NETWORK PROGRAMMING
13	Simple applet programs
14	Simple network programs using Java

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SEMESTER V

Course Code	Title of the Course
13051	**OPERATING SYSTEMS**

Unit No	ntents
	BLOCK 1 INTRODUCTION
1	Introduction, components and goals, operating system architecture
2	Process concepts: Introduction, process states, process management
3	Interrupts, Interprocess communication
	BLOCK 2 : ASYNCHRONOUS CONCURRENT EXECUTION
4	Introduction, mutual exclusion, implementing mutual exclusion primitives
5	Software solution to the mutual exclusion problem, hardware solution to mutual exclusion problem, semaphores
6	Concurrent Programming, introduction, monitors
	BLOCK 3 : DEADLOCK AND INFINITE POSTPONEMENT
7	Introduction : Examples of deadlock, Related problem indefinite postponement, resource concepts
8	**Conditions for Deadlock:** Deadlock solution, prevention, avoidance with Dijkstra's banker algorithm, Deadlock detection, Recovery
9	Processor scheduling: Introduction, scheduling levels, preemptive vs nonpreemptive scheduling priorities, scheduling criteria, scheduling algorithms
	BLOCK 4 REAL MEMORY AND VIRTUAL MEMORY MANAGEMENT
10	Introduction, memory organization, memory management, hierarchy, management strategies
11	Contiguous vs non-contiguous memory allocation, fixed partition multiprogramming, variable partition multiprogramming
12	Virtual memory management Introduction, page replacement, strategies, page fault frequency, page replacement, page release, page size
	BLOCK 5 DISK PERFORMANCE & FILE, DATABASE SYSTEMS
13	Introduction, disk scheduling strategies, rotational optimization
14	File and database system introduction, data hierarchy, files, file systems, file optimization, file allocation, free space management, file access control

Text Book:

1. Operating Systems, Deital&Deital, Pearson Education,Third Edition, 2008

Reference Books

1. An Introduction to operating system concepts and practice, Pramod Chandra, PHI,2008

2. Operating system concepts, Abraham silberschatz peter Galvin, Wiley India, 2007.

Course Code	Title of the Course
13052	RELATIONAL DATABASE MANAGEMENT SYSTEMS (RDBMS)

Unit No	ntents
	BLOCK 1 INTRODUCTION
1	**Data base System Applications**, data base System VS file System – View of Data – Data Abstraction –Instances and Schemas – data Models – the ER Model
2	**Model :**Relational Model – Other Models – Database Languages – DDL – DML – database Access for applications Programs – data base Users and Administrator – Transaction Management – data base System Structure – Storage Manager – the Query Processor.
3	**History of Data base Systems -** Data base design and ER diagrams – Beyond ER Design Entities, Attributes and Entity sets – Relationships and Relationship sets – Additional features of ER Model – Concept Design with the ER Model – Conceptual Design for Large enterprises.
	BLOCK 2 : RELATIONAL MODEL
4	**Introduction**– Integrity Constraint Over relations – Enforcing Integrity constraints – Querying relational data – Logical data base Design – Introduction to Views – Destroying / altering Tables and Views.
5	**Relational Algebra** – Selection and projection set operations – renaming – Joins – Division – Examples of Algebra overviews –
6	**Relational calculus** – Tuple relational Calculus – Domain relational calculus – Expressive Power of Algebra and calculus.
	BLOCK 3 : SQL QUERY
7	**Form of Basic SQL Query** – Examples of Basic SQL Queries – Introduction to Nested Queries – Correlated Nested Queries Set – Comparison Operators – Aggregative Operators – NULL values – Comparison using Null values – Logical connectivity's – AND, OR and NOT – Impact on SQL Constructs – Outer Joins – Disallowing NULL values – Complex Integrity Constraints in SQL Triggers and Active Data bases. Schema refinement
8	**Normal forms :**Problems Caused by redundancy – Decompositions – Problem related to decomposition – reasoning about FDS – FIRST, SECOND, THIRD Normal forms – BCNF–
9	**Join:** Lossless join Decomposition – Dependency preserving Decomposition – Schema refinement in Data base Design – Multi valued Dependencies – FORTH Normal Form.
	BLOCK 4 TRANSACTION
10	**Introduction :**Transaction Concept- Transaction State- Implementation of Atomicity and Durability – Concurrent – Executions – Serializability- Recoverability – Implementation of Isolation – Testing for serializability
11	**Protocols :** Lock Based Protocols – Timestamp Based Protocols- Validation- Based Protocols – Multiple Granularity.
12	**Recovery and Atomicity** – Log – Based Recovery – Recovery with Concurrent Transactions – Buffer Management – Failure with loss of nonvolatile storage-Advance Recovery systems- Remote Backup systems
	BLOCK 5 STORAGE
13	**Data on External Storage** – File Organization and Indexing – Cluster Indexes, Primary and Secondary Indexes – Index data Structures – Hash Based Indexing – Tree base Indexing – Comparison of File Organizations – Indexes and
14	**Performance Tuning-** Intuitions for tree Indexes – Indexed Sequential Access Methods (ISAM) – B+ Trees: A Dynamic Index Structure.

Text Books:

1. Raghurama Krishnan, Johannes Gehrke, Data base Management Systems, 3rd Edition, TATA McGrawHill.2003.
2. Silberschatz, Korth, Data base System Concepts, 6th Edition, Tata McGraw Hill, 2011.

Reference Books:

1. Relational Database Principles 2nd Edition, Colin Ritchie, 2004
2. Sharad Maheswari and Ruchin Jain, Database management systems Complete Practical Approach,Firewall media, 2006
3. Peter Rob & Carlos Coronel, Data base Systems design, Implementation, and Management, 7th Edition.
4. Elmasri Navrate , Fundamentals of Database Systems, Pearson Education.

Course Code	Title of the Course
13053	COMPUTER ARCHITECTURE

Unit No	Contents
	BLOCK 1 INTRODUCTION
1	Fundamentals: Measuring and reporting performance, quantitative principles of computer design, classifying instruction set architecture
2	Memory addressing, addressing modes, types and size of operands, operations in the instruction set, operands and operations for media and signal processing
3	Instructions for control flow, Encoding an instruction set, Example architecture, MIPS and TM32
	BLOCK 2 : INSTRUCTION LEVEL PARALLELISM
4	Instruction Level Parallelism: Pipelining and Hazards - Concepts of ILP - Dynamic scheduling
5	Dynamic Hardware prediction - Multiple issues - Hardware based speculation
6	Limitations of ILP - Case studies: lP6 Micro architecture
	BLOCK 3 : ILP WITH SOFTWARE APPROACH
7	Instruction Level Parallelism With Software Approaches: Compiler techniques for exposing ILP - Static branch prediction
8	Static multiple issue : VLIW - Advanced compiler support - Hardware support for exposing parallelism
9	Hardware Vs software speculation. Mechanism - IA 64 and Itanium Processor.
	BLOCK 4 MEMORY AND I/O
10	Memory And I/O: Cache performance - Reducing cache miss penalty and miss rate - Reducing hit time - Main memory and performance - Memory technology
11	Types of storage devices - Buses - RAID - Reliability, availability and dependability
12	I/O performance measures - Designing I/O system.
	BLOCK 5 MULTIPROCESSOR AND THREAD LEVEL PARALLELISM
13	Multiprocessors And Thread Level Parallelism: Symmetric and distributed shared memory architectures - Performance issues - Synchronization
14	Models of memory consistency - Multithreading.

TEXT BOOKS

1. John L. Hennessey and David A. Patterson," Computer Architecture: A Quantitative Approach", Third Edition, Morgan Kaufmann, 2003.
2. D. Sima, T. Fountain and P. Kacsuk, " Advanced Computer Architectures: A Design Space Approach", Addison Wesley, 2000.

REFERENCE BOOKS

1. Kai Hwang "Advanced computer architecture Parallelism Scalability Programmability" Tata Mcgraw Hill Edition 2001.
2. Vincent P.Heuring, Harry F.Jordan, “Computer System Design and Architecture”, Addison Wesley, 2nd Edition 2004.

Course Code	Title of the Course
13054	**RELATIONAL DATABASE MANAGEMENT SYSTEMS (RDBMS) LAB**

Unit No.	Contents
	BLOCK 1 : TABLE MANIPULATION
1	**Table** creation, Renaming a Table, Copying another table, Dropping a Table
2	**Table Description:** Describing Table Definitions, Modifying Tables, Joining tables, Number and Date functions.
	BLOCK 2 : SQL QUERIES AND SUB QUERIES
3	**SQL Queries:** Queries, Sub Queries, and aggregate functions
4	**DDL:** Experiments using database DDL SQL statements
5	**DML:** Experiment using database DML SQL statements
6	**DCL:** Experiment using database DCL SQL statements
	BLOCK 3 : INDEX AND VIEW
7	**Index :** Experiment using database index creation, Renaming a index, Copying another index, Dropping a index
8	**Views:** Create Views, Partition and locks
	BLOCK 4 : EXCEPTION HANDLING AND PL/SQL
9	**Exception Handling:** PL/SQL Procedure for application using exception handling
10	**Cursor:** PL/SQL Procedure for application using cursors
11	**Trigger:** PL/SQL Procedure for application using triggers
12	**Package:** PL/SQL Procedure for application using package
13	**Reports:** DBMS programs to prepare report using functions
	BLOCK 5 : APPLICATION DEVELOPMENT
14	**Design and Develop Application:** Library information system, Students mark sheet processing, Telephone directory maintenance, Gas booking and delivering, Electricity bill processing, Bank Transaction, Pay roll processing. Personal information system, Question database and conducting Quiz and Personal diary

Reference Books:

1. Raghurama Krishnan, Johannes Gehrke, Data base Management Systems, 3rd Edition, TATA McGrawHill.2003.
2. Silberschatz, Korth, Data base System Concepts, 6th Edition, Tata McGraw Hill, 2011.
3. Relational Database Principles 2nd Edition, Colin Ritchie, 2004
4. Sharad Maheswari and Ruchin Jain, Database management systems Complete Practical Approach,Firewall media, 2006
5. Peter Rob & Carlos Coronel, Data base Systems design, Implementation, and Management, 7th Edition.
6. Elmasri Navrate , Fundamentals of Database Systems, Pearson Education.

SEMESTER VI

Course Code	Title of the Course
13061	COMPUTER NETWORKS

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	**Introduction** : computer networks applications, line configuration, topology, transmission modes
2	**Categories of Networks:** LAN, WAN,MAN, OSI layers
3	**Physical Layer:** analog and digital signals performance, transmission media
	BLOCK 2 : DATA LINK LAYER
4	**Introduction:** Error detection and correction, block coding, cyclic redundancy check, framing, flow and error control
5	**Data link layer protocols:** stop and wait protocol, sliding window protocol, ARQ, go-back-n ARQ, selective-repeat ARQ
6	**Multiple access protocols:** ALOHA, CSMA, CSMA/CD, CSMA/CA
	BLOCK 3 : NETWORK LAYER
7	**Introduction:** Circuit switching, Packet switching, Message switching, virtual circuit and datagram subnets
8	**Routing Algorithms:** state routing, shortest path routing, dynamic routing, distance vector routing
9	**Multicast Routing: algorithms, congestion, control algorithms**
	BLOCK 4 : TRANSPORT LAYER
10	**introduction:** process to process delivery, UDP, TCP, connection oriented vs connection less services
11	**Application and Services:** Domain Name system, Remote login, Mail exchange, File transfer, RPC, Remote file access, WWW and HTTP, SNMP
	BLOCK 5 : NETWORK SECURITY
12	Introduction: cryptography, Encryption models, Transposition and substitution chipers, Cryptographic principles
13	**Symmetric key cryptography:** DES, AES
14	**Asymmetric key cryptography:** RSA, security services

Text Books:

1. Computer Networks, 3rd Edition, Andrew S Tanenbaum, 2010
2. Data Communication and Networking, 4th edition, Behrouz A. Forouzan, 2008

Reference Books:

1. Data and computer communication , 8th edition, William stallings, prentice Hall
2. An Engineering approach to computer networks, 2nd edition, S.Keshav, Pearson education,2008

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Course Code	Title of the Course
13062	VISUAL BASIC PROGRAMMING

Unit No	CONCEPTS
	BLOCK 1: VISUAL BASIC CONCEPTS
1	**Introduction to GUI** - Visual Basic : Starting and Exiting Visual Basic Project Explorer Working with Forms Properties Window
2	**Using the Toolbox** Toolbars Working with Projects Programming Structure of Visual Basic applications Event and Event driven Procedures
3	**Program Design** - Form and Controls - Writing the Code - Saving, Running and Testing - Making EXE File - Printouts
	BLOCK 2 : VISUAL BASIC CODE,EVENTS AND CONTROLS
4	**Adding code and using events**: Using literals data types - declaring and using variables using the operator subroutines and functions
5	**Looping and decision control structures:** if then else, structure select structure, for next, do.. loop and while.. wend.
6	**Using intrinsic Visual basic Controls with methods and Properties:** Label ,Text box, Command button, Frame, Checkbox, option button, List box, Combo box, Drive List box, directory List box and file list box Formatting controls control arrays, Tab order
	BLOCK 3 : VISUAL BASIC PROCEDURES, FUNCTIONS AND ARRAYS
7	**Creating Procedures, functions** - String functions, date and Time function , numeric functions- Recursive Functions
8	**Multiple Forms** - Startup Forms - SubMain Procedure
9	**Arrays** - Control Arrays - Indexing and Event Handling - Graphics
	BLOCK 4 : MENUS AND MDI FORMS
10	**Menus:** creating menus, adding code to menus
11	**Using MDI forms** - MDI form basic building MDI form creating MDI Child Forms
	BLOCK 5: DATABASE OBJECT (DAO) AND PROPERTIES
12	Database object (DAO) and properties -accessing Recordset objects- Move first, MoveLast, MovePrevious and MoveNext methods Begin, Commit and Rollback transaction accessing Microsoft Access files.
13	Active Data Objects (ADO) ADO and OLE DB and ADO Primer What are OLE DB and ADO? ADO object Model Converting DAO Code to Use ADO.
14	Connecting to the database Retrieving a recordset Creating a query dynamically Using a parameterized query using action queries - Adding records Editing records closing the database connection**.**

Text Books

1. Gary Cornwell Visual basic 6 , Tata McGraw Hill

Reference Books:

1. Scott warner Teach yourself Visual basic 6 , Tata McGraw-Hill
2. Noel Jerke The Complete Reference , Tata McGraw-Hill
3. Eric A. Smith, Valar Whisler, and Hank Marquis Visual Basic 6 programming

Course Code	Title of the Course
13063	SOFTWARE ENGINEERING

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	**Software:** Role of software, Software myths. Generic view of process: A layered technology, a process framework, The Capability Maturity Model Integration (CMMI)
2	**Process patterns**, Process assessment, Personal and Team process models.
3	**Process model:** The waterfall model, Incremental process models, Evolutionary process models, The Unified process.
	BLOCK 2 : REQUIREMENT ENGINEERING:
4	Design and Construction, Requirement Engineering Tasks, Requirements Engineering Process, Validating Requirements.
5	**Building the Analysis Model:** Requirement analysis, Data Modeling concepts, Object-Oriented Analysis
6	**Modeling:** Scenario-Based Modeling, Flow-Oriented Modeling Class-Based Modeling, Creating a Behavioral Model.
	BLOCK 3 : SYSTEM DESIGN
7	**Design Engineering:** Design process and quality, Design concepts, the design model.
8	**Architectural Design:** Software architecture, Data design, Architectural styles and patterns, Architectural Design.
9	**User interface design:** The Golden rules, User interface analysis and design, Interface analysis, Interface design steps, Design evaluation.
	BLOCK 4 : SYSTEM TESTING
10	**Testing Strategies:** Approach to Software Testing, Unit Testing, Integration Testing, Test strategies for Object-Oriented Software, Validation Testing, System Testing, the art of Debugging, Black-Box and White-Box testing.
11	**Product Metrics:** Software Quality, Product Metrics, Metrics for Analysis Model, Design Model, Source code and Metrics for testing, Metrics for maintenance. Metrics for Process and Projects Domains: Software Measurement, Metrics for Software Quality and Software Process.
	BLOCK 5 : RISK and QUALITY MANAGEMENT
12	Risk Strategies: Reactive vs. Proactive Risk strategies, software risks, Risk identification
13	**Risk Protection and refinement:** Risk projection, Risk refinement, Risk Mitigation, Monitoring and Management, RMMM Plan.
14	**Quality Management:** Quality concepts, Software quality assurance, Software Reviews, Formal Technical reviews, Statistical Software quality Assurance, Software reliability, The ISO 9000 quality standards.

TEXT BOOK:

1. Roger S. Pressman Software Engineering - A practitioner's Approach McGraw-Hill 6th Edition (2010)

REFERENCE BOOKS:

1. Richard Fairlay Software Engineering Concepts McGraw Hill Book Company (2005)
2. Pankaj Jalote An Integrated Approach to Software Engineering Narosa Publishing House 3rd Edition (2005)
3. Software Engineering, Somzerville, 8th Edition, Pearson Education 2007.
4. Software Engineering K.K. Agarwal & Yogesh Singh, 3rd Edition New Age International Publishers 2007.
5. Software Engineering an Engineering Approach James F. Peters, Witold Pedrycz - John Wiley & Sons 2000.
6. Software Engineering Principles and Practice Waman S Jawadekar, , Tata McGraw-Hill 2004.

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Course Code	Title of the Course
13064	**LAB : VISUAL BASIC PROGRAMMING**

Unit No.	Contents
1	Building simple applications
2	Working with intrinsic controls ,Control Arrays
3	Application with multiple forms
4	Application with dialogs
5	Application with Menus
6	Application using data controls
7	Application using Common Dialogs
8	Drag and Drop Events
9	Database Management
10	Creating ActiveX Controls
11	Database object (DAO) and properties
12	Active Data Objects (ADO) ADO and OLE DB
13	Connecting to the database ,Retrieving a record set Creating a query dynamically Using a parameterized query using action queries - Adding records Editing records closing the database connection
14	**Simple Application development:** 1. Library information system 2. Students mark sheet processing 3. Telephone directory maintenance 4. Gas booking and delivering 5. Electricity bill processing 6. Bank Transaction 7. Pay roll processing 8. Personal information system 9. Question database and conducting Quiz 10. Personal diary

Text Books

1. Gary Cornwell Visual basic 6 , Tata McGraw Hill

Reference Books:

1. Scott warner Teach yourself Visual basic 6 , Tata McGraw-Hill
2. Noel Jerke The Complete Reference, Tata McGraw-Hill
3. Eric A. Smith, Valar Whisler, and Hank Marquis Visual Basic 6 programming

9. BCA (Bachelor of Computer Application)

Duration – Three years
Eligibility – A pass in HSC (or) 3 year Diploma
Medium – English
<u>Lateral Entry to II year BCA:</u>

3 year Diploma in Computer Science Engineering, EEE / ECE / IT /Computer Hardware Maintenance / Computer Technology / Electronics Engineering.

S.No	Course Code BCA	Course Code *BCA(LE)*	Title of the Course	CIA Max.	ESE Max.	TOT Max.	C Max.
	FIRST YEAR						
	SEMESTER I						
1	10111A 10111B		**Part I:** Tamil - Paper I Communication Skills – I	25	75	100	4
2	10112		**Part II:** English – Paper I	25	75	100	4
3	10113		Part III : Core Course: **C and Date Structure**	25	75	100	4
4	10114		Course: **C and Data Structure Lab**	25	75	100	4
			Total	**100**	**300**	**400**	**16**
SEMESTER II							
5	10121A 10121B		**Part I:** Tamil Paper II: Communication Skills – II	25	75	100	4
6	10122		**Part II:** Paper I: English – II	25	75	100	4
7	10123		**Part III : Core Course:** Programming in C++	25	75	100	4
8	10124		**Part III : Core Course:** Programming in C++ Lab	25	75	100	4
			Total	**100**	**300**	**400**	**16**
SECOND YEAR							
SEMESTER III							
9	10131A 10131B	12731A 12731B	**Part I:** Tamil Paper III Human Skills Development - I	25	75	100	4
10	10132	12732	**Part II:** Paper I: English – III	25	75	100	4
11	10133	12733	**Part III : Core Course:** Relational Database Management Systems (RDBMS)	25	75	100	4
12	10134	12734	**Part III : Core Course:** RDBMS Lab	25	75	100	4
			Total	**100**	**300**	**400**	**16**
SEMESTER IV							
13	10141A 10141B	12741A 12741B	**Part I:** Tamil Paper IV Human Skills Development - II	25	75	100	4

14	10142	12742	**Part II:** Paper I: English – IV	25	75	100	4
15	10143	12743	**Part III : Core Course:** Internet and Java Programming	25	75	100	4
16	10144	12744	**Part III : Core Course:** Internet and Java Programming Lab	25	75	100	4
			Total	**100**	**300**	**400**	**16**
THIRD YEAR							
SEMESTER V							
17	10151	12751	**Part III : Core Course:** Accounting Fundamentals	25	75	100	4
18	10152	12752	**Part III : Core Course:** Computer Graphics	25	75	100	4
19	10153	12753	**Part III : Core Course:** Operating Systems	25	75	100	4
20	10154	12754	**Part III : Core Course:** Unix & Shell Programming Lab	25	75	100	4
			Total	**100**	**300**	**400**	**16**
SEMESTER VI							
21	10161	12761	**Part III : Core Course:** Management Principles and Techniques	25	75	100	4
22	10162	12762	**Part III : Core Course:** System Analysis and Design	25	75	100	4
23	10163	12763	**Part III : Core Course:** Visual Basic Programming	25	75	100	4
24	10164	12764	**Part III : Core Course:** Visual Basic Programming Lab	25	75	100	4
			Total	**100**	**300**	**400**	**16**
			Grand Total	**600**	**1800**	**2400**	**96**

CIA: Continuous Internal Assessment, **ESE:** End Semester Examination,
TOT: Total, **C:** Credit Points **Max:** Maximum

FIRST YEAR
SEMESTER I

Course Code	Title of the Course
10111A	**PART 1 : TAMIL - PAPER -1**

கூறு 1

3. கண்ணதாசன் - ஸ்ரீ கிருஷ்ண கானம்

5. புல்லாங்குழல் கொடுத்த
6. குருவாயூருக்குவாருங்கள்

கூறு 2

7. கோகுலத்துபசுக்கள்
8. கோகுலத்தில்ஒருநாள் ராதை
9. ஆயர்பாடிமாளிகையில்

கூறு 3

பட்டுக்கோட்டைகல்யாணசுந்தரம்

5. நெஞ்சில் குடியிருக்கும்
6. செய்யும் தொழிலேதெய்வம்

கூறு 4

4. பாரதியார்

கண்ணன் என் விளையாட்டுப்பிள்ளை
பாரதமாதாதிருப்பள்ளிஎழுச்சி

கூறு 5

1.	**பாரதிதாசன்**	-	உலகப்பன்பாட்டு (5)
2.	**நாமக்கல் கவிஞர்**	-	நோயற்றவாழ்வு7 பாட்டு
3.	**பெ.தூரன்**	-	நிலாபிஞ்சு

கூறு 6

10. **வல்லிக் கண்ணன்**	-	வெறும் புகழ்
11. **கு.ப.இராஜகோபாலன்**	-	எதற்காக?
12. **மீரா**	-	பதினைந்து

கூறு 7

7. **சிற்பி**	-	சர்ப்பயாகம்
8. **ஞானக்கூத்தன்**	-	தோழர் மோசிகீரனார்

கூறு 8

7. **அப்துல் ரகுமான்**	-	கண்ணும் எழுதேம்
8. **சண்முகசுப்பையா**	-	வயிறு

கூறு 9

7. **சிலப்பதிகாரம்**	-	வழக்குரைகாதை
8. **கம்பராமாயணம்**	-	அயோத்தியாகாண்டம்

கூறு 10

4. **சீறாப்புராணம்**	-	ஈத்தங்குலைவரவழைத்தபடலம் (1)

கூறு 11

தேம்பாவணி - காட்சிப்படலம்
பாடல்எண்(ஒவ்வொருபாடலின் முதல்வரி)

1. இன்னவாயில்
2. கொழுந்துறும்
3. பஞ்(ச) அரங்கில்

கூறு 12

தேம்பாவணி - காட்சிப்படலம்
பாடல்எண் (ஒவ்வொருபாடலின் முதல்வரி)

1. எண்ணுளே
2. ஒண்தலங்கள்
3. இரவியேந்தகஞ்சக்

கூறு 13

தேம்பாவணி - காட்சிப்படலம்
பாடல்எண் (ஒவ்வொருபாடலின் முதல்வரி)

1. கன்னியாயதாயும்
2. ஏந்திஓங்குஉளத்து
3. ஆவ தேமுனர்
4. கொல்லும் வேலொடும்

கூறு 14

தேம்பாவணி - காட்சிப்படலம்
பாடல்எண் (ஒவ்வொருபாடலின் முதல்வரி)

1. என்றவாசகம்
2. அம்பினால்
3. வேண்டும் ஓர் வினை

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Course Code	Title of the Course
10111B	**PART 1 : COMMUNICATION SKILLS - I**

Unit - I Communication – Meaning – Types- Importance

Unit – II Barriers to Effective Communication – Principles – Principles of Effective Communication

Unit – III Oral Communication – Meaning – Importance- Forms of Oral Communication

Unit – IV Intonation –Meaning – Function- Types
Preparation of Speech- Steps Involved

Unit – V Principles of Effective Oral Communication

Unit – VI Written Communication – Meaning –Steps – Importance- Advantages
Use of words and Phrases

Unit – VII Sentence – Meaning –Sentence formation- Characteristics of an Effective Sentence

Unit – VIII Paragraph Writing –Essay Writing –Steps Involved –Outline-Layout –Contents-Drafting-Correction- Final Draft

Unit – IX Application for Employment and Curriculum Vitae –Steps involved

Unit – X Non –Verbal Communication – Meaning –Types –Body Language –Postures-Gestures –Facial Expressions –Eye Contact

Unit – XI Report Writing –Report –Types of Reports –Format of a Report

Unit – XII Essentials of a Good Report –Preparation of Report-Procedure Involved

Unit – XIII Meetings-Purpose of the Meeting – Procedure

Unit – XIV Group Discussion –Quality of Content-Participation –Logical Presentation –Behavioural Skills

References:

1. Krishna Mohan & Meera Banerjee, Developing Communication Skills, 2005.
2. Geetha Nagaraj, Write to Communicate, 2004.
3. Wren & Martin, English Grammar and Composition, 2002.
4. Dale Carnegie, How to Win Friends and Influence People, 1981.
5. Dale R Jordan, Language Skills and Use.
6. Gartside L. Bahld, Nagammiah and McComas, Satterwhite, Modern Business Correspondence.
7. Rajendra Pal and Kortahalli J S, Essentials of Business Communication.
8. Wallace, Michael J, Study Skills in English.
9. Editors of Readers Digest, Super Word Power.

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Course Code	Title of the Course
10112	**PART II : ENGLISH – I**

Prose

Unit – I Water-the Elixir of life - C.V. Raman
Unit – II Mrs. Packletide's Tiger - SAKI
Unit – III A Deed of Bravery - Jim Carbett
Unit – IV The Cat - Catharine M.Willson
Unit – V On Letter Writing - Alpha of the Plough
Unit – VI Our Ancestors - Carl Sagan
Unit – VII Our Civilization - C.E.Foad
Unit – VIII A Hero on Probation - B.R. Nanda
Unit – IX Dangers of Drug Abuse - Hardin B. Fones
Unit – X Food - J.B.S. Haldane

Grammar

Unit – XI - Articles-Gerunds-Participles-Infinitives-Modals-Proposition –Tenses.
Unit – XII - Direct and Indirect Speech-Transformation of sentences- Active and passive voice.

Composition

Unit – XIII - Letter writing - Precis writing - Developing hints.
Unit – XIV - Dialogue writing - Paragraph writing.

References:

4. Sebastian D K, *Prose for the Young Reader*, Macmillan.
5. *Active English Grammar*, Ed. by the Board of Editors, Macmillan.
6. Modern English – *A Book of Grammar Usage and Composition* by N.Krishnaswamy, Macmillan Publishers.

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Course Code	Title of the Course
10113	**C AND DATA STRUCTURE**

Unit no.	Contents
	BLOCK 1 : INTRODUCTION TO C
1	**Program Development styles and Basics of C**. Introduction to C – Character set – Identifiers and keywords – Data types – constants – Variables – declarations – Declaring variables, - Rules for defining variables. Initializing variables - Type conversion. Operator and Expressions
2	**Data input, output** and Preliminaries – single character input and output – Entering input data – Writing output data – gets and puts functions –
3	**control statements:** Branching and looping – Nested control structures – Switch – Break – Continue and Goto.
	BLOCK 2 : FUNCTIONS,ARRAYS AND POINTERS
4	**Function:** defining a function – Accessing a function – Passing arguments to a function – Recursion – Library function – Macros – C preprocessor – Program structure: Storage classes – Automatic variables – Global variables – Static variables– Multiple programming – Bitwise operation.
5	**Arrays** – Array initialization, Definition of Array, Characteristic of Array, One dimensional array, Two dimensional array, Multidimensional arrays, Character array and strings – string handling functions.
6	**Pointers** – Features of Pointers, Pointer declaration, Arithmetic operation with pointers, Pointers and Arrays, Pointers and two dimensional arrays, Array of Pointers, Pointers to Pointers, Pointers and strings.
	BLOCK 3 : STRUCTURE UNION AND FILES
7	Structures and Unions: defining a structure – Processing a structure – Structures and pointers– Passing structures to functions – Self referential structures – Bit fields – Unions – Enumerations.
8	Data file: Opening and Closing a data file – Creating a data file – Processing a data file – Unformatted data file – Command line parameters.
	BLOCK 4 : LINEAR DATA STRUCUTRE
9	**Introduction to Data Structure,** Stack, Stack related terms, operation on a stack, Representation of Stack, Implementation of a stack - Polish notation.
10	**Queues,** Various Positions of Queue, Circular Queues. Operations on Queue , Representation of Queues. Applications of Queue..
11	**List,** Merging lists, Linked list, Single linked list, Double Linked List, Header Linked list, Insertion and Deletion of linked list, Traversing a linked list. Representation of Linked list
	BLOCK 5 : NON-LINEAR DATA STRUCTURE
12	**Introduction** – Trees, Binary Trees, Types of Binary trees,
13	**Binary Tree Representation**, Traversing Binary Trees,
14	**Binary Search tree**, Insertion and Deletion operations, trees and their applications Hashing Techniques.

TEXT BOOKS:

1. Programming in ANSI C, Fifth Edition, E.Balagurusamy, Tata McGraw-Hill Publishing Company Ltd, 2011
2. Data Structures, Seymour Lipschutz, G.A.Vijayalakshmi Pai, Second Edition , Schaum's Outlines, Tata Mc-Graw Hill Private Ltd., 2006.

REFERENCE BOOKS :

1. Fundamentals of Data structures in C, Second edition, Ellis Horowitz and Sartaj Sahini, Universities press, 2007.
2. Programming and Data Structure, Pearson Edition, Ashok N Kamthane, 2007.

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Course Code	Title of the Course
10114	**C AND DATA STRUCTURE LAB**

Experiments based on C programming and Data Structures

Unit No.	Contents
	BLOCK 1 : C PROGRAM FUNDAMENTALS
1	Simple C Programs
2	Using if and switch constructs Programs
3	Looping statements Problems
	BLOCK 2 : FUNCTIONS,ARRAYS,STRINGS,FILEAND POINTERS
4	Functions and Recursive programs
5	Arrays ,Strings and Matrices Programs
6	File Handling Programs
7	Pointers and Arrays Programs programs
	BLOCK 3 – STRUCTURE , UNION AND FILES
8	Structure and union : Programs using structure and union
9	Files : Programs based on file handling
	BLOCK 4 : LINEAR DATA STRUCTURE PROGRAMS
10	Stacks, queues ,expression evaluation programs
11	Infix to postfix conversion
12	Linked List programs: List, Merging lists, Linked list, Single linked list, Double Linked List, Header Linked list, Insertion and Deletion of linked list, Traversing a linked list.
	BLOCK 5 : NON LINEAR DATA STRUCURE EXPERIMENTS
13	Tree Programs : Trees, Binary Trees, Types of Binary trees, Binary Tree Representation,
14	Traversing Binary Trees, Binary Search tree, Insertion and Deletion operations,

REFERENCE BOOKS:

3. Programming in ANSI C, Fifth Edition, E.Balagurusamy, Tata McGraw-Hill Publishing Company Ltd, 2011
4. Data Structures, Seymour Lipschutz, G.A.Vijayalakshmi Pai, Second Edition , Schaum's Outlines, Tata Mc-Graw Hill Private Ltd., 2006.
5. Fundamentals of Data structures in C, Second edition, Ellis Horowitz and Sartaj Sahini, Universities press, 2007.
6. Programming and Data Structure, Pearson Edition, Ashok N Kamthane, 2007.

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SEMESTER II

Course Code	Title of the Course
10121A	PART 1 : TAMIL - PAPER II

கூறு 1

தேம்பாவணி - காட்சிப்படலம்

பாடல்எண் (ஒவ்வொருபாடலின் முதல்வரி)

69. சொல் தவிர்ந்த
70. அன்னை
71. அஞ்சுவார்
72. சொல்லக் கேட்டனள்
73. மற்செய்கை
74. மண்கனியப்
75. அழுதுஆர்ந்த

கூறு 2

தேம்பாவணி - காட்சிப்படலம்

பாடல்எண் (ஒவ்வொருபாடலின் முதல்வரி)

76. பொய் பொதுளும்
77. இன்புஅருந்தி
78. வழுதாயின இன்பு
79. மறம் ஏவினர்

கூறு 3

தேம்பாவணி - காட்சிப்படலம்

பாடல்எண் (ஒவ்வொருபாடலின் முதல்வரி)

1. மண்ணோர்கள்
2. பொய்யாவிதியோய்
3. விடியா இருள்
4. அழுவார் எவரும்

கூறு 4

சிறுகதை - நீலபத்மநாபனின் ‘‘வானவீதியில்’’

கூறு 5

உரைநடை - கம்பன் புறத்திணை - தி.சொக்கலிங்கம்

இலக்கணம் - எழுத்தும் சொல்லும்

கூறு 6

1. முதலெழுத்துகள்,சார்பெழுத்துகள்
2. மொழிமுதலெழுத்துகள்,மொழி இறுதிஎழுத்துகள்

கூறு 7

1. ஒற்றெழுத்துமிகலும் மிகாமையும்,
2. ஆகு பெயர்,அன்மொழித் தொகை.
3. வினா-விடை வகைகள்

கூறு 8

9. தமிழ்ச் சொல்லமைப்பின் சிறப்பு–பெயர்,வினை, இடை,உரிவடிவங்கள்,
10. பிறமொழிச் சொற்களைத் தமிழில் ஆளும் முறைகள்

கூறு 9

11. அல் வழி,வேற்றுமைப் புணர்ச்சிகள்
12. திணை,பால்,எண், இட இயைபு.

தமிழ் இலக்கியவரலாறு

கூறு 10

3. இக்கால இலக்கிய வகைகள்

அ) மரபுக் கவிதை
ஆ) புதுக் கவிதையின் தோற்றமும் வளர்ச்சியும்

கூறு 11

1. உரைநடை இலக்கியங்கள் - தோற்றமும் வளர்ச்சியும்

அ) கட்டுரை ஆ) சிறுகதை
இ) புதினம் ஈ) நாடகம்

கூறு 12

3. இக்கால இலக்கியக் களங்கள்

திரைப்படம்,தொலைக்காட்சி,வானொலி, இதழ்கள் தமிழுக்குஆற்றிவரும் பணிகள்

கூறு 13

3. **தமிழும் சமயமும் :**

அ) சைவம் ஆ)வைணவம் இ)சமணம் ஈ)பௌத்தம் உ)இசுலாம்
ஊ) கிறித்துவம்

கூறு 14

1. பிற்காலக் காப்பியங்கள் :

அ) கம்பராமாயணம் **ஆ)** பெரியபராணம்

2. இணையம் - பற்றியசெய்திகள்

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Course Code	Title of the Course

Course Code	Title of the Course
10121B	**PART 1 : COMMUNICATION SKILLS - II**

Unit – I Code and Content of Communication Skills
Unit – II Stimulus and Response of Communication Skills
Unit – III Effective Speaking Guidelines
Unit – IV Pronunciation Etiquette of Communication Skills
Unit – V Phonetics in Communication Skills
Unit – VI A self Assessment of Communicating Soft Skills
Unit – VII Language Skills –Ability –Skill Selected Need- Learner Centre activities
Unit – VIII Listening Skills –Importance –Types of Listening- Interview Skills
Unit – IX Conversation Skills –Modes
Unit – X Presentation Skills - Preparing –Planning-Presentation
Unit – XI Written Communication –Structure of Effective Sentences –Paragraph
Unit – XII Technical Writing-Creative Writing- Editing and Publishing
Unit – XIII Corporate Communication Skills-Internal –Effective business writing –Letters, Proposals, Resume
Unit – XIV Corporal Communication Skills-External - Press release - Newsletters- Interviewing skills

References:

1. Dutt. Kiranmai & Geeta Rajjevan. Basic Communication Skills. Rev.ed. Foundation Books Pvt.Ltd. Cambridge House, New Delhi 2006.
2. Bill R. Swetmon. Communication Skills for the 21st Century. Chennai: Eswar Press. First South Asian Edition 2006.
3. Glass. Lillian. Talk to Win. New York: Perigee Books,1987.
4. Pease. Alan. Signals: How to Use Body Language for Power, Success and Love, New York: Bantam Books, 1981.
5. Walters. Lilly. Secrets of Successful Speakers. New York: McGraw-Hill, Inc., 1993.
6. Mandal. S.K. How to Succeed in Group Discussions & Personal Interviews. Mumbai: JAICO Publishing House.
7. Rogoff. Leonard and Ballenger. Grady. Office Guide to Business Letters, Memos & Reports. New York: Macmillan, 1994.
8. Krishna Mohan & Meera Banerjee, Developing Communication Skills, 2005.
9. Geetha Nagaraj, Write to Communicate, 2004.
10. Wren & Martin, English Grammar and Composition, 2002.
11. Rajendra Pal and Kortahalli J S, Essentials of Business Communication.

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10122	PART II : ENGLISH – II

Poetry

Unit – I	Sonnet	- William Shakespeare
Unit – II	Lines Composed upon Westminster Bridge	-William Wordsworth
Unit – III	Grecian Urn	- John Keats (1795-1827)
Unit – IV	Andrea Del Sarto	- Robert Browning (1812-1889)
Unit – V	The Road Not Taken	- Robert Frost (1874-1963)
Unit – VI	Strange Meeting	- Wilfred Owen (1813-1918)
Unit – VII	Gitanjali	- Rabindranath Tagore (1861-1946)
Unit – VIII	The Coromandel Fishers	- Sarojini Naidu
Unit – IX	The Express	- Stephen Spender
Unit – X	**Shakespeare :** The Merchant of Venice	

Language Use:

Unit – XI	Essay writing
Unit – XII	Note Making
Unit – XIII	Report writing
Unit – XIV	Comprehension

References:

4. *The Golden Quill*, P.K. Seshadri, Macmillan.
5. *The Merchant of Venice*, Shakespeare. (Any overseas edition).
6. *Active English Grammar*, Ed. by the Board of Editors, Macmillan.
4. Modern English – *A Book of Grammar Usage and Composition* by N.Krishnaswamy, Macmillan Publishers.

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Course Code	Title of the Course
10123	PROGRAMMING IN C++

Unit No	Contents
	BLOCK I : PRINCIPLES OF OBJECT-ORIENTED PROGRAMMING
1	**Principles of Object-Oriented Programming:** Software Crisis – Software Evolution – Basic Concepts of Object-Oriented Programming – Benefits of OOP – Object-Oriented Languages - Applications of OOP – Application of C++ -
2	**Structure of a C++ Program** – Tokens – Keywords – Identifiers – Basic Data Types – User defined Data types – Derived data types – Symbolic constants – Type compatibility – Declaration of variables – Dynamic initialization of variables –Reference variables – Operators in C++ - Manipulators – Type cast operator – Expressions and their types-Implicit conversions
3	**Control structures and function** – The main function – Function prototyping – inline functions – Function overloading.
	BLOCK 2 : CLASSES AND OBJECTS
4	**Introduction** - Specifying a Class -Defining Member Function - C++ Program with Class - Making an Outside Function Inline - Nesting of Member Functions - Arrays within a Class
5	**Memory Allocation for Objects** - Static Data Members - Static Member Functions - Arrays of Objects - Objects as Function Arguments - Friendly Functions - Returning Objects.
6	**Constructors and Destructors**: Introduction - Constructors - Parameterized Constructors - Multiple Constructors in Class - Constructors with Default Arguments - Dynamic Initialization of Objects - Copy Constructor - Dynamic Constructors - Constructing Two Dimensional Arrays - Destructors.
	BLOCK 3 : INHERITANCE, POLYMORPHISM AND POINTERS
7	**Inheritance**: Introduction - Defining Derived Classes - Single Inheritance - Making a Private Member Inheritable - Multilevel Inheritance - Multiple Inheritance - Hierarchical Inheritance - Virtual Base classes - Abstract Classes - Constructors in Derived Classes - Member Classes: Nesting of Classes.
8	**Pointers Virtual Functions and Polymorphism**: Introduction - Pointers to Objects - this Pointer - Pointers to Derived Classes - Virtual Functions - Pure Virtual Functions.
9	**Managing Console I/O Operations** - C++ Streams - C++ Stream Classes - Unformatted I/O Operations, Formatted Console I/O Operations - Managing Output with Manipulators.
	BLOCK 4 : WORKING WITH FILES
10	**Introduction** - Classes for File Stream Operations - Opening and Closing a File - Detecting End of File - More About Open (): File Modes
11	**File Pointers and their Manipulations** - Sequential Input and output Operations - Updating a File - Random Access - Error handling During File Operations - Command Line Arguments.
	BLOCK 5 : TEMPLATES AND EXCEPTION HANDLING
12	**Templates**: Introduction - Function Templates - Overloaded Function Templates - Nesting of Function Calls - Multiple Arguments Function Template - User Defined Templates.
13	**Exception Handling**: **Introduction** - Error Handling - Exception Handling Model - Exception handling Constructs - Handler Throwing the Same Exception Again -
14	**Other Exception Handling:** List of Exceptions - Catch All Exceptions - Exceptions in Constructors and Destructors - Handling Uncaught Exceptions - Ten Rules for Handing Exceptions Successfully.

Text Book:

1. E.Balagurusamy, Object Oriented Programming with C++, Tata McGraw Hill, New Delhi, Sixth Edition, 2013.

Books for Reference:

1. Object Oriented Program in C++ – Nabajyoti Barkakati, A prentice Hall of India Private Limited, New Delhi 1997.
2. Mastering C++ – K R Venugopal, T. Ravishankar, RajKumar, Tata Mc Graw-Hill Publishing Company Limited, New Delhi, 2006.

Course Code	Title of the Course
10124	**PROGRAMMING IN C++ LAB**

Syllabi Based On Course : Object Oriented Programming And C++

Unit No.	Contents
	BLOCK 1
1.	Simple Programs using decisions, loops and arrays
2.	Simple functions & Inline functions
	BLOCK 2
3.	Usage of classes and Objects
4.	this pointer and Static functions
5.	Constructors and Destructors
	BLOCK 3
6.	Function overloading
7.	Operator Overloading
8.	Friend functions
	BLOCK 4
9.	Inheritance & Multiple Inheritance
10.	Pointers
11.	Polymorphism
12.	Virtual Functions
	BLOCK 5
13.	Files
14.	I/O Streams

Books for Reference:

1. E.Balagurusamy, Object Oriented Programming with C++, Tata McGraw Hill, New Delhi, Sixth Edition, 2013.
2. Object Oriented Program in C++ – Nabajyoti Barkakati, A prentice Hall of India Private Limited, New Delhi 1997.
3. Mastering C++ – K R Venugopal, T. Ravishankar, RajKumar, Tata Mc Graw-Hill Publishing Company Limited, New Delhi, 2006.

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SECOND YEAR
SEMESTER III

Course Code	Title of the Course
10131A / 12731A	**PART 1 : TAMIL - PAPER III**

கூறு 1 :பத்துப்பாட்டு–முல்லைப்பாட்டு

கூறு 2 :எட்டுத்தொகை–ஐங்குறுநூறு

கூறு 3 :கபிலர் - குறிஞ்சித்திணை

கூறு 4 :மஞ்ஞைப்பத்து–முதல் மூன்றுபாடல்கள்

கூறு 5 :குறுந்தொகை–பரணர் பாடல்கள் பா. எண். 19, 24, 36, 128, 399

கூறு 6 : நற்றிணை– பெருங்குன்றூர்கிழார் - பா. எண். 5

பெருவழுதியார் - பா. எண். 55

பெருங்கௌசிகனார் - பா. எண். 139

கூறு 7 :நற்றிணை– கருவூர்க்கோசிகனார் - பா. எண். 214

உலோச்சனார் - பா. எண் 249

கூறு 8 :அகநானூறு –சேந்தம்பூதனார் பாடல்கள் பா.எண். 84, 207

கூறு 9 :புறநானூறு –மறோக்கத்துநப்பசலையார் பாடல்கள்

பா. எண். 37, 39, 126, 226, 280

கூறு 10 :பதினெண் கீழ்க்கணக்கு–திருக்குறள் - வாழ்க்கைத்துணைநலம் (6),

அறிவுடைமை (43),பிரிவாற்றாமை (116)

கூறு 11: நான்மணிக்கடிகை–எள்ளற்க (3),பறைபடவாழா (4),

கூறு 12:நான்மணிக்கடிகை - மண்ணயறிப (5),கள்ளிவயிற்றில் (6),கல்லிற்பிறக்கும்(7)

கூறு 13: நாடகம் - இராசராசசோழன் - அரு. இராமநாதன்

கூறு 14: நாவல் - சுவடுகள் - இரா. பாலசுப்பிரமணியன்,சத்யாவெளியீடு,மதுரை

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Course Code	Title of the Course
10131B / 12731B	**PART 1 : HUMAN SKILL DEVELOPMENT- I**

Unit – I Human Skills –Developing skills-Types

Unit – II Mind-Levels of functions
Habits-Meaning-Types-Merits of good habits - Interpersonal Relationship-Features-Interpersonal Behaviour

Unit – III Thinking ahead- Significance of thinking ahead

Unit – IV Developing Personality-Meaning -Need- Factors influencing personality, Ways of developing personality -Building positive personality

Unit – V Self-concept-Self-esteem-Meaning-Importance - Self- efficacy-Self-acceptance-Meaning-Importance - Etiquette-Meaning-Etiquettes in using mobile, telephones-Dais Etiquette

Unit – VI Goal-setting Skills-Meaning-Types-Importance-

Unit – VII Decision-making skills-Meaning-Types-Steps in decision-making

Unit – VIII Negotiating Skills-Styles-Structure-Creating negotiation-Competitive Negotiation

Unit – IX Attitudes-Meaning-Types-Importance-Developing positive attitudes

Unit – X **C**oping with Change – Meaning – Characteristics - Importance of change Resistance to change - Dealing with change

Unit – XI Leadership-Meaning- Importance-Characteristics-Styles-

Unit – XII Human Relations Skill-Need-Canons of good human relations

Unit – XIII **C**ounselling-Meaning-Importance-Forms- Conflicts-Meaning-Types-Causes-Effects-Managements of conflicts

Unit – XIV Stress-Meaning-Types-Causes-Effects-Managing the stress - Anger-Meaning-Causes-Consequences-Anger Management

References:

13. Les Giblin, Skill with People, 1995.
14. Shiv Khera, You Can Win, 2002.
15. Christian H Godefroy, Mind Power.
16. Dale Carnegie, How to Enjoy Your Life and Your Job, 1985.
17. Natalie H Rogers, How to Speak without Fear, 1982.
18. Dale Carnegie, How to Develop Self-Confidence and Influence People by Public Speaking.

Course Code	Title of the Course
10132/ 12732	PART II : ENGLISH – III

Short Stories

Unit – I A Hero - R.K. Narayanan

Unit – II The Diamond Necklace - Guy de Maupassant

Unit – III The Verger - Somerset Maugham

Unit – IV The Postmaster - Rabindranath Tagore

One Act Plays

Unit – V The Proposal - Anton Chekhou

Unit – VI The Boy Comes Home - A.A. Milne

Unit – VII The Silver Idol - James R. Waugh

Unit – VIII Progress - St. John Ervine

Unit – IX The Pie and the Tart - Huge Chesterman

Unit – X Reunion - W.st. Joh Tayleur

Unit – XI A kind of Justice - Margaret Wood

Unit – XII The Refugee - Asif Currimbhoy

Grammar

Unit – VIII Parts of speech-Noun- Pronoun- Adjective

Degrees of Comparison- Verb- Adverb

Composition

Unit – XIV Agenda- Minutes- Notice- Descriptive Writing

References:

2. *Aroma*, Ed. by the Board of Editors, Publishers- New Century Book House, Chennai.
2. *Six Short Stories*, Ed. by the Board of Editors, Harrows Publications, Chennai.
2. *One Act Plays*, Ed. by the Board of Editors, Harrows Publications, Chennai.
3. Modern English – *A Book of Grammar Usage and Composition* by N.Krishnaswamy, Macmillan Publishers.
4. *English for Communication*, Ed. by the Board of Editors, Harrows Publications, Chennai.

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Course Code	Title of the Course
10133/ 12733	RELATIONAL DATABASE MANAGEMENT SYSTEM (RDBMS)

it No	Contents
	BLOCK 1 INTRODUCTION
1	**Data base System Applications**, data base System VS file System – View of Data – Data Abstraction –Instances and Schemas – data Models – the ER Model
2	**Model :**Relational Model – Other Models – Database Languages – DDL – DML – database Access for applications Programs – data base Users and Administrator – Transaction Management – data base System Structure – Storage Manager – the Query Processor.
3	**History of Data base Systems -** Data base design and ER diagrams – Beyond ER Design Entities, Attributes and Entity sets – Relationships and Relationship sets – Additional features of ER Model – Concept Design with the ER Model – Conceptual Design for Large enterprises.
	BLOCK 2 : RELATIONAL MODEL
4	**Introduction**– Integrity Constraint Over relations – Enforcing Integrity constraints – Querying relational data – Logical data base Design – Introduction to Views – Destroying / altering Tables and Views.
5	**Relational Algebra** – Selection and projection set operations – renaming – Joins – Division – Examples of Algebra overviews –
6	**Relational calculus** – Tuple relational Calculus – Domain relational calculus – Expressive Power of Algebra and calculus.
	BLOCK 3 : SQL QUERY
7	**Form of Basic SQL Query** – Examples of Basic SQL Queries – Introduction to Nested Queries – Correlated Nested Queries Set – Comparison Operators – Aggregative Operators – NULL values – Comparison using Null values – Logical connectivity's – AND, OR and NOT – Impact on SQL Constructs – Outer Joins – Disallowing NULL values – Complex Integrity Constraints in SQL Triggers and Active Data bases. Schema refinement
8	**Normal forms :**Problems Caused by redundancy – Decompositions – Problem related to decomposition – reasoning about FDS – FIRST, SECOND, THIRD Normal forms – BCNF–
9	**Join:** Lossless join Decomposition – Dependency preserving Decomposition – Schema refinement in Data base Design – Multi valued Dependencies – FORTH Normal Form.
	BLOCK 4 TRANSACTION
10	**Introduction :**Transaction Concept- Transaction State- Implementation of Atomicity and Durability – Concurrent – Executions – Serializability- Recoverability – Implementation of Isolation – Testing for serializability
11	**Protocols :** Lock Based Protocols – Timestamp Based Protocols- Validation- Based Protocols – Multiple Granularity.
12	**Recovery and Atomicity** – Log – Based Recovery – Recovery with Concurrent Transactions – Buffer Management – Failure with loss of nonvolatile storage-Advance Recovery systems- Remote Backup systems
	BLOCK 5 STORAGE
13	**Data on External Storage** – File Organization and Indexing – Cluster Indexes, Primary and Secondary Indexes – Index data Structures – Hash Based Indexing – Tree base Indexing – Comparison of File Organizations – Indexes and
14	**Performance Tuning-** Intuitions for tree Indexes – Indexed Sequential Access Methods (ISAM) – B+ Trees: A Dynamic Index Structure.

Text Books:

1. Raghurama Krishnan, Johannes Gehrke, Data base Management Systems, 3rd Edition, TATA McGrawHill.2003.
2. Silberschatz, Korth, Data base System Concepts, 6th Edition, Tata McGraw Hill, 2011.

Reference Books:

1. Relational Database Principles 2nd Edition, Colin Ritchie, 2004
2. Sharad Maheswari and Ruchin Jain, Database management systems Complete Practical Approach,Firewall media, 2006
3. Peter Rob & Carlos Coronel, Data base Systems design, Implementation, and Management, 7th Edition.
4. Elmasri Navrate , Fundamentals of Database Systems, Pearson Education.

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Course Code	Title of the Course
10134/ 12734	RELATIONAL DATABASE MANAGEMENT SYSTEM (RDBMS) - LAB

Unit No.	Contents
	BLOCK 1 : TABLE MANIPULATION
1	**Table** creation, Renaming a Table, Copying another table, Dropping a Table
2	**Table Description:** Describing Table Definitions, Modifying Tables, Joining tables, Number and Date functions.
	BLOCK 2 : SQL QUERIES AND SUB QUERIES
3	**SQL Queries:** Queries, Sub Queries, and Aggregate functions
4	**DDL:** Experiments using database DDL SQL statements
5	**DML:** Experiment using database DML SQL statements
6	**DCL:** Experiment using database DCL SQL statements
	BLOCK 3 : INDEX AND VIEW
7	**Index :** Experiment using database index creation, Renaming a index, Copying another index, Dropping a index
8	**Views:** Create Views, Partition and locks
	BLOCK 4 : EXCEPTION HANDLING AND PL/SQL
9	**Exception Handling:** PL/SQL Procedure for application using exception handling
10	**Cursor:** PL/SQL Procedure for application using cursors
11	**Trigger:** PL/SQL Procedure for application using triggers
12	**Package:** PL/SQL Procedure for application using package
13	**Reports:** DBMS programs to prepare report using functions
	BLOCK 5 : APPLICATION DEVELOPMENT
14	**Design and Develop Application:** Library information system, Students mark sheet processing, Telephone directory maintenance, Gas booking and delivering, Electricity bill processing, Bank Transaction, Pay roll processing. Personal information system, Question database and conducting Quiz and Personal diary

Reference Books:

1. Raghurama Krishnan, Johannes Gehrke, Data base Management Systems, 3rd Edition, TATA McGrawHill.2003.
2. Silberschatz, Korth, Data base System Concepts, 6th Edition, Tata McGraw Hill, 2011.
3. Relational Database Principles 2nd Edition, Colin Ritchie, 2004
4. Sharad Maheswari and Ruchin Jain, Database management systems Complete Practical Approach,Firewall media, 2006
5. Peter Rob & Carlos Coronel, Data base Systems design, Implementation, and Management, 7th Edition.
6. Elmasri Navrate , Fundamentals of Database Systems, Pearson Education.

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SEMESTER IV

Course Code	Title of the Course
10141A/ 12741A	PART 1 : TAMIL – PAPER IV

கூறு 1 : செய்யுள் உறுப்புகள் - யாப்பு -எழுத்து,அசை,சீர்,

கூறு 2 :செய்யுள் உறுப்புகள் - யாப்பு -தளை,அடி,தொடை

கூறு 3 :வெண்பா,ஆசிரியப்பா,கலிப்பா,வஞ்சிப்பா,

கூறு 4 :புதியயாப்புவடிவங்கள் - சிந்து,கண்ணி,கீர்த்தனை

கூறு 5 :புதுக்கவிதையில் குறியீடு–படிமம்.

கூறு 6 :அகப்பொருள் - புறப்பொருள் - ஐந்திணைவிளக்கம்

கூறு 7 :அகப்பொருள் துறைகள் - வரைவுகடாதல்,அறத்தொடுநிற்றல்,
உடன்போக்கு

கூறு 8 :புறப்பொருள் துறைகள் - வஞ்சினக்காஞ்சி,கையறுநிலை,செவியறிவுறூஉ

கூறு 9 :அணி இலக்கணம் - உவமை,உருவகம்,வேற்றுமை,பிறிதுமொழிதல்,
தற்குறிப்பேற்றம்,சிலேடை,பின்வருநிலை

கூறு 10 :நிறுத்தல் குறிகள்

கூறு 11 : தொல்காப்பியம் - சங்கஇலக்கியம் - எட்டுத்தொகை,பத்துப்பாட்டு,

கூறு 12 :பதினெண்கீழ்க்கணக்கு ஐம்பெருங்காப்பியங்கள் - பிற்காலக் காப்பியங்கள் - கம்பராமாயணம் - பெரியபுராணம்

கூறு 13 :இக்காலக் காப்பியங்கள் - பாரதியின் பாஞ்சாலிசபதம் -
பாரதிதாசனின் பாண்டியன் பரிசு

கூறு 14. :கண்ணதாசனின் இயேசுகாவியம் ,சிற்பியின் -மௌனமயக்கங்கள்.

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Course Code	Title of the Course
10141B/ 12741B	**PART 1 : HUMAN SKILL DEVELOPMENT- II**

Unit – I Guidance & Counselling – Role of Counsellor - Importance and Techniques of counselling

Unit – II Managerial skill- Need – Importance

Unit – III Human relational skills-Communication-Attention

Unit – IV Conceptual skills-Meaning-Importance

Unit – V Technical skills-Techniques-Practices-Tools-Procedures

Unit – VI Presentation skills-Planning-Preparation-Delivery

Unit – VII Organization skills-Meaning-Nature-Importance-Types

Unit – VIII Multi-Tasking skills Responsibilities-Causes

Unit – IX Leader- Qualities of a good leader

Unit – X Understanding Skills -Human systems: Individual, Group, organization, and their major interactions

Unit – XI Understanding Skills -Human systems: Community and Society, and their major interactions

Unit – XII Problem solving skills – Handling –Facing - Importance

Unit – XIII Cooperative Learning Skills

Unit – XIV Making Social Responsibilities-Causes

References:

1. Les Giblin, Skill with People, 1995.
2. Shiv Khera, You Can Win, 2002.
3. Christian H Godefroy, Mind Power.
4. Dale Carnegie, How to Enjoy Your Life and Your Job, 1985.
5. Natalie H Rogers, How to Speak without Fear, 1982.
6. Dale Carnegie, How to Develop Self-Confidence and Influence People by Public Speaking.

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Course Code	Title of the Course
10142 / 12742	**PART II : ENGLISH – IV**

Short Stories

Unit – I Lalajee - Jim Corbelt
Unit – II A Day's Wait - Hemmingway
Unit – III Two old Men - Leo Tolstoy
Unit –IV Little Girls wiser than - Men Tolstoy
Unit – V Boy who Wanted more Cheese - William Elliot Griffir

Drama

Unit – VI Pygmalion - G.B. Shaw

Fiction

Unit – VII Swami and Friends - R.K. Narayanan

Tales from Shakespeare

Unit – VIII - The Merchant of Venice
Unit – IX - Romeo and Juliet
Unit – X - The Winter's Tale

Biographies

Unit – XI - Martin-Luther king - R.N. Roy
Unit – XII - Nehru - A.J. Toynbee

Grammar

Unit – XIII - Concord- Phrases and Clauses-Question Tag

Composition

Unit – XIV - Expansion of Proverbs
- Group Discussion
- Conversation (Apologizing, Requesting, Thanking)

References:

1. *Sizzlers*, by the Board of Editors, Publishers-:Manimekala Publishing House, Madurai.
2. *Pygmalion* – G.B. Shaw
3. *Swami and Friends* – R.K. Narayan
4. *Tales from Shakespeare* Ed. by the Board of Editors, Harrows Publications, Chennai.
5. Modern English – *A Book of Grammar Usage and Composition* by N.Krishnaswamy, Macmillan Publishers.

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Course Code	Title of the Course
10143 / 12743	INTERNET AND JAVA PROGRAMMING

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	**Basic Internet Concepts:** Connecting to the Internet – Domain Name System – E-mail
2	**The World Wide Web** – Internet Search Engines – Web Browsers – Chatting and conferencing on the Internet
3	**Online Chatting** –Messaging – Usenet Newsgroup – Internet Relay chat (IRC) – FTP – Telnet.
	BLOCK 2 : FUNDAMENTALS OF OBJECT-ORIENTED PROGRAMMING
4	**Basic concepts** of OOP – Benefits – Applications . Java Evolution: Features – how java differs from C and C++ - java and internet- java support system – java environment
5	**Overview of Java Language** –Introduction – Simple Java Program – Comments – Java Program Structure – Tokens – Java Statements – Implementing a Java Program – JVM – Command Line Arguments. Constants – Variables – Data Types – Type Casting..
6	**Operators and Expressions:** Arithmetic Operators – Relational, Logical, Assignment, Increment and Decrement, Conditional, Bitwise, Special Operators – Arithmetic Expressions, Evaluation of Expression – Precedence of Arithmetic Operators – Type Conversions – Operator Precedence and associativity – Mathematical Functions. **Decision Making and Branching:** If –if…..else –Nesting of if……. Else – else if–switch. Decision Making and Looping: While – do – for – jump in loops – labeled loops.
	BLOCK 3 : CLASSES, OBJECTS AND METHODS
7	**class:** Defining a class –fields –methods –creating objects – accessing class members – constructors – methods overloading –static members –nesting of methods – Inheritance –overriding methods –final variables-classes –methods
8	**Arrays, Strings and Vectors** :One dimensional Arrays –creating of array – Two dimensional arrays- strings –vectors –Wrapper classes – Enumerated Types - Interfaces: Multiple Inheritance
9	**Packages:** Defining interface –Extending interfaces – Implementing Interfaces - Putting Classes Together
	BLOCK 4: MULTITHEREADING , EXCEPTION AND APPLETS
10	**Multithreaded Programming** – Creating Threads –Extending the thread class – Stopping and Blocking a thread –Life cycle of a thread –using thread methods – Thread Exceptions –Priority –Synchronization –Implementing the 'Runnable' Interface
11	**Managing Error and Exceptions:** Types of errors –Exceptions –Syntax of Exception Handling code – Multiple Catch statements –using finally statement – Throwing our own Exceptions – using exceptions for Debugging - Graphics Programming: The Graphics Class – Lines and Rectangles – Circles and Ellipses – Drawing Arcs – Drawing Polygons – Line Graphs – Using Control Loops in Applets – Drawing Bar Charts.
12	**Applet Programming:** How applets differ from Applications – preparing to write applets – Building Applet Code – Applet life cycle – creating an Executable

	Applet – Designing a Web Page – Applet Tag – Adding Applet to HTML file – Running the Applet – Passing parameters to Applets – Displaying Numerical values – Getting input from the user
	BLOCK 5 : MANAGING INPUT/OUTPUT FILES IN JAVA
13	**Introduction** – concept of streams –stream classes – byte stream classes – character Stream
14	**I/O classes**: –using stream –using the file class –Input / output Exceptions – creation of files – Reading / writing characters – reading writing bytes Random access files- Interactive input and output –Other stream classes

Text Books:

1. R. Krishnamoorthy and S. Prabhu, Internet and Java Programming, New Age International Publishers, 2004 (Unit I).
2. Programming with Java, 4e, E. Balagurusamy, Tata McGraw-Hill, 2010.

Reference books:

1. Deitel, Deitel and Nieto, Internet and World Wide Web – How to program, Pearson Education, 2000.
2. Naughton and H.Schildt, Java 2 - The complete reference, Tata McGraw-Hill, Fourth edition, 2006.
3. Elliotte Rusty Harold, Java Network Programming, O'Reilly Publishers, 2000.
4. B.Mohamal Ibrahim , Java : J2SE – A Practical Approach, Firewall media, 2006.
5. Cay S. Horstmann, Gary Cornell, Core Java, Volume I and II, 5th Edition, Pearson Education, 2003.
6. Topley, J2ME in A Nutshell, O'Reilly Publishers, 2002.
7. Hunt, Guide to J2EE Enterprise Java, Springer Publications, 2004.
8. Ed Roman, Enterprise Java Beans, Wiley Publishers, 1998.

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Course Code	Title of the Course
10144 / 12744	**INTERNET AND JAVA PROGRAMMING - LAB**

Experiments based on Internet Programming Theory

Unit No.	Contents
	BLOCK 1 : JAVA FUNDAMENTAL PROBLEMS:
1	Simple Java Problems
2	class and objects
3	Conditional control using java
4	Looping using java
	BLOCK 2 : OOP CONCEPTS
5	Function overloading programs
6	Operator overloading programs
7	Inheritance programs, Packages
8	Polymorphism programs Message passing programs
	BLOCK 3 : THREAD & VIRTUAL FUNCTION
9	Threads
10	Virtual function programs
	BLOCK 4 : I/O AND EXCEPTION HANDLING
11	Exception handling programs
12	I/O manipulation programs,
	BLOCK 4 :NETWORK PROGRAMMING
13	Applet programs
14	Implementation of simple network programs using java

Reference books:

1. R. Krishnamoorthy and S. Prabhu, Internet and Java Programming, New Age International Publishers, 2004 (Unit I).
2. Programming with Java, 4e, E. Balagurusamy, Tata McGraw-Hill, 2010.
3. Deitel, Deitel and Nieto, Internet and World Wide Web – How to program, Pearson Education, 2000.
4. Naughton and H.Schildt, Java 2 - The complete reference, Tata McGraw-Hill, Fourth edition, 2006.
5. Elliotte Rusty Harold, Java Network Programming, O'Reilly Publishers, 2000.
6. B.Mohamal Ibrahim , Java : J2SE – A Practical Approach, Firewall media, 2006.
7. Cay S. Horstmann, Gary Cornell, Core Java, Volume I and II, 5th Edition, Pearson Education, 2003.
8. Topley, J2ME in A Nutshell, O'Reilly Publishers, 2002.
9. Hunt, Guide to J2EE Enterprise Java, Springer Publications, 2004.
10. Ed Roman, Enterprise Java Beans, Wiley Publishers, 1998.

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THIRD YEAR
SEMESTER V

Course Code	Title of the Course
10151 / 12751	**ACCOUNTING FUNDAMENTALS**

Unit No.	Contents
	BLOCK 1 : ACCOUNTING FUNDAMENTALS
1	**Introduction:** Accounting - Definition - Accounting for historical function and managerial function
2	**Scope of accounting** - Financial accounting - Cost accounting and management accounting
3	Managerial uses - Differences.
	BLOCK 2 : FINANCIAL ACCOUNTING:
4	**Accounting concepts** - Conventions - Principles
5	**Accounting standards** - International Accounting standards.
	BLOCK 3 : DOUBLE ENTRY SYSTEM OF ACCOUNTING
6	Double entry system of accounting - Accounting Books - Preparation of Journal and Ledger
7	Subsidiary books Errors and rectification
8	Preparation of a Trial balance and Final accounts
	BLOCK 4 : ACCOUNTING FROM INCOMPLETE RECORDS
9	**Introduction:** Accounting from incomplete records – Statement of affairs methods – Conversion method
10	Preparation of Trading, Profit and Loss Account from incomplete records.
11	Balance Sheet from incomplete records.
	BLOCK 5 : RATIO ANALYSIS
12	Ratio Analysis - Types - Profitability ratios - turnover ratios - liquidity ratios - propriety ratios - Market Earnings Ratios
13	Factors affecting efficiency of ratios - How to make effective use of ratio analysis - Uses and limitation of ratios.
14	Construction of Profit & Loss account and Balance sheet with ratios and relevant figures - Inter-firm, Intra-firm comparisons.

REFERENCE BOOKS:

1. M.A.Arulanandam & K.S.Raman - Advanced Accounting
2. R.C.Gupta & Radaswamy - Advanced Accounting
3. M.C.Shukla & T.S.Grewal - Advanced Accounting
4. Jain & Narang - Advanced Cost Accounting
5. P. Das Gupta - Advanced Studies in Cost Accounting
6. S.N.Maheswari - Management Accounting & Financial Control
7. Manmohan & Goyal - Principles of Management Accounting
8. N.K.Prasad - Advanced Cost Accounting

Course Code	Title of the Course
10152 / 12752	COMPUTER GRAPHICS

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	**Introduction** Application areas of Computer Graphics, overview of graphics systems, video-display devices, raster-scan systems, random scan systems, graphics monitors and work stations and input devices.
2	**Output primitives:** Points and lines, line drawing algorithms, mid-point circle and ellipse algorithms.
3	**Filled area primitives:** Scan line polygon fill algorithm, boundary-fill and flood-fill algorithms.
	BLOCK 2 : 2 D TRANSFORM AND CLIPPING
4	**2-D geometrical transform:** Translation, scaling, rotation, reflection and shear transformations
5	**2D Matrix representations:** homogeneous coordinates, composite transforms, transformations between coordinate systems.
6	**2-D viewing:** The viewing pipeline, viewing coordinate reference frame, window to view-port coordinate transformation, viewing functions,
7	**Clipping Algorithms:** Cohen-Sutherland and Cyrus-beck line clipping algorithms, Sutherland –Hodgeman polygon clipping algorithm.
	BLOCK 3 : 3D OBJECT REPRESENTATION
8	**Introduction:** Polygon surfaces, quadric surfaces, spline representation,
9	**Curve and surfaces:** Hermite curve, Bezier curve and B-Spline curves, Bezier and B-Spline surfaces. Basic illumination models, polygon rendering methods.
	BLOCK 4 : 3D GEMETRIC TRANSFORMATION
10	**3-D Geometric transformations:** Translation, rotation, scaling, reflection and shear transformations, composite transformations.
11	**3-D viewing:** Viewing pipeline, viewing coordinates, view volume and general projection transforms and clipping.
	BLOCK 5 : VISIBLE SURFACE DETECTION METHODS AND ANIMATION
12	**Classification**, back-face detection, depth-buffer, scan-line, depth sorting, BSP-tree methods, area sub-division and octree methods.
13	**Computer animation:** Design of animation sequence, general computer animation functions, raster animation,
14	**Other Animation Techniques:** Computer animation languages, key frame systems, motion specifications.

Text Books

1. Donald Hearn and M.Pauline Baker, Computer Graphics C version, Pearson Education, 2007.

Reference Books:

1. M. Newman and F. Sproull, Interactive Computer Graphics, McGraw Hill 2004
2. Foley, VanDam, Feiner and Hughes, Computer Graphics Principles and Practice, 2nd Edition in C, Pearson Education, 2004.
3. Plastok and Gordon Kalley, Computer, McGraw Hill 2000.

Course Code	Title of the Course
10153 / 12753	OPERATING SYSTEMS

it No.	ntents
	BLOCK 1 : INTRODUCTION
1	**Introduction:** Definition of Operating Systems – Computer System Organization
2	**Computer System Architecture** – Operating System Structure – Operating System Operations
3	**System Structures:** Operating System Services – System Calls – System Programs – Operating System Design and Implementation.
	BLOCK 2 : PROCESS CONCEPT
4	**Process Concept:** Process Scheduling – Operations on Processes – Inter Process Communication
5	**Process Scheduling:** Scheduling Concepts – Scheduling Criteria – Scheduling Algorithms – Multiple Processor Scheduling
	BLOCK 3 : SYNCHRONIZATION
6	**Synchronization:** The Critical Section Problem – Synchronization Hardware – Semaphores – Classic Problems of Synchronization – Monitors
7	**Deadlocks:** Deadlocks Characterization – Methods for Handling Deadlocks
8	**Deadlock** Prevention – Avoidance – Detection –Recovery from Deadlock.
	BLOCK 4 : MEMORY MANAGEMENT
9	**Memory Management Strategies:** Swapping – Contiguous Memory Allocation – Paging – Segmentation
	BLOCK 5 : FILE SYSTEM
10	**File Concept** – Access Methods – Directory
11	**Structure** – File System Mounting – File Sharing – Protection.
12	**Implementing File Systems:** File System Structure – File System Implementation
13	**Directory Implementation** – Allocation Methods – Free Space Management
14	**Secondary Storage Structure:** Overview of Mass Storage Structure – Disk Structure – Disk Attachment – Disk Scheduling – Disk Management.

TEXT BOOK:

1. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, “Operating System Principles”, 7th Edition, Wiley India Edition, 2008.

REFERENCE BOOKS:

1. Andrew S.Tanenbaum, Operating Systems Design and Implementation, 3rd edition, Prentice Hall, 2006.
2. Harvey M. Deitel, An Introduction to Operating Systems, 3rd Edition, Addison Wesley 2007.

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Course Code	Title of the Course
10154 / 12754	UNIX & SHELL PROGRAMMING LAB

Unit No.	Contents
	BLOCK 1 INTRODUCTION
1	**Introduction** Operating system ,objective, History, Features of Unix
2	**Kernal and shell**
3	**Unix file system:** File and common commands-Shell-More about files-Directories-Unix system-Basics of file-Directories and filenames-
	BLOCK 2 : UNIX PERMISSIONS
4	**Permissions**- Inodes-Directory hierarchy-Devices-the grep family-Other filters
5	**Stream editor** sed - awk pattern scanning and processing language-files and good filters.
6	**Wild card characters**
	BLOCK 3 : UNIX COMMANDS
7	**Unix commands with syntax:** Syntax and unix commands
8	**Unix shells:** History of unix shells
9	**Deciding on a shell**
	BLOCK 4 : SHELL COMMADS - FILES
10	**Shell Command files**
11	**Bourne shell**: Bourne shell programming
12	**Shell programming files:** Shell programming on files
	BLOCK 5 : MENU DRIVEN SHELL PROGRAM:
13	**Menu Driven File handling**
14	**Menu Driven shell program – file:** edit, create and delete file using menu

REFERENCE BOOKS:

1. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, “Operating System Principles”, 7th Edition, Wiley India Edition, 2008.
2. Andrew S.Tanenbaum, Operating Systems Design and Implementation, 3rd edition, Prentice Hall, 2006.
3. Harvey M. Deitel, An Introduction to Operating Systems, 3rd Edition, Addison Wesley 2007.

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SEMESTER VI

Course Code	Title of the Course
10161 / 12761	MANAGEMENT PRINCIPLES AND TECHNIQUES

Unit No.	Contents
	BLOCK 1 : INTRODUCTION TO MANAGEMENT
1	**Introduction :** Concept and Definition of Management - Management Functions
2	**Planning :** Principles - Planning - Planning Process - Decision-making
3	**Organising** - Structure - Delegation - Staffing - Direction - communication - Motivation - Leadership - Control.
	BLOCK 2 : LINEAR PROGRAMMING
4	**Introduction** - History of OR - Meaning of OR - Principles of Modelling - Application of OR
5	Formulation of LP models - Graphical solution in Maximization problem and Minimization Problem - Algebraic solutions
6	**Simplex method** - Feasibility - Optimality - Artificial variables - Duality - Dual - simplex Algorithm
7	**Transportation problem** - finding Optimal solution - Assignment problem - Least Time Transportation problems.
	BLOCK 3 : PERT / CPM
8	PERT/CPM : Arrow (network) diagram representations - Time estimates - critical path - Floats
9	Construction of Time chart and Resource Levelling
10	Probability and cost considerations in project scheduling - Shortest Route Problem - Project control.
	BLOCK 4 : REPLACEMENT THEORY
11	**Replacement Theory : Introduction** - Various replacement situations
12	**Replacement policy** - Variable maintenance costs and fixed money value - Variable maintenance costs and Variable money value
	BLOCK 5 ; REPLACEMENT POLICY
13	Individual replacement policy - Group replacement policy
14	Reliability - Applications of Replacement problems in real life problem.

REFERENCE BOOKS:

1. Hamdy A Taha, Operations Research An Introduction, Macmillan Publishing Company (1982)
2. Don.T.Philps, A.Ravindran, James.J.Solberg, Operations Research - Principles and Practice, John Wiley & Sons (1976).
3. Richard I Lenin, Charles A.Kirkpatrick, David S Rubin, Quantitative Approach to Management.

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Course Code	Title of the Course
10162 / 12762	SYSTEM ANALYSIS AND DESIGN

Unit No.	Contents
	BLOCK 1 : SYSTEM CONCEPTS AND SYSTEM DEVELOPMENT LIFE CYCLE
1	**System Concepts** - Characteristics - Elements of a system - Types of Systems: Abstract, Physical, Open, Closed and Man-made Information system - Computer Based Information Systems: MIS, DSS, TPS and OAS
2	**System Development Life Cycle** - Problem Definition - Feasibility Study - Analysis - Design - Development - Implementation - Post Implementation and Maintenance
3	**System Analyst :** Interpersonal Skills - Technical Skill - Communication Skills - Role of Systems Analyst.
	BLOCK 2 : SYSTEM ANALYSIS
4	**System Analysis :** Bases for planning in System Analysis - Preliminary Investigation - Determining the User's information requirements, Case Scenario, Problem Definition and Project Initiation, Background Analysis
5	**Fact Finding Techniques :** Interview - Questionnaire - Record Review - Observation. Systems Analysis: Analysing Systems data - Feasibility Study: Technical, Economical and Operational - Steps in Feasibility Analysis, Feasibility Report, Oral Presentation
6	**Systems Costs & Benefits:** Categories of Cost - Benefits - Cost Benefit Analysis: Break Even, Present Value, Pay Back and Cash Flow. **Analysis Tools :** Data flow concept - Data Flow Diagram - Data Dictionary - Decision Table - Decision Tree - Structured English.
	BLOCK 3 : SYSTEM DESIGN
7	**System Design :** Process and stages of System Design : Logical and Physical Design. Design Methodologies: Structured design - Form Driven Methodology - Major Development Activities
8	**Input Output and Form Design:** Input Design : Capturing Data for input - Input Validation - Input Design of on-line systems. Output Design - Printed, Display and Audio.
9	**Forms Design :** Definition - Classification of Forms, Requirements of Forms Design - Types of Forms - Forms Control.
	BLOCK 4 : FILE AND DATABASE DESIGN:
10	**File concepts** - Types of Files - Methods of File Organization - Sequential - Direct - Indexed - Database Design: Database concept
11	**Types of Databases :** Hierarchical, Network and Relational.
12	**System Development**: Software Design - Top Down Approach - Flow Chart: System Flow Chart - Program Flow Chart - HIPO - IPO - VTOC - Warnier Orr Diagram - Structured Walkthrough - Quality Assurance - Levels of Assurance - System Testing - Special Systems Tests
	BLOCK 4 : SYSTEM EVALUATION,IMPLEMENTATION AND MAINTENANCE
13	**System Evaluation and Implementation** Training Personnel - Training Methods - Conversion: Conversion Methods - Parallel, Direct, Pilot and Phase-in. Conversion Plan - Site Preparation - Data and File Preparation - Post Implementation Review -
14	**System Maintenance :** Corrective - Adaptive - Hardware and Software Selection : Computer Industry - Software Industry - Procedure of Hardware and Software

	Selection: Major phases in Hardware and Software selection - Evaluation Process - Financial considerations.

Text Book

Elias M.Awad, Systems Analysis and Design, 1990, Galgotia Publication Pvt. Ltd.

Reference Book:

1. James A. Sen, Analysis and Design of Information System, 1985, McGraw Hill.

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COURSE CODE	TITLE OF THE COURSE
10163 / 12763	VISUAL BASIC PROGRAMMING

Unit No	Concepts
	BLOCK 1: VISUAL BASIC CONCEPTS
1	**Introduction to GUI** - Visual Basic : Starting and Exiting Visual Basic Project Explorer Working with Forms Properties Window
2	**Using the Toolbox** Toolbars Working with Projects Programming Structure of Visual Basic applications Event and Event driven Procedures
3	**Program Design** - Form and Controls - Writing the Code - Saving, Running and Testing - Making EXE File - Printouts
	BLOCK 2 : VISUAL BASIC CODE,EVENTS AND CONTROLS
4	**Adding code and using events**: Using literals data types - declaring and using variables using the operator subroutines and functions
5	**Looping and decision control structures:** if then else ,structure select structure , for next , do.. loop and while.. wend.
6	**Using intrinsic Visual basic Controls with methods and Properties:** Label ,Text box, Command button, Frame, Checkbox, option button, List box, Combo box, Drive List box, directory List box and file list box Formatting controls control arrays, Tab order
	BLOCK 3 : VISUAL BASIC PROCEDURES, FUNCTIONS AND ARRAYS
7	**Creating Procedures, functions** - String functions, date and Time function , numeric functions- Recursive Functions
8	**Multiple Forms** - Startup Forms - SubMain Procedure
9	**Arrays** - Control Arrays - Indexing and Event Handling - Graphics
	BLOCK 4 : MENUS AND MDI FORMS
10	**Menus:** creating menus, adding code to menus
11	**Using MDI forms** - MDI form basic building MDI form creating MDI Child Forms
	BLOCK 5: DATABASE OBJECT (DAO) AND PROPERTIES
12	Database object (DAO) and properties -accessing Recordset objects- Move first, MoveLast, MovePrevious and MoveNext methods Begin , Commit and Rollback transaction accessing Microsoft Access files.
13	Active Data Objects (ADO) ADO and OLE DB and ADO Primer What are OLE DB and ADO? ADO object Model Converting DAO Code to Use ADO.
14	Connecting to the database Retrieving a recordset Creating a query dynamically Using a parameterized query using action queries - Adding records Editing records closing the database connection**.**

Text Books

1. Gary Cornwell Visual basic 6 , Tata McGraw Hill

Reference Books:

1. Scott warner Teach yourself Visual basic 6 , Tata McGraw-Hill
2. Noel Jerke The Complete Reference , Tata McGraw-Hill
3. Eric A. Smith, Valar Whisler, and Hank Marquis Visual Basic 6 programming

COURSE CODE	TITLE OF THE COURSE
10164 / 12764	VISUAL BASIC PROGRAMMING LAB

Unit No.	Contents
	BLOCK 1
1	Building simple applications
2	Working with intrinsic controls ,Control Arrays
	BLOCK 2
3	Application with multiple forms
4	Application with dialogs
5	Application with Menus
6	Application using data controls
7	Application using Common Dialogs
	BLOCK 3
8	Drag and Drop Events
9	Database Management
10	Creating ActiveX Controls
11	Database object (DAO) and properties
12	Active Data Objects (ADO) ADO and OLE DB
	BLOCK 4
13	Connecting to the database ,Retrieving a recordset Creating a query dynamically Using a parameterized query using action queries - Adding records Editing records closing the database connection
	BLOCK 5
14	**Simple Application development:** 1. Library information system 2. Students mark sheet processing 3. Telephone directory maintenance 4. Gas booking and delivering 5. Electricity bill processing 6. Bank Transaction 7. Pay roll processing 8. Personal information system 9. Question database and conducting Quiz 10. Personal diary

Text Books

1. Gary Cornwell Visual basic 6 , Tata McGraw Hill

Reference Books:

1. Scott warner Teach yourself Visual basic 6 , Tata McGraw-Hill
2. Noel Jerke The Complete Reference , Tata McGraw-Hill
3. Eric A. Smith, Valar Whisler, and Hank Marquis Visual Basic 6 programming

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10. B.Sc PSYCHOLOGY

Duration – Three years
Eligibility – A pass in HSC (or) 3 year Diploma
Medium – English

Sl. No.	Course Code	Title of the Course	CIA Max.	ESE Max.	TOT Max	C Max.
FIRST YEAR						
I Semester						
1.	11911 A / B	Part 1 Tamil –I / Communication Skills-I	25	75	100	4
2.	11912	Part 2 English-I	25	75	100	4
3.	11913	Introduction to Psychology	25	75	100	4
4	11914	Human Development	25	75	100	4
		Total	**100**	**300**	**400**	**16**
II Semester						
5	11921 A/ B	Part 1 Tamil –II / Communication Skills-II	25	75	100	4
6	11922	Part 2 English-II	25	75	100	4
7	11923	Social Psychology	25	75	100	4
8	11924	Industrial Psychology	25	75	100	4
		Total	**100**	**300**	**400**	**16**
SECOND YEAR						
III Semester						
9.	11931 A/B	Part 1 Tamil – III / Human Skills Development-I	25	75	100	4
10	11932	Part 2 English-III	25	75	100	4
11	11933	Psychological Statistics	25	75	100	4
12	11934	Psychopathology	25	75	100	4
		Total	**100**	**300**	**400**	**16**
IV Semester						
13	11941 A/B	Part 1 Tamil –IV / Human Skills Development-II	25	75	100	4
14	11942	Part 2 English-IV	25	75	100	4
15	11943	Cross Cultural psychology	25	75	100	4
16	11944	Educational Psychology	25	75	100	4
		Total	100	300	400	16
THIRD YEAR						
V Semester						
17	11951	Human Resource Management	25	75	100	4
18	11952	Environmental Psychology	25	75	100	4
19	11953	Stress Management	25	75	100	4
20	11954	Practical	25	75	100	4
		Total	**100**	**300**	**400**	**16**
VI Semester						
21	11961	Health Psychology	25	75	100	4
22	11962	Guidance and Counselling	25	75	100	4
23	11963	Consumer Behavioural	25	75	100	4
24	11964	Psychological Adjustment	25	75	100	4

		Total	**100**	**300**	**400**	**16**
		Grand Total	**600**	**1800**	**2400**	**96**

CIA: Continuous Internal Assessment, **ESE:** End Semester Examination,
TOT: Total, **C:** Credit Points **Max:** Maximum

SEMESTER – I

Course Code	Title of the Course
11911A	PART- I TAMIL - I

கூறு 1

4. கண்ணதாசன் - ஸ்ரீ கிருஷ்ண கானம்

1. புல்லாங்குழல் கொடுத்த
2. குருவாயூருக்கு வாருங்கள்

கூறு 2

3. கோகுலத்து பசுக்கள்
4. கோகுலத்தில் ஒரு நாள் ராதை
5. ஆயர்பாடி மாளிகையில்

கூறு 3

பட்டுக்கோட்டை கல்யாண சுந்தரம்

1. நெஞ்சில் குடியிருக்கும்
2. செய்யும் தொழிலே தெய்வம்

கூறு 4

5. பாரதியார்

கண்ணன் என் விளையாட்டுப்பிள்ளை
பாரத மாதா திருப்பள்ளி எழுச்சி

கூறு 5

10. பாரதிதாசன் - உலகப்பன் பாட்டு (5)
11. நாமக்கல் கவிஞர் - நோயற்ற வாழ்வு 7 பாட்டு
12. பெ.தூரன் - நிலா பிஞ்சு

கூறு 6

13. வல்லிக் கண்ணன் - வெறும் புகழ்
14. கு.ப.இராஜகோபாலன் - எதற்காக?
15. மீரா - பதினைந்து

கூறு 7

9. சிற்பி - சர்ப்ப யாகம்
10. ஞானக்கூத்தன் - தோழர் மோசிகீரனார்

கூறு 8

9. அப்துல் ரகுமான் - கண்ணும் எழுதேம்
10. சண்முக சுப்பையா - வயிறு

கூறு 9

9. சிலப்பதிகாரம் - வழக்குரை காதை
10. கம்பராமாயணம் - அயோத்தியா காண்டம்

கூறு 10

5. சீறாப்புராணம் - ஈத்தங்குலை வரவழைத்த படலம் (1)

கூறு 11

தேம்பாவணி - காட்சிப்படலம்
பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)
80. இன்னவாயில்

81. கொழுந்துறும்
82. பஞ்(ச) அரங்கில்

கூறு 12

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

83. எண்ணுளே
84. ஒண்தலங்கள்
85. இரவியேந்த கஞ்சக்

கூறு 13

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

86. கன்னியாயதாயும்
87. ஏந்தி ஓங்கு உளத்து
88. ஆவ தேமுனர்
89. கொல்லும் வேலொடும்

கூறு 14

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

90. என்ற வாசகம்
91. அம்பினால்
92. வேண்டும் ஓர் வினை

Course Code	Title of the Course
11911B	**PART-I : COMMUNICATION SKILLS - I**

Unit - I Communication – Meaning – Types- Importance

Unit – II Barriers to Effective Communication – Principles – Principles of Effective Communication

Unit – III Oral Communication – Meaning – Importance- Forms of Oral Communication

Unit – IV Intonation –Meaning – Function- Types Preparation of Speech- Steps Involved

Unit – V Principles of Effective Oral Communication

Unit – VI Written Communication – Meaning –Steps – Importance- Advantages Use of words and Phrases

Unit – VII Sentence – Meaning –Sentence formation- Characteristics of an Effective Sentence

Unit – VIII Paragraph Writing –Essay Writing –Steps Involved –Outline-Layout – Contents-Drafting-Correction- Final Draft

Unit – IX Application for Employment and Curriculum Vitae –Steps involved

Unit – X Non –Verbal Communication – Meaning –Types –Body Language – Postures-Gestures –Facial Expressions –Eye Contact

Unit – XI Report Writing –Report –Types of Reports –Format of a Report

Unit – XII Essentials of a Good Report –Preparation of Report-Procedure Involved

Unit – XIII Meetings-Purpose of the Meeting – Procedure

Unit – XIV Group Discussion –Quality of Content-Participation –Logical Presentation –Behavioural Skills

References:

1. Krishna Mohan & Meera Banerjee, Developing Communication Skills, 2005.
2. Geetha Nagaraj, Write to Communicate, 2004.
3. Wren & Martin, English Grammar and Composition, 2002.
4. Dale Carnegie, How to Win Friends and Influence People, 1981.
5. Dale R Jordan, Language Skills and Use.
6. Gartside L. Bahld, Nagammiah and McComas, Satterwhite, Modern Business Correspondence.
7. Rajendra Pal and Kortahalli J S, Essentials of Business Communication.
8. Wallace, Michael J, Study Skills in English.
9. Editors of Readers Digest, Super Word Power.

Course Code	Title of the Course
11912	**PART-II : ENGLISH PAPER - I**

Prose

UNIT – 1	Water-the Elixir of life	- C.V. Raman
UNIT – 2	Mrs. Packletide's Tiger	- SAKI
UNIT – 3	A Deed of Bravery	- Jim Carbett
UNIT – 4	The Cat	- Catharine M.Willson
UNIT – 5	On Letter Writing	- Alpha of the Plough
UNIT – 6	Our Ancestors	- Carl Sagan
UNIT – 7	Our Civilization	- C.E.Foad
UNIT – 8	A Hero on Probation	- B.R. Nanda
UNIT – 9	Dangers of Drug Abuse	- Hardin B. Fones
UNIT – 10	Food	- J.B.S. Haldane

Grammar

UNIT – 11 - Articles-Gerunds-Participles-Infinitives-Modals-Proposition – Tenses.

UNIT – 12 - Direct and Indirect Speech-Transformation of sentences- Active and passive voice.

Composition

UNIT – 13 - Letter writing - Precis writing - Developing hints.

UNIT – 14 - Dialogue writing - Paragraph writing.

References:

1. Sebastian D K, Prose for the Young Reader, Macmillan.
2. Active English Grammar, Ed. by the Board of Editors, Macmillan.
3. Modern English – A Book of Grammar Usage and Composition by N.Krishnaswamy, Macmillan Publishers.

Course Code	Title of the Course
11913	**INTRODUCTION TO PSYCHOLOGY**

UNIT 1: Introduction

Psychology Nature – Scope - Origin of psychology , Definition, Psychology- as a Science, Specialities within Psychology, Psychology and other Disciplines - Perspectives in Psychology:

UNIT 2: Biology of behaviour

Perspectives in Psychology: Psychodynamic - behavioural Humanistic - Bio Psycholog

UNIT 3: Work of Psychologists

Typical behaviour Patterns Brain ad Behaviour Neurons and synapses A guide to the nervous system Association context Behaviour and experience test and right hemisphere functions

UNIT 4: Methods of Assessment In Psychology

Goals of Psychological Enquiry, Introspective method Observation method Experimental method and correlation method

UNIT 5: Methods of Assessment In Psychology

Case Study Method, Clinical Method, Genetic Method, Interview Method, Survey Method, Rating Scales, Checklists, Questionnaires, Psychological Tests, Cross-cultural Method.

UNIT 6: Sensory Process and perception

Sensory channels sensory processes vision learning smell taste skin sense from sensory process the perception process: Attending from perception visual depth perception constancy movement perception individual differences

UNIT 7: Foundations of Behaviour

(Nervous system The neuron the neural impulse the central nervous system the brain location of the brain functions Peripheral nervous system The endocrine system Impart of the functions of endoergic glands and summary 9Ref SK Mangal)

UNIT 8: Heredity and enrolment behaviour

Biological bases behaviour (heredity) nature environment behaviour (environment) Hormones on behaviour Genetics and behaviour sociological bass of behaviour Environment Behaviour Natural internal behaviour (Physical or material environment) and behaviour Man made environment subjective environment How these two affect behaviour Heredity Vs Environment

UNIT 9: Principle of learning

Classical Conditioning Instrumental conditioning cognitive Learning and learning some things are easier to learn than others

UNIT 10: States of consciousness

Nature of Consciousness Brain and consciousness nature of Consciousness normal walking consciousness directed consciousness Flowing consciousness divide consciousness Fantasy and Day dreaming Sleep stages of sleep renoland Non REM Sleep dreams The content of dreams functions and meaning of dreams circadian Rhythms

UNIT 11: Nature of consciousness

Altered states of consciousness characteristics of Altered states consciousness Hypothesis Depersonalization and near death experiences Meditation Altering consciousness with drugs

UNIT 12: Attention in psychology

Attention definition types of attention Overt and covert attention factors influencing attention neural correlates of attention characteristics of attention Assessment of attention , determinants of Attention

UNIT 13: Perception in psychology

Principles of perceptual organization constancies in perception size shape form space movement etc depth perception psychologists life span changes in perception extra sensory perception (ESP) illusion Plasticity of perception

UNIT 14: Aptitudes

meaning and nature of aptitudes - Aptitude ability and achievement - Intelligence - aptitude and Interest measurement of aptitudes and Utility of Aptitudes – Aptitudes tests (ref Sk Mangal)

References

1. Baron, R.A. (1996). Psychology. 3ed. New Delhi: Prentice Hall.
2. Lahey, B. B. (1998). Psychology: An Introduction. New Delhi: Tata Mc Graw Hill.
3. Feldman, R. S. (2002). Understanding Psychology. New Delhi: Tata Mc Graw Hill.
4. Bootzin, R. R., Bower, G. H., Crocker, J., & Hall, E. (1991). Psychology Today. London: Mc Graw Hill.
5. NCERT (2002). Introduction to Psychology- Part-I. New Delhi: NCERT.
6. Balachandran, M. (2003). Psychology for Nursing Students. Thiruvananthapuram: Maanas
7. Parameshwaran, E. G. & Beena, C. (2006). An Invitation to Psychology. Hyderabad: Neelkamall

Course Code	Title of the Course
11914	**Human Developmental**

UNIT 1: Introduction

Introduction – Concept of human Development- Basic Concepts: Growth, Development, Stages Life span development and their characteristics

UNIT 2: Maturation Aspects of development

Maturation Aspects of development, Significant facts, Physical , social cognitive Moral principles of human development

UNIT 3: Factors influencing human

Ecological factors, heredity factors chromosomal abnormalities

UNIT 4: Parental Development and Infancy

Stages of parental development – teratogens and parental involvement New born appearance reflexes assessment and states of infancy

UNIT 5: Physical and Motor development

Cognitive and language development, psychological development- emotions , attachment and temperament

UNIT 6: Physical Development

Patterns of physical growth – the reflexes – structural and systematic changes – growth rate – Aging process – health factors

UNIT 8: Childhood

Early and middle childhood Physical and motor development, cognitive development language development , emotions, play, aggression and altruism

UNIT 9: Intellectual Development

Approaches: Psychometric, Piagetian and Information Processing. Moral Development : Kohlberg's Levels of Morality, Language development and Acquisition of language

UNIT 10: Emotional and self Development

Emotions – The emerging sense of self – Emotions throughout life span development – Temperament Self Concept – socialization – identity crisis in adolescence – relationship with parents, peers, life partner and aging parents

UNIT 11: Puberty and Adolescents

Meaning biological changes , sexual maturation growth spurt, primary and secondary sexual characteristics, responses to physical changes, development of identity, Erikson and Marcia's vies, Adolescents relationship family , Peers Adult society

UNIT 12: Early Adult hood

Early Adulthood- Vocational adjustment foundation of intimate relationship and sexuality Marriage : Marital adjustment and conditions influencing it parenthood adjustment to parenthood

UNIT 13: Middle Adult hood

Psychical development – Occupational adjustment stable and unstable patterns, preparation for retirement psychological changes, coping with mid life crisis, changes in relationships marriage relationship with maturing children ageing parents siblings, grandparent hood , friendship

UNIT 14: Late Adulthood

Primary and secondary ageing, theories of ageing psychological aspects models of coping , models of successful ageing, lifestyle and social issues, work retirement and leisure. Personal relationship with adult children , siblings, great – grandparenthood stages and patterns of grieving

References

1. Hurlock, E. B., (2001) Developmental Psychology, New Delhi: McGraw Hill
2. Srivastava, A. K., (2002) Child Development: An Indian Perspective, New Delhi, NCERT.
3. Diane E. Papalia and Sally Wendkos Olds (1998) Human Development, McGraw Hill, INC
4. Papalia D.E (2004) Human Development 9th Edition New delhi Tata McGrawHill
5. Berk L.c. (2008) child Development New Delhi: prentice Hall of India (Pvt) Ltd

SEMESTER –II

Course Code	Title of the Course
11921A	**PART-I TAMIL-II**

கூறு 1

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

93. சொல் தவிர்ந்த
94. அன்னை
95. அஞ்சுவார்
96. சொல்லக் கேட்டனள்
97. மற்செய்கை
98. மண்கனியப்
99. அழுது ஆர்ந்த

கூறு 2

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

100. பொய் பொதுளும்
101. இன்பு அருந்தி
102. வழுதாயின இன்பு
103. மறம் ஏவினர்

கூறு 3

தேம்பாவணி - காட்சிப்படலம்

பாடல் எண் (ஒவ்வொரு பாடலின் முதல்வரி)

104. மண்ணோர்கள்
105. பொய்யா விதியோய்
106. விடியா இருள்
107. அழுவார் எவரும்

கூறு 4

சிறுகதை - நீலபத்மநாபனின் ‘‘வான வீதியில்’’

கூறு 5

உரைநடை - கம்பன் புறத்திணை - தி.சொக்கலிங்கம்

இலக்கணம் - எழுத்தும் சொல்லும்

கூறு 6

5. முதலெழுத்துகள், சார்பெழுத்துகள்
6. மொழி முதலெழுத்துகள் , மொழி இறுதி எழுத்துகள்

கூறு 7

7. ஒற்றெழுத்து மிகலும் மிகாமையும்,
8. ஆகு பெயர் , அன்மொழித் தொகை.
9. வினா-விடை வகைகள்

கூறு 8

13. தமிழ்ச் சொல்லமைப்பின் சிறப்பு – பெயர் , வினை, இடை, உரி வடிவங்கள் ,
14. பிற மொழிச் சொற்களைத் தமிழில் ஆளும் முறைகள்

கூறு 9

15. அல் வழி, வேற்றுமைப் புணர்ச்சிகள்
16. திணை, பால் , எண் , இட இயைபு.

தமிழ் இலக்கிய வரலாறு

கூறு 10

4. **இக்கால இலக்கிய வகைகள்**
   அ) மரபுக் கவிதை

   ஆ) புதுக் கவிதையின் தோற்றமும் வளர்ச்சியும்

கூறு 11

1. **உரை நடை இலக்கியங்கள் - தோற்றமும் வளர்ச்சியும்**

   அ) கட்டுரை ஆ) சிறுகதை

   இ) புதினம் ஈ) நாடகம்

கூறு 12

4. **இக்கால இலக்கியக் களங்கள்**
   திரைப்படம் , தொலைக்காட்சி, வானொலி, இதழ்கள் தமிழுக்கு ஆற்றி வரும் பணிகள்

கூறு 13

1. **தமிழும் சமயமும் :**
   அ) சைவம் ஆ)வைணவம் இ)சமணம் ஈ)பௌத்தம் உ)இசுலாம்

   ஊ) கிறித்துவம்

கூறு 14

1. **பிற்காலக் காப்பியங்கள் :**

   அ) கம்பராமாயணம் **ஆ)** பெரியபராணம்

2. **இணையம் - பற்றிய செய்திகள்**

Course Code	Title of the Course
11921B	**PART-I : COMMUNICATION SKILLS - II**

Unit – I Code and Content of Communication Skills

Unit – II Stimulus and Response of Communication Skills

Unit – III Effective Speaking Guidelines

Unit – IV Pronunciation Etiquette of Communication Skills

Unit – V Phonetics in Communication Skills

Unit – VI A self Assessment of Communicating Soft Skills

Unit – VII Language Skills –Ability –Skill Selected Need- Learner Centre activities

Unit – VIII Listening Skills –Importance –Types of Listening- Interview Skills

Unit – IX Conversation Skills –Modes

Unit – X Presentation Skills - Preparing –Planning-Presentation

Unit – XI Written Communication –Structure of Effective Sentences –Paragraph

Unit – XII Technical Writing-Creative Writing- Editing and Publishing

Unit – XIII Corporate Communication Skills-Internal –Effective business writing – Letters, Proposals, Resume

Unit – XIV Corporal Communication Skills-External - Press release - Newsletters- Interviewing skills

References:

1. Dutt. Kiranmai & Geeta Rajjevan. Basic Communication Skills. Rev.ed. Foundation Books Pvt.Ltd. Cambridge House, New Delhi 2006.
2. Bill R. Swetmon. Communication Skills for the 21st Century. Chennai: Eswar Press. First South Asian Edition 2006.
3. Glass. Lillian. Talk to Win. New York: Perigee Books,1987.
4. Pease. Alan. Signals: How to Use Body Language for Power, Success and Love, New York: Bantam Books, 1981.
5. Walters. Lilly. Secrets of Successful Speakers. New York: McGraw-Hill, Inc., 1993.
6. Mandal. S.K. How to Succeed in Group Discussions & Personal Interviews. Mumbai: JAICO Publishing House.
7. Rogoff. Leonard and Ballenger. Grady. Office Guide to Business Letters, Memos & Reports. New York: Macmillan, 1994.
8. Krishna Mohan & Meera Banerjee, Developing Communication Skills, 2005.
9. Geetha Nagaraj, Write to Communicate, 2004.
10. Wren & Martin, English Grammar and Composition, 2002.
11. Rajendra Pal and Kortahalli J S, Essentials of Business Communication.

Course Code	Title of the Course
11922	**PART-II : ENGLISH PAPER - II**

Poetry

UNIT – 1 Sonnet - William Shakespeare UNIT – 2 Lines Composed upon Westminster Bridge -William Wordsworth

UNIT – 3 Grecian Urn - John Keats (1795-1827)

UNIT – 4 Andrea Del Sarto - Robert Browning (1812-1889)

UNIT – 5 The Road Not Taken - Robert Frost (1874-1963) UNIT – 6 Strange Meeting - Wilfred Owen (1813-1918)

UNIT – 7 Gitanjali - Rabindranath Tagore (1861-1946)

UNIT – 8 The Coromandel Fishers - Sarojini Naidu

UNIT – 9 The Express - Stephen Spender UNIT – 10 Shakespeare : The Merchant of Venice

Language Use:

UNIT – 11 Essay writing UNIT – 12 Note Making UNIT – 13 Report writing UNIT – 14 Comprehension

References:

1. The Golden Quill, P.K. Seshadri, Macmillan.
2. The Merchant of Venice, Shakespeare. (Any overseas edition).
3. Active English Grammar, Ed. by the Board of Editors, Macmillan.
4. Modern English – A Book of Grammar Usage and Composition by N.Krishnaswamy, Macmillan Publishers.

Course Code	Title of the Course
I11923	**SOCIAL PSYCHOLOGY**

UNIT 1: Social Psychology

Scope and Methods of Social Psychology: Field of social psychology – Relation of psychology – Sociology and Anthropology – Practical Vs. Scientific knowledge – Social interaction and social influence and some variables under study – Society and culture – Interpersonal response traits –

UNIT 2: Methods of social psychology

Methods of social psychology – Observation -experiment - correlational method – Situational observation. Development of Social Behavior: Social behavior of Animal – Social life among insects – Territoriality, Dominance and status – Mammals – Social development in children – Infant social relation in early childhood.

UNIT 3: Individual Society and Culture*:*

The individual – Society – Social reality – Socio-economic status – Closed society and open society and culture primary and secondary groups – Rural and urban language – Child rearing practices – Values.

UNIT 4: Perceptional Process

Social Perception: Perceptional process in infancy – Cognitive theories – personal factors- influence of wants and goals.

Person Perception: Nature of person perception – Impression formation – Halo effect – Stereotype – Order effect – Attribution process – Accuracy of judgements – Two traditions of person perception in India.

UNIT 5: Language and Communication:

Communication behavior – Language and communication-signs, signals and symbols –
Language a cultural product – Meaning of meaning – Language and socialization.

UNIT 6: Sociometry

Interpersonal Attraction: Sociometry – Theories of interpersonal attraction – Attraction and social interaction.

Social Motivation: Behavior is biogenic and sociogenic – Social drives – Need for affection, dependence, aggression, self-assertion – Achievement motivation – Maslow's theory – Social incentives – Rewards and punishments praise and reproof – Competition and co-operation.

UNIT 7: Social Attitudes

Definition – Components – Beliefs, attitudes and values – formation of attitudes – Measurement – Method of social distance – Thurston's method – Likert's method – Osgood's method of semantic differential – Processes of attitude change – Persuation radicalism and conservatism – Diffusion of innovations – Resistance to change – Attitude and behavior – Theories – Indian studies – Gandhian methods.

UNIT 8: Groups Formation of Groups

Kinds of groups – Togetherness – Small groups – properties of group – Formal and informal groups – Group structure – Group norms – Experimental studies – Village factions – Group effectiveness, cohesiveness – Reference group – Group morale – Group dynamics.

UNIT 9: Conformity and Deviation*:*

Conformity – Deviation – Social mobility – Crime and delinquency.

UNIT 10: Leadership and Social Power

Leadership and Social Power: Leadership and dominance – Institutional leaders – Dominant leaders – Persuasive leaders – Expert – Experimental studies – Traits – situational test – Factorial studies – Democratic Vs. authoritarian leaders – Leaders and the led – Theories of leadership – Emerging pattern of rural leadership in India – Social Power – Bases of power – Exchange theory.

UNIT 11: Public Opinion

Propaganda and Prejudice: Public opinion – Meaning – Process – Steps in opinion formation – Role of leaders – Gauging public opinion – Survey method – Propaganda – Its relation to education – Language and propaganda – Role of suggestion – Techniques – Prejudice – Characteristics – Growth – Forms of hostile action – Methods to reduce prejudice.

UNIT 12: Collective Behavior

Mass society – Audience – Mob behavior – Kinds of mobs – Theories of crowd behavior
– Rumours and mass hysteria, fads and crazes – Social movements.

UNIT 13: Social Change

Definition – Psychological processes involved – Barriers to change – Attitude toward innovation – Values and norms – Problems of motivation.

UNIT 14: Social Conflicts and their resolution*:*

Tensions and conflicts – Various problems in Indian society leading to conflicts –
Gandhian techniques resolution.National Integration: Characteristics of a nation – Barriers to the growth of integration –Religion not a factor – Role of social learning.

References

1. Baron R A, and Byrne D, Social Psychology, 7th edn, New Delhi: Prentice Hall of India Private Ltd., 1999.

2. Feldman R S, Social Psychology, 2nd Edn, New Jersey: Prentice Hall Inc., 1998.

3. Krech D, Crutchfield S and Ballachey E L, Individual in Society: Text Book of Social Psychology, Tokyo: Hill Kiogakuha, Ltd.,1962.

4. Kuppuswamy B, An Introduction to Psychology, Bombay, Media Promoters & Publishers Pvt. Ltd, 1980.

Course Code	Title of the Course
11924	**INDUSTRIAL PSYCHOLOGY**

UNIT 1: Industrial psychology

Industrial psychology: Definition – Nature – Scope – Objectives – Areas covered in industrial psychology

UNIT 2: Contribution of industrial psychology – Basic Concepts – Foundations of industrial psychology.

UNIT 3: Individual Behaviour

Individual Behaviour: Individual Differences: Meaning – Nature – Dimensions – Values of individual differences in industries

UNIT 4: Factors influencing Individual Behaviour

Factors influencing Individual Behaviour: Personality – Learning – Ability – Perception – Attitude – Job Satisfaction: Factors influencing job satisfaction.

UNIT 5: Morale

Morale: Morale: Meaning – Types – Aspects – Characteristics of high and low morale – Essential requirements for high morale – Psychological requirements for high morale.

UNIT 6: Motivation*:*

Meaning of Motivation – Importance – Types – Motivation in industry – Monetary and non-monetary incentives. Fatigue, Boredom and Monotony: Meaning – Causes – Remedies.

UNIT 7: Group Behaviour

Group Behaviour: Groups: Meaning – Definition of Group - Types – Group dynamics – Group norms – Group cohesiveness –Advantages of group

UNIT 8: Leadership

Leadership: Definition of Leadership- Meaning – Types – Importance of leadership – style of leadership- Qualities of a successful leader.

UNIT 9: Leadership Skill

Organisation of culture- Role model – Behaviour Adjustment- team Work-Performance- Creativity –Interpersonal Intra personal skill-cultural flexibility –communication –Self assessment of learning

UNIT 10:Stress:

Meaning – Sources of stress – Causes of stress – Managing stress – Nature of stress- Stress management –Ethical dilemmas- Work Environment Factors

UNIT11: Maladjusted Worker

Maladjusted Worker: Meaning – Causes of Maladjustment –Readjustment. Of Maladjustment worker

UNIT 12: Counselling

Meaning – Significance – Types and process – Employee health, safety and security – Industrial accidents – Accident proneness – Prevention of accidents – Quality of Work

UNIT 13 : Life

Meaning – Importance – Factors affecting quality of work life

UNIT 14: Work Motivation

Attitude of working culture –Behaviourism -Work Place –Psychological Health

References

1. Ghosh P K & Ghorpade M B, Industrial Psychology, Himalaya Pub, Mumbai.
2. Ramnath Sharma, Industrial Psychology, Rajhans Prakashan Mandir, Meerut.
3. Arnold John et-al, Work Psychology: Understanding human behaviour in the work place, Macmillan, Delhi.

SEMESTER –III

Course Code	Title of the Course
11931A	**PART -1 TAMIL -III**

கூறு 1: பத்துப்பாட்டு – முல்லைப்பாட்டு

கூறு 2: எட்டுத்தொகை – ஐங்குறுநூறு

கூறு 3: கபிலர் - குறிஞ்சித்திணை

கூறு 4: மஞ்ஞைப்பத்து – முதல் மூன்று பாடல்கள்

கூறு 5: குறுந்தொகை – பரணர் பாடல்கள் பா. எண். 19, 24, 36, 128, 399

கூறு 6: நற்றிணை – பெருங்குன்றூர்கிழார் - பா. எண். 5
பெருவழுதியார் - பா. எண். 55
பெருங்கௌசிகனார் - பா. எண். 139

கூறு 7: நற்றிணை – கருவூர்க்கோசிகனார் - பா. எண். 214
உலோச்சனார் - பா. எண் 249

கூறு 8: அகநானூறு – சேந்தம்பூதனார் பாடல்கள் பா.எண். 84, 207

கூறு 9: புறநானூறு – மறோக்கத்து நப்பசலையார் பாடல்கள்
பா. எண். 37, 39, 126, 226, 280

கூறு 10: பதினெண் கீழ்க்கணக்கு – திருக்குறள் - வாழ்க்கைத் துணை நலம் (6), அறிவுடைமை (43), பிரிவாற்றாமை (116)

கூறு 11: நான்மணிக்கடிகை – எள்ளற்க (3), பறைபடவாழா (4),

கூறு 12: நான்மணிக்கடிகை - மண்ணயறிப (5),கள்ளிவயிற்றில் (6), கல்லிற்பிறக்கும்(7)

கூறு 13: நாடகம் - இராசராசசோழன் - அரு. இராமநாதன்

கூறு 14: நாவல் - சுவடுகள் - இரா. பாலசுப்பிரமணியன், சத்யா வெளியீடு, மதுரை

Course Code	Title of the Course
11931B	**PART-I : HUMAN SKILLS DEVELOPMENT - I**

Unit – I Human Skills –Developing skills-Types

Unit – II Mind-Levels of functions Habits-Meaning-Types-Merits of good habits - Interpersonal Relationship- Features- Interpersonal Behaviour

Unit – III Thinking ahead- Significance of thinking ahead

Unit – IV Developing Personality-Meaning -Need- Factors influencing personality, Ways of developing personality -Building positive personality

Unit – V Self-concept-Self-esteem-Meaning-Importance - Self- efficacy-Self- acceptance-Meaning-Importance - Etiquette-Meaning-Etiquettes in using mobile, telephones-Dais Etiquette

Unit – VI Goal-setting Skills-Meaning-Types-Importance-

Unit – VII Decision-making skills-Meaning-Types-Steps in decision-making

Unit – VIII Negotiating Skills-Styles-Structure-Creating negotiation-Competitive Negotiation

Unit – IX Attitudes-Meaning-Types-Importance-Developing positive attitudes

Unit – X **C**oping with Change-Meaning-Characteristics-Importance of change Resistance to change-Dealing with change

Unit – XI Leadership-Meaning-Importance-Characteristics-Styles-

Unit – XII Human Relations Skill-Need-Canons of good human relations

Unit – XIII **C**ounselling-Meaning-Importance-Forms- Conflicts-Meaning-Types-Causes-Effects-Managements of conflicts

Unit – XIV Stress-Meaning-Types-Causes-Effects-Managing the stress - Anger-Meaning-Causes-Consequences-Anger Management

References:

1. Les Giblin, Skill with People, 1995.
2. Shiv Khera, You Can Win, 2002.
3. Christian H Godefroy, Mind Power.
4. Dale Carnegie, How to Enjoy Your Life and Your Job, 1985.
5. Natalie H Rogers, How to Speak without Fear, 1982.
6. Dale Carnegie, How to Develop Self-Confidence and Influence People by Public Speaking.

Course Code	Title of the Course
11932	**PART-II : ENGLISH PAPER - III**

Short Stories

UNIT – 1	A Hero	- R.K. Narayanan
UNIT – 2	The Diamond Necklace	- Guy de Maupassant
UNIT – 3	The Verger	- Somerset Maugham
UNIT – 4	The Postmaster	- Rabindranath Tagore

One Act Plays

UNIT – 5	The Proposal	- Anton Chekhou
UNIT – 6	The Boy Comes Home	- A.A. Milne
UNIT – 7	The Silver Idol	- James R. Waugh
UNIT – 8	Progress	- St. John Ervine
UNIT – 9	The Pie and the Tart	- Huge Chesterman
UNIT – 10	Reunion	- W.st. Joh Tayleur
UNIT – 11	A kind of Justice	- Margaret Wood
UNIT – 12	The Refugee	- Asif Currimbhoy

Grammar

UNIT – 13 Parts of speech-Noun- Pronoun- Adjective
Degrees of Comparison- Verb- Adverb

Composition

UNIT – 14 Agenda- Minutes- Notice- Descriptive Writing

References:

1. Aroma, Ed. by the Board of Editors, Publishers- New Century Book House, Chennai.
2. Six Short Stories, Ed. by the Board of Editors, Harrows Publications, Chennai. One Act Plays, Ed. by the Board of Editors, Harrows Publications, Chennai.
3. Modern English – A Book of Grammar Usage and Composition by N.Krishnaswamy, Macmillan Publishers.
4. English for Communication, Ed. by the Board of Editors, Harrows Publications, Chennai.

Course Code	Title of the Course
11933	**Psychological Statistics**

UNIT 1: Statistics

Definitions of statistics Importance of Statics in Psychological and Research Collections of Data Primary and Secondary-classification of Data Formation of Frequency distribution

UNIT 2: Diagrammatic and Graphical

Diagrammatic and Graphical representation of Statistical Data Simple Multiple sub divided, percentage Bar Diagram and Pie diagram – Histogram Frequency polygon Frequency Curve, Ogives

UNIT 3: Measures of Central Tendency

Measures of Central Tendency Mean Median Mode Geometric Mean and Harmonic Mean

UNIT 4 : Measure of Dispersion

Measure of Dispersion range Quartile Deviation, Mean Deviation (about mean) Standard Deviation and Co efficient of variance –concept of Skeweness –Karl Pearson and Bowley's coefficient of Skeweness –Simple Problems- Kurtosis 9concpet only)

UNIT 5: Correlation

Correlation-Scatter diagram – Karl Pearson co efficient of correlation Spearmen's Rank correlation – regression equation –Properties of regressive coefficient simple problems

UNIT 6: Concept of probability

Concept of probability basic Definition- mathematical and statistical approach – Addition and multiplication Theorems(without proof) simple Problems

UNIT 7 : Normal Distribution

Concept of Probability, Meaning, Importance and Characteristics of Normal Probability Curve, Deviations from Normal Probability Curve

UNIT 8: Binomial Distributions

Skewness and Kurtosis, Applications of NormalProbability Curve, Binomial Distributions.

UNIT 9: Analysis Of Variance

Purpose and Assumptions of Analysis of Variance, One way and Two way Analysis of Variance

UNIT 10: Non-Parametric Tests

Nature and Assumptions, Distribution Free Statistics, Chi-square, Contingency Co- efficient, Median and Sign test, Friedman test

UNIT 11: Preparation of Data for Computer Analysis

Familiarisation with Software Packages for Statistical Analysis and their Applications.

UNIT 12: Inferential Statistics

Introduction to Inferential Statistics and Hypothesis Testing about the Difference between Two Independent Means: The meaning of Statistical Inference and Hypothesis Testing; Hypothesis Testing about the difference between Two Independent means; Null and the Alternative Hypotheses;

UNIT 13: The Random Sampling Distribution

The Random Sampling Distribution of the Difference between Two Sample Means; Properties of the Sampling Distribution of the Difference between Means; Choice of HA: One-Tailed and Two-Tailed Tests;

UNIT 14: Interpreting the Results of Hypothesis Testing

A Statistically Significant Difference versus a Practically Important Difference; Errors in Hypothesis Testing; Power of a Test; Levels of Significance versus p-Values.

References

1. Garrett, H. E. (2004). Statistics in Psychology and Education, 6th ed. New Delhi: Paragon InternationalPublishers.
2. Mangal, S. K. (2004). Statistics in Psychology and Education. 2nd ed. New Delhi: Prentice Hall.
3. Guilford, J. P., & Fruchter. (1987). Fundamental Statistics in Psychology and Education. 6th ed. Singapore:McGraw Hill.
4. Girija, M., Sasikala, L., & Girija. (2004). Introduction to Statistics.1st ed. New Delhi: Vrinda Publications.
5. Bhandarkar, K. M. (2006). Statistics in Education. 1st ed. Hyderabad: Neelkamal.
6. Kumar, P. (2006). Psychological Statistics. 1st ed. Jaipur: Aavishkar Publishers.

Course Code	Title of the Course
11934	**Psychopathology**

UNIT 1: Abnormal Behaviour

Meaning- Historical conceptions of Abnormal behaviour; Biological viewpoint, Psychosocial viewpoint and Sociocultural viewpoint

UNIT 2: Anxiety Disorders: panic disorders & agoraphobia: Specific phobias, Social phobia; Generalized anxiety disorder; Obsessive compulsive disorders

UNIT 3: Somatofoim & Dissociative disorders

Somatization disorder, hypochondriasis, Pam disorder, conversion disorder; Dissociative Amnesia, dissociative fugue, dissociative trance disorder, dissociative identity disorder, multiple personality disorder

UNIT 4: Personality Disorders

Clinical Features, Types, Causal factors in Personality disorder, Treatment: Anti-Social (psychopathic) Personality - clinical picture, casual factors & treatment

UNIT 5: Mental Retardation

Definition of mental retardation, levels of mental retardation, Causes - biological, psychosocial deprivation Types of Mental Retardation,

UNIT-6 Mental Illness: Signs, symptoms and syndromes. Causes of abnormal behaviour: biological, psychological, social, and cultural factors.

UNIT-7 Organic Mental Disorders

Treatment Organic mental disorders - delirium, dementias associated with Age - Alzheimer's disorder, causal factors & treatment

UNIT 8 : Conversion and dissociative disorder

UNIT 9: Psychological models of psychopathology
Biological, psychodynamic, behavioural, cognitivebehavioural.

UNIT 10:Psychotic Disorder

Symptoms, types, etiology, and treatment: Schizophrenia and delusional disorder.

UNIT 11: Classification systems in psychopathology

ICD-10 and DSM–5; Approaches to psychopathology: Biological, psychodynamic, behavioral, cognitive, and socio-cultural, social constructionist.

UNIT 12: Trauma and stress related disorders

PTSD, Dissociative disorders: Dissociative identity disorders, derealization disorders.

UNIT 13: Eating and sleep disorders

Anorexia nervosa, Bulmia nervosa, Insomnia and Narcolepsy.

UNIT 14: Schizophrenia spectrum and other psychotic disorders

Schizophrenia, delusional disorder, brief psychotic disorder. Bipolar and related disorder: Bipolar I, &II, cyclothymic disorder. Depressive disorder: Disruptive mood dysregularities, major depressive disorder.

References

1. Barlow David H. & Durand, V.Mark (1995) Abnormal Psychology, Brooks / Cole Publishing Co.
2. Carson, Robert, Butcher, James V., Coleman, James (1988): Abnormal Psychology and Modern Life, VIII edition, Scott, Frismand & Co.
3. Lamm, A. Introduction to Psychopathology. New York: Sage
4. Buss, A.H. (1999). Psychopathology. New York: John Wiley
5. Carson, R.C., Butcher, J.N. & Mineka, S. (2010). Abnormal Psychology and Modern Life. Pearson Education, Inc. and Dorling Kindersley Publication Inc.
6. Sarason, G.I. & Serason, R.V. (2007). Abnormal Psychology: The Problem of Maladaptive Behaviour (II Edition). Pearson Education, Inc. and Dorling Kindersley Publication Inc.
7. Singh , A.K. Adhunik Asamanya Manovigyan. N. Delhi: Motilal Banarasidas
8. Krochin, S.J. (2004). Modern Clinical Psychology. CBS Publishsers abd Distributors Pvt. Ltd.
9. American Psychiatric Association (2013). Diagnostic and statistical manual of mental disorders:DSM-5. American Psychiatric Pub.
10. Barlow, D. H. & Durand, V. M. (1999).Abnormal psychology: An integrative approach (2nd Ed). Pacific Grove: Brooks/Cole.
11. Buss. A. H. (1999). Psychopathology. New York: John Wiley.
12. Butcher, J.N. (2014). Abnormal Psychology. New Delhi: Pearson Education
13. Carson, R. C., Butcher, J. N., &Mineka, S. (2001). Abnormal psychology and modern life (11thEd.). New York. Allyn and Bacon
14. Kaplan, H. J., &Sadock, B. J. (2004).Synopsis of comprehensive textbook of psychiatry (10thEd.). Baltimore: Williams &Wlkins.
15. Sarason. I. G. &Sarason, B. R. (2006). Abnormal psychology. (11 thEd). Delhi: Prentice Hall India.

SEMESTER –IV

Course Code	Title of the Course
11941A	**PART 1 TAMIL IV**

கூறு 1: செய்யுள் உறுப்புகள் - யாப்பு - எழுத்து, அசை, சீர்,

கூறு 2: செய்யுள் உறுப்புகள் - யாப்பு - தளை, அடி, தொடை

கூறு 3: வெண்பா, ஆசிரியப்பா, கலிப்பா, வஞ்சிப்பா,

கூறு 4: புதிய யாப்பு வடிவங்கள் - சிந்து, கண்ணி, கீர்த்தனை

கூறு 5: புதுக்கவிதையில் குறியீடு – படிமம்.

கூறு 6: அகப்பொருள் - புறப்பொருள் - ஐந்திணை விளக்கம்

கூறு 7: அகப்பொருள் துறைகள் - வரைவு கடாதல், அறத்தொடு நிற்றல், உடன்போக்கு

கூறு 8: புறப்பொருள் துறைகள் - வஞ்சினக்காஞ்சி, கையறுநிலை, செவியறிவுறூஉ

கூறு 9: அணி இலக்கணம் - உவமை, உருவகம், வேற்றுமை, பிறிது மொழிதல், தற்குறிப்பேற்றம், சிலேடை, பின்வருநிலை.

கூறு 10: நிறுத்தல் குறிகள்.

கூறு 11: தொல்காப்பியம் - சங்கஇலக்கியம் - எட்டுத்தொகை, பத்துப்பாட்டு,

கூறு 12: பதினெண்கீழ்க்கணக்கு.

கூறு 13: ஐம்பெருங்காப்பியங்கள் - பிற்காலக் காப்பியங்கள் - கம்பராமாயணம் - பெரியபுராணம்.

கூறு 14: இக்காலக் காப்பியங்கள் - பாரதியின் பாஞ்சாலி சபதம் - பாரதிதாசனின் பாண்டியன் பரிசு - கண்ணதாசனின் இயேசு காவியம் , சிற்பியின் - மௌன மயக்கங்கள்.

Course Code	Title of the Course
11941B	**PART-I : HUMAN SKILLS DEVELOPMENT - II**

Unit – I Guidance & Counselling – Role of Counsellor - Importance and Techniques of counselling

Unit – II Managerial skill- Need – Importance

Unit – III Human relational skills-Communication-Attention

Unit – IV Conceptual skills-Meaning-Importance

Unit – V Technical skills-Techniques-Practices-Tools-Procedures

Unit – VI Presentation skills-Planning-Preparation-Delivery

Unit – VII Organization skills-Meaning-Nature-Importance-Types

Unit – VIII Multi-Tasking skills Responsibilities-Causes

Unit – IX Leader- Qualities of a good leader

Unit – X Understanding Skills -Human systems: Individual, Group, organization, and their major interactions

Unit – XI Understanding Skills -Human systems: Community and Society, and their major interactions

Unit – XII Problem solving skills – Handling –Facing - Importance

Unit – XIII Cooperative Learning Skills

Unit – XIV Making Social Responsibilities-Causes

References:

1. Les Giblin, Skill with People, 1995.
2. Shiv Khera, You Can Win, 2002.
3. Christian H Godefroy, Mind Power.
4. Dale Carnegie, How to Enjoy Your Life and Your Job, 1985.
5. Natalie H Rogers, How to Speak without Fear, 1982.
6. Dale Carnegie, How to Develop Self-Confidence and Influence People by Public Speaking.

Course Code	Title of the Course
11942	**PART-II : ENGLISH PAPER - IV**

Short Stories

UNIT – 1 Lalajee - Jim Corbelt

UNIT – 2 A Day's Wait - Hemmingway

UNIT – 3 Two old Men - Leo

Tolstoy UNIT –4 Little

Girls wiser than - Men Tolstoy

UNIT – 5 Boy who Wanted more Cheese - William Elliot Griffir

Drama

UNIT – 6 Pygmalion - G.B. Shaw

Fiction

UNIT – 7 Swami and Friends - R.K. Narayanan

Tales from Shakespeare

UNIT – 8 - The Merchant of

Venice UNIT – 9 - Romeo and Juliet

UNIT – 10 - The Winter's Tale

Biographies

UNIT – 11 - Martin-Luther king - R.N. Roy

UNIT – 12 - Nehru - A.J.

Toynbee

Grammar

UNIT – 13 - Concord- Phrases and Clauses-Question Tag

Composition

UNIT – 14 - Expansion of Proverbs

- Group Discussion

- Conversation (Apologizing, Requesting, Thanking)

References:

1. Sizzlers, by the Board of Editors, Publishers-:Manimekala Publishing House, Madurai.
2. Pygmalion – G.B. Shaw
3. Swami and Friends – R.K. Narayan
4. Tales from Shakespeare Ed. by the Board of Editors, Harrows Publications, Chennai.
5. Modern English – A Book of Grammar Usage and Composition by N.Krishnaswamy, Macmillan Publishers.

Course Code	Title of the Course
11943	**Cross Cultural Psychology**

UNIT 1: Introduction

Definition of culture Approaches to culture Interface between psychology and culture Pancultural principles verses culture

UNIT 2 : specific Differences of culture

Specific Differences: Etics and Emics Methods of understanding culture, scope of cultural psychology.

UNIT 3: Culture and perception

Culture and perception – perception and experience cultural influence on visual perception Culture and Cognition culture,

UNIT 4: Categorization of culture

Categorization and concept formation, culture and memory, Culture and consciousness- culture and dreams, culture and time, culture and perception of pain

UNIT 5: Culture and intelligence

Culture and intelligence – cross cultural research on intelligence, cultural differences in the meaning and concept of intelligence

UNIT 6: Culture and health

Difference in the definitions of health, sociocultural influence on physical health and medical disease process Cultural influences on attitudes and beliefs related to health and diseases Culture,

UNIT 7: Self culture and personality

Self culture and personality Culture and gender –culture and gender stereotype, How does culture influence gender ?

UNIT 8: Culture in India

Indigenous healing traditions in India. History of mental health practices in India Current status of mental health practice in India Key issues in the theory and practice of mental health in India

UNIT 9: Eastern perspectives in Psychology (15 hrs)

The concept of mind in Samkhya and Nyaya-Vaiseshika philosophy Bhagavad Gita and mental

health Concepts of health and illness in Ayurveda Personality –implications from Yoga Indian perspective on emotion

UNIT 10: Philosophical Antecedents

Cultural Psychology definition and meaning; Platonism, positivism, relativism, historicism; Methods of cross-cultural research; Cultural Narratives; Methodological Issues

UNIT 11: Socialization & Development; Cultural Learning

Attribution; Notions of Individuality and relatedness; Intergroup Behaviour Cultural Ideals; Acculturation, Enculturation and Socialization; Child Rearing, parenting and family in Indian setting; Play, use of symbols and communication

UNIT 12: Cognitive and Cultural factors

Notion of g and s; Genetic epistemology; Cultural factors in memory and information processing; Ability testing;

UNIT 13: Intellectual Processes

Emotional Intelligence; Metacognition; Theories of mind and metacognition

UNIT 14: Suggestive Assignment

Report on cultural influences on parenting in personal local setting; identification of cultural ideals

References

1. Matsumoto,D and Juang,L(2004).Culture and Psychology. Third edition. UNITed States: Thomson wadsworth .
2. Misra,G and Mohanthy .A.k(2002) .Perspectives on indigenous Psychology .
3. Valsiner,J(2007).Culture in minds and societies . Foundations of cultural Psychology New Delhi :Sage publications
4. Misra ,G (2003).implications of culture for psychological knowledge .in J.W .Berry R.C Mishra and R.C Tripathi.(Eds).Psychology in human and social development .New Delhi : Sage publications
5. Dalal,A.K and Misra ,G (eds) (2002).New directions in Indian Psychology . New delhi: Sage publications NewDelhi.concept publishing company
6. Basu, A.M. (2004). A new knowledge of madness-nineteenth century Asylum Psychiatry in Bengal. Indian Journal of History of Science, 39 (3), 247-277.
7. Basu, A.M. (2004). Emergence of a marginal science in a colonial city: Reading psychiatry

in Bengali periodicals. The Indian Economic and Social History Review, 41 (2), 103-141.

8. Basu, A.M. (2005). Historicizing Indian Psychiatry. Indian Journal of Psychiatry. 47, 126-129.

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19. Goldberger, N.R. & Veroff, J.B. (Eds.) (1995). The culture and psychology reader. NY: New York University Press.

20. Kakar, S. (1997). Culture and Psyche. Delhi: Oxford University Press.

21. Greenfield, P. M. (1997). You can't take it with you: Why ability assessments don't cross cultures. American Psychologist, 52, 1115-1124. (CP

Course Code	Title of the Course
11944	**EDUCATIONAL PSYCHOLOGY**

UNIT 1: Introduction

Definition, Historical Background, Role and Scope of Educational Psychology, Effective Teaching Methods.

UNIT 2: Educational Psychology

Educational Psychology: Meaning - Definition - Nature and scope - Relation between psychology and educations

UNIT 3 :Nature of educational psychology

Scope of educational psychology - Methods of educational psychology

UNIT 4: Hunan Development - Growth and development - Principles of developemnt- Theories of growth and development - Psycho-analytic theory

UNIT 5: Erickson's theory of psychosocial development

Kohlberg's theory of moral - Jean Piaget's theory of development - Stages of Development of adolescent - Educational measures

UNIT 6: Nature of Learning

Nature of learning - Learning and maturation - Factors affecting learning- Theories of learning Plateau of learning - Individual Differences – Memory- Remembering and forgetting - Motivation.

UNIT 7: Intelligence

Definition of intelligence - Types of intelligence - Factor theories of intelligence - Measuring intelligence - Classification of intelligence tests Intelligence test in India - Creativity - Theories of Creativity - Creativity and education

UNIT 8: Personality

Characteristics of Personality - Theories of Personality -Assessment of personality - The Rorschach Ink Blot test - Thematic Apperception Test (TAT) - Children's apperception test (CAT) - Psychodrama - Draw a man test - Doll play - Weakness of personality tests.

UNIT 9: Exceptional Children

Types of Exceptional Children - Need for Special Education - Gifted Children - The Mentally Retarded - Hearing Impairment -Blindness - The Orthopedically handicapped - Educational Guidance - Vocational guidance - Guidance personnel - Problems of educational guidance

UNIT-10: Student Development and Diversity

General Principles of Development, Importance of Development in Education, Brain and Cognitive Development, Language Development, Importance of Personal, Social and Emotional Development, Importance of Culture, CommUNITy and Gender.

UNIT 11: Understanding Learning Process

Importance of Behavioural Approaches to Learning, Importance of Information Processing Approach, Complex Cognitive Processes, Importance of Social, Cognitive and Constructivistic

Views of Learning

UNIT 12: Motivational and learning :

Nature and conditions of learning – Laws of effect and readiness – Role of transfer in education – Motivating the students – Achievement motivation – Extrinsic reward – Punishment and failure – Effects of punishment in the classroom

UNIT 13: Classroom Management: Designing Physical Environment of the Classroom, Creating Positive Environment for Learning, Classroom Assessment.

UNIT 14: Special education:

Readiness for learning: The role of maturation in learning – Special education for exceptional children – Intellectually gifted – Mentally retarded – Culturally deprived – Physically handicapped.

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2. Santrock, J. W. (2004). Educational Psychology. 2nded. International Edition: McGraw Hill.
3. Ormrod, J. E. (2000). Educational Psychology: Developing Learners. ed. New Jersey: Merrill.
4. Mangal S K, Advanced Educational Psychology.
5. Charles E Skinner, Educational Psychology, Prentice Hall of India.
6. Chatterjee S K, Advanced Educational Psychology, Books and Allied (1).
7. Kakkar S B, Educational Psychology, Prentice Hall of India (P) Ltd.
8. John W Santrock, Educational Psychology, Tata McGraw Hill.
9. Rather A R, Creativity: Its Recognition and Development, Sarup and Son! Course Material prepared by — Dr. J. Sujathamalini Lecturer, Alagappa University College of Education, Karaikudi.

Semester –V

Course Code	Title of the Course
11951	**Human Resource Management**

UNIT 1 **Introduction to Human Resource Management**: Definition – Objectives and functions

UNIT 2 Role and structure of Human Resource Function in organizations- Challenges in Human Resource Management

UNIT 3 Approaches to Human Resource Management

UNIT 4 **Human Resource Planning**: Personnel Policy - Characteristics

UNIT 5 **HR Planning**: Need, Scope and Process – Job analysis – Job description – Job specification- Succession Planning.

UNIT 6 **Recruitment and Selection Process**: Sources of recruitment- internal Vs. External; Domestic Vs. Global sources- Selection process

UNIT 7 Placement and Induction – Training and Development: Need Assessment- Training methods for Operatives and Supervisors

UNIT 8 Executive development: Need and Programs.

UNIT 9 **Employee Compensation and Retention**: Wages and Salary Administration – Bonus – Incentives – Fringe Benefits –Flexi systems

UNIT 10 Sweat Equity- Job evaluation systems –Promotion – Demotions – Transfers- Labour Attrition: Causes and Consequences

UNIT 11 **Employee Retention**: Need and Programs.

UNIT 12 Employee Welfare, Separation: Welfare and safety – Accident prevention – Employee Grievances and their Redressal – Industrial Relations

UNIT 13 Trade Unions - Multiplicity of Trade Unions – Workers Participation in Management- Separation: Need and Methods.

UNIT 14 Human Resource Information System- Personnel Records/ Reports- e- Record on Employees – Personnel research and personnel audit – Objectives – Scope and importance.

References

1. Mathis and Jackson, Human Resource Management, South-Western College, 2004.
2. Nkomo, Fottler and McAfee, Human Resource Management, South-Western College, 2007.
3. R. Wayne Mondy, Human Resource Management, Prentice Hall, 2011.
4. Venkataraman & Srivastava, Personnel Management & Human Resources
5. Arun Monappa, Industrial Relations
6. Yodder & Standohar, Personnel Management & Industrial Relations
7. Edwin B. Flippo, Personnel Management , McGraw-Hill, 1984
8. Pigors and Myers, Personnel Administration
9. R.S. Dwivedi, Manpower Management
10. Lynton & Pareek, Training and Development, Vistaar Publications, 1990.

Course Code	Title of the Course
11952	**Environmental psychology**

UNIT 1: Environmental Psychology

Introduction – Nature – Characteristics – Research – Research methods in environmental psychology – Data collection methods.

UNIT 2: Environmental Perception

Introduction –Environmental perception conventional approaches to perception – Environmental perception – Movement, habituation and the perception of change – Perception of movement – Habituation or adaptation – Perception of change – Perception of natural cognition and Cognitive mapping

UNIT 3: Cognition and Attitudes

Components of cognitive maps –Familiarity, gender and socioeconomic class – Functions of cognitive maps – Memory and cognitive maps – Cognitive organization of the environment –

UNIT 4 : Assessing the Scenic Environment

Landscape assessment, Aesthetics and preference – Descriptive systems of scenic value – Physical perceptual approach to scenic value – Forming attitudes toward the environment – Classical conditioning of attitudes – Instrumental conditioning of attitudes social learning of attitudes – Measuring attitudes toward the environment: The search for the PEQI – Changing attitudes toward the environment: a prelude – Environmental attitudes predict environmental behavior.

UNIT 5: Theories of Environment

Behavior Relationships: Introduction – Nature and function of theory in environmental psychology – Environment–Behavior theories: Fledgling theories in a fledgling field – Arousal approach – Environmental load approach – Under stimulation approach – adoption level theory: Optimal stimulation – behavior constraint approach – Barker's Ecological psychology

UNIT 6: Environmental Stress

Environmental stress approach Stress – Characteristics of stressors – Appraisal – Characteristics of the stress response –Natural disaster – Characteristics of natural disasters – Effects of natural disaster

UNIT 7: Environmental Noise pollution

Technological catastrophe – Characteristics of technological catastrophe noise – Defining,

measuring, and perceiving noise – Important noise variables – Sources of noise – Psychological effects of noise – Effects of noise on performance – Noise and social behavior – Commuting – Impedance.

UNIT 8 : Weather, Air Pollution, and Behavior

Introduction – Heat and behavior –Perception of and psychological reaction to ambient temperatures – Heat and performance – Heat and social behavior – Cold temperatures and behavior – Cold temperatures and health – Cold extremes and performance – Cold extremes and social behavior – Wind and behavior – Perception of wind – Behavior effects of wind – Barometric pressure and altitude – Physiological effects – Acclimatization to high attitudes – Behavior effects of air pressure performance – Air pollution and social behavior – Integrating weather and pollution effects: A final note.

UNIT 9: Personal Space and Territoriality

Personal space – Functions of personal space – Methods for studying personal space – Situational determinants of personal space: Research evidence – Individual difference determinants of personal space: Research Evidence – Interpersonal positioning effects
– Spatial zones that facilitate goal fulfillment – Consequences of too much of too little personal space – Consequences of personal space invasion – Territorial behavior: Keep off my turf – Functions of territoriality – Research on territoriality in humans.

UNIT 10: Crowding

Introduction – Effects of population density on animals –Psychological consequences of high density for animals – Conceptual perspectives: Attempts to understand high density effects in animals – Effects of high density onhumans – Methodologies used to study high density in humans feeling the effects of density: consequences for affect, arousal, and illness effects of density on social behavior – Effects of high density on task performance – Putting the pieces together: Conceptualizations of density effects on humans – Eliminating the causes and effects of crowding – Look to the future. The City
– Perceiving and experiencing the city – Relationship between environmental conditions and imagery – Socioeconomic status and cognitive mapping – Race and cognitive mapping – Gender and cognitive mapping – Relevant experience and cognitive mapping – Effects of urban life on the city dweller: Research evidence – Stress – coping
– Affiliative behavior – Performance – Crime – long-term behavioral effects health – Environmental solutions to urban problems – defensible space – Land use – Social factors – Urban renewal – Escaping from the city.

UNIT 11 : Architecture

Introduction – Extent of architectural influence – An historical overview – Architectural determinism – Environmental possibilism – Environmental probabilism – Behavior and elements of architectural design – Lighting – Windows – Color – Aesthetics – Furnishings – Privacy – Design process – Congruence – Design alternatives – Stages in the design process – Awareness of design

alternatives selection of behavioral criteria

UNIT 12: Behaviorally based research

Implementing the design process: models for the future. Design in Selected Environments: Residential setting – Preferences – use of space in the home satisfaction with the home environment – Propinquity: the effect of occupying nearby territories – Learning environments – Classroom environments – Libraries – Museum environments
– Pedestrian environments: Shopping Malls, plazas and crosswalks hospital setting residential care facilities for the aged – Non institutional residences for the aged.

UNIT 13: Environments for Work, Leisure and the Future*:*

Introduction – Work environments – Study the design of the work environment – Designing the office landscape – Personalization – Territoriality and status in the work environment – Leisure and recreation environments – Wilderness and camping areas – Recreation environments affording exercise – Recreational environments for children –Future environments – Living in space – Experimental undersea environments. Changing behavior to Save the Environment – Environmental psychology and saving the environment – Environmental education – Reinforcement techniques – Positive reinforcement – Encouraging good behavior – Negative reinforcement and punishment
– Alternatives to positive reinforcement – Feedback – Letting us know we're doing – Integrating and evaluating the various approaches to eliminating environmentally destructive behavior – approaches to specific environmental problem – Littering – Saving energy at home – Residential energy conservation – Energy conversation and transportation – Vandalism – Curbing environmentally destructive acts – Assessment of the present and the future.

UNIT 14: Mental health and Social learning*:*

Fostering Mental Health and School Development – Characteristics of mental health –
Home and mental health – School and mental health – Social learning in the classroom
– Social development – Self concept and behaviour – Teachers guidance of self and social development.

References

1. Bell P A, Fisher J D and Loomis R J, Environmental Psychology, Philadelphia:
2. W.E. Saunders Co., 1978.
3. Fisher J D, Bell P A and Baum A, Environmental Psychology, 2nd Edn.
4. NewYork: Holt, Rinchar and Winstorn, 1984.
5. Pajeons J D, Environment and Behavior, Massachusetts: Addison and Wesley
6. Publishing Co., 1977.

Course Code	Title of the Course
115953	**Elective: STRESS MANAGEMENT**

UNIT 1: Stress Management: Stress

Definitions- Models of Stress – Theories of Stress - Stress reactions – Coping and Stress Management techniques- Pain and its management - Psychological reactions of a patient to loss – Stages of Acceptance by Kubler-Ross.

UNIT 2: Organizational stress

Concept- different perspectives, symptoms, individual factors influencing job stress.

UNIT 3: Causes of job stress

Personal characteristics, organizational structure and change, properties of work and work setting, job role, machine pacing and shift work.

UNIT 4: Consequences of job stress

Job behavior- job satisfaction- performance-absenteeism-health strains- physical illness-mental/ psychological symptoms.

UNIT 5: Stress management interventions

Individual centered- cognitive, behavioural and physical. Organization centered-preventing- mitigating and moderating organizational stressors.

UNIT 6: Coping with stress:

Types of coping strategies – Minimizing stress

UNIT 7: Therapeutic Approaches to stress management

Counselling, hypno-suggestive-philosophical hypnosis-self- talk, music therapy. Stress: Concept, different perspectives, and symptoms; framework of stress; Occupational stress, P-E fit model of job stress.

UNIT 8: Causes of occupational stress:

Personal characteristics, organizational structure, properties of work and work setting, job roles.

UNIT 9: Effects of occupational stress

Effect on job behavior, job satisfaction, performance, absenteeism. Effects on employee's physical and psychological health.

UNIT 10 : Individual-centered stress management techniques

Cognitive, hypnosuggestive, behavioural and physical (yoga and aerobic exercises) interventions; Coping strategies.

UNIT 11: Organization

Centered stress management interventions: Preventing, mitigating and moderating stress at organizational level.

UNIT 12 Work Related Stress and Stress Management

What is Stress?- Stressors – Individual differences in stress – Consequences of Distress – Managing Work-Related Stress – Any two stress reduction techniques –Case studies

UNIT 13 Areas of counselling

Educational, career, marital and gerontological; stress management oriented counselling; Counselling for terminal disease patients: cancer and HIV /AIDS.

UNIT 14 Stress Management Interventions

Individual centered- cognitive, behavioural and physical. Organization centered- Preventing, mitigating and moderating organizational stressors.

References

- Matteson, M.T. & Ivancevich, J.M. (1987). Controlling work stress: Effective human resources and management strategies. San Francisco: Josey Bass.
- Pestonjee, D.M. (1992). Stress and coping. New Delhi: Sage Publications.
- Ross, R. R. & Altmair, E.M. (1994). Interventions in occupational stress. New Delhi: Sage Publications.
- Srivastava, A.K. (1999) Management of occupational stress: Theories and practice. New Delhi: Gyan Publishing House.

Course Code	Title of the Course
11954	**Practical -I**

Historical background of Experimental Psychology and relationship with other sciences; Scope of Experimental Psychology and its Methods List of experiments (Any 15)

1. Distraction
2. Span
3. Division
4. Muller Lyer Illusion
5. Horizontal vertical Illusion
5. Organization in perception
6. Set in Perception
7. Motor learning
8. Insight & Trial & Error Learning
9. Transfer of learning- Habit Interference
10. Retroactive Inhibition & Proactive inhibition
11. Memory for names & faces
12. Paired Associate Learning
13. Facial expressions (group experiment)
14. Student stress Scale
15. Anger Expression
16. Knowledge of results
17. Level of Aspiration

Reference

Chaube. S.P.(1985): Experimental Psychology, Laxmi Narain Publishers

Semester-VI

Course Code	Title of the Course
11961	**Health Psychology**

UNIT 1: Health Psychology:

Definition of health psychology - Illness today and the fast – personality and illness Need - Functions of health psychologists.

UNIT 2 : Mind body

Definition Mind body relationship functions and need of help psychologist Bio psychosocial model

UNIT 3 :Stress

Definition - Three faces of stress - High-risk behaviours and health-stress models - Symptoms - Sources - Stress coping strategies.

UNIT 4 :Substance Abuse

Kicking harmful habits: Definition - Abuse dependence and tolerance - Quiting smoking: The promise of perseverance.

UNIT 5 : Use and Misuse of Alcohol

Alcoholism - .Definition - Dependence and abuse - Effects of alcohol - Epidemiology of alcoholism - Alcoholism and health risk Etiology of alcoholism - Clinical treatment.

UNIT 6 : The Problem of Pain

Headache and low back pain: Definition and issues-Epidemiology of pain - Problems - Prevalence and costs The physiology of pain-Etiology of pain - The psychology of pain:' Learning stress and depression - Two pain disorders: Headaches and low back pain - Treatment of pain.

UNIT 7: Coping with chronic or catastrophic illness

Silent killers: Hypertension - 7 The hurry sickness - Biomedical background - Epidemiology of coronary disease - (Unmodifiable risk factors - Modifiable risk factors - Psychosocial factors - Biomedical intervention - Cognitive behavioural programmes.)

UNIT 8: The Healthy Child

Healthy lifestyle and prevention - Chronically ill children: Habits - Lifestyle and prevention -

Children with asthma - Children witJ;1 cystic fibrosis - Children with diabetes mellitus - Intervention.

UNIT 9: Concept of Mind

Definition – Concept of mind- The modern era of Brain imaging Ct Scan MRT Scan, PET Scan CBF Approaches to psychological psychology Philosophic Approaches Clinical Approaches and Experimental method

UNIT 10: Organisation of the Nervous system

Organisation of the Nervous system – Evolution of the brain Structures in the nervous system – The peripheral nervous system –The Spinal cord The brain

UNIT 11: Brain disorder

Factors Implementation –Psychological Deficit in the Brian injured localization of psychological Functions in the Brain- The effects of brain lesions on psychological function

UNIT 12: Neuron and groups of neurons

The anatomy of neuron –Supporting cells in the nervous system what neurons do the neuron at rest the Neuron in Action 0Synaptic communication – What happen when neurons go wild?

UNIT 13 : Emotion and Measurements

The identification and measurement of emotions- Introspective reports Observation of facial expression Measurement in terms of psychological changes Physiology of emotions Theories of emotions

UNIT 14: Aging

Definition - Biology of aging - Lifestyle and aging: Stress - Nutrition and exercise - Death and dying: Grief and preparation.

References

1. Rice P L, Health Psychology, Brooks/Cole Publishing Co., New York, 1998.
2. Sarafino E P, Health Psychology, John Wiley & Sons, Inc, New York, 1990.
3. Course Material Prepared by - Mrs. Vijayalakshmi Reader, Dept. of Psychology Sri Avinashilingam University Coimbatore-641 043.
4. S.K.mangal genral Psychology Sterling Publication 2008

Course Code	Title of the Course
11962	**Guidance and Counselling Psychology**

UNIT 1: Perspectives of peace psychology

Peace psychology: nature, scope and relevance - Psychological causes and effects of violence - Psychological causes and effects of non-violence - Nonviolence through the Indian perspective: thoughts of Gandhi, Ambedkar, Phule

UNIT 2: Understanding violence

Genocide and mass killings: roots and preventions (case study of 1984 sikh riots, kandhamal riots, gujrat riots, caste discrimination and anti-dalit violence) - Psychological approaches to understanding terrorism -. Intimate violence, women and children as victims of structural violence -. Human rights violations as structural violence

UNIT 3: Intrapersonal and interpersonal perspectives of nonviolence

Intrapersonal non violence from the individualistic perspective - Intrapersonal non violence from a collectivistic perspective -. Personality and behavioural tendencies of peaceful persons -. Nonviolent communication Approaches

UNIT 4: Peace building & peace –making interventions

Mediation: peacemaking, peace keeping and peace building. -. Assessing conflict, understanding and dealing with conflict escalation. -. Specific conflict resolution approaches: negotiation, mediation, conciliation, arbitration and adjudication. -. Communicating and practicing forgiveness.

UNIT 5: Anti Social behaviour

Juvenile Delinquency meaning difference causes classification Factors, prevention –

Crime Meaning Diffeecne , causes, classification, factors, prevention

UNIT 6: Counselling:

Meaning and Definition — Importance — Merits — Appeacites — Special areas in Counselling: Education— Family — Delinquents Marriage counselling

UNIT 7: Counselling Process

Steps in Counselling Process — Psychological Tr•iwe and Diagnosis — Factors affecting counselling process — Qualities of counsellors

UNIT-8: Counselling Interview

Essential Aspects — Counsellor — Counselee relationship — Interview Techniques — Structuring the counselling relationship

UNIT 9: Group Counselling:

Meaning — Types of Groups — Value of Group Counselling — Process of Group Counselling — Assumptions and Limitations of Group Counselling —

UNIT-10 Individual Counselling

Meaning —similarities and Dissimilarities between Individual and group counselling

UNIT 11 : Professional Training for counselling

Counsellor Preparation and professional issues — academic preparation — Practical skills - Selection and training of counsellors.

UNIT 12: Modern trends in Counselling — Counselling and Psychotherapy -Trends in counselling — Career. Counselling— Ethical and Legal considerations in counselling.

UNIT 13: Counseling relationship

Working in a counseling Relationship – counselor skills in the understanding and action phases – Transference and counter Transference – The real relationship –Termination of counseling relationship –Functions of termination-Timing of termination –issues of termination resistance to termination –Premature termination- counselor initiated termination

UNIT 14:Theoreis of counseling

Pshcyanaltic Adlerian and Humanistic Theories of counseling-Psychoanalytic theories – psychoanalysis Adlerian theory Adlerian counseling-Humanistic theories –Person centred Counselling Existential and Gestall therapy

References

1. Narayana Rao S, Counselling Psychology, Tata Mc-Graw Hill Pub.Co., New Delhi
2. Bordin S, Edward, Psychological Counselling, ed2, Meredith Corpn., New York, 1968
3. Blum L., Milton and Balinsky Benjamin, Counselling and psychology. Asia Pub., House, New Delhi, 1961.
4. MacNair, R. M. (2003). The psychology of peace: An introduction. Westport, CT: Praeger .
5. Christie, D. J., Wagner, R. V., & winter, D. D. (Eds.) (2001). Peace, Conflict, and Violence: Peace Psychology for the 21st Century. Upper Saddle River, NJ: Prentice-Hall.
6. Mayton, D.M. (2009). Nonviolence and Peace Psychology: Intrapersonal, Interpersonal,

Societal, and World Peace. New York: Springer.

7. Waldron, V.R. & Kelley, D.L. (2008). Communicating Forgiveness. Arizona: SAGE
8. Anstey, M. (1993). *Practical Peacemaking: A Mediator's Handbook.* Cape: Juta & Co.
9. Victoroff, J. (2005). The Mind of the Terrorist: A Review and Critique of Psychological Approaches. The Journal of Conflict Resolution, 49 (1), 3-42.
10. Woolf, L. M., & Hulsizer, M. R. (2005). Psychosocial roots of genocide: Risk, prevention, and intervention, Journal of Genocide Research, 7(1), 101-128.
11. Calloway-Thomas, C. (2010).Empathy in Global World: An intercultural perspective. Indiana: SAGE
12. Jeong, Ho-Won. (2008). Understanding Conflict and Conflict Analysis. Virginia: SAGE
13. Puniyani, R. (2009). Dalit and social justice. Mythri Books. Kerala. India

Course Code	Title of the Course
11963	**Consumer Behavioural**

UNIT I: Introduction

Definition, scope, consumer roles, history of consumer behaviour and the marketing concept, contributing disciplines and application of consumer behaviour.

UNIT2: Market Segmentation

Market segmentation: need, types – geographic, demographic, psychographic and life style. Product positioning: need and strategy.

UNIT 3: Consumer Motivation

Consumer motivation: needs, goals, motive arousal, reactions to frustration.

UNIT4: Consumer Personality

Consumer Personality: nature, influences on consumer behaviour. Consumer Emotions Consumer emotions: nature, uses in advertising. Consumer Perception and its implications.

UNIT5: Consumer Learning

Consumer Learning: Classical and Instrumental theories in the context of consumer behaviour.

UNIT 6: Consumer Attitude

Attitude: functions, Tri-component attitude model and Katz's models of attitude and attitude change.

UNIT 7: Attitude Change

Post purchase attitude change: cognitive dissonance theory and attribution theory.

UNIT 8: Marketing Communication

Marketing communication: process, barriers, types of communication systems, Source, Message and Medium of Communication.

UNIT 9: Consumers In Their Social And Cultural Settings

Reference group: Nature, types and influences on consumers.Family life cycle stages,

UNIT-10: Decision Making

Nature of household and purchases and family decision making and resolving conflict.

UNIT-11: Social Class

Social class: Nature of social class, symbols of status, concept of money and social class, social class categories and consumer behaviour.

UNIT- 12: Consumer Decision Making

Consumer decision: Stages in consumer decision process – situational influence, problem recognition,

UNIT-13: Purchase and Action

Information search, evaluation of alternatives and selection, outlet selection and purchase and post purchase action.

UNIT-14: Organizational Buyer

Organizational Buyer: nature, market structure and pattern of demand, characteristics, decision approach, purchase pattern and organizational buyer decision process.

References

1. Kumar, A and Singh, K. (2013). Consumer Behaviour and Marketing Communication: An Indian Perspective1st Edition. Dreamtech Press, New Delhi.
2. Schiffman LG and Kanuk LL (2007), Consumer Behaviour, 9th edition, Prentice- Hall of India Pvt Ltd, New Delhi, India
3. Batra Satish K and S.H.H. Kazmi (2004), Consumer Behaviour – Text and Cases, Excel Books, A-45, Naraina, Phase I, New Delhi, India.

Course Code	Title of the Course
11964	**Psychological Adjustment**

UNIT-I: The Dynamics Of Adjustment

Adjustment: Meaning, Nature and Dynamics – The Scientific Approach to behaviour – The roots of happiness

UNIT-2: Improving Academic Performance

Improving Academic Performance: Developing Sound Study Habits – Improving Your Reading – Getting More out of Lectures – Applying Memory Principles.

UNIT –3: Personality And Adjustment

Personality: Meaning, Nature and Characteristics - Personality traits – Psychodynamic Perspectives – Behavioural Perspectives – Humanistic Perspectives – Biological Perspectives

UNIT-4 Assessment of Personality

Assessment of Personality- Definition-Meaning-Types- Psychological Testing – Self– Report – Inventories – Projective tests.

UNIT – 5: Stress and Adjustment

The Nature of Stress – Major Types of Stress: Frustration – Conflict – Change – Pressure. Responding to Stress – The Potential Effects of Stress - Factors influencing stress tolerance – Monitoring Your Stress.

UNIT-6: Coping Patterns

Common coping patterns – Types of Coping: Appraisal-focused constructive coping – Problem-focused constructive coping – Emotion-focused constructive coping.

UNIT – 7: Communication

Process of Interpersonal Communication – Non–Verbal Communication – Toward More Effective Communication- Communication Problems –

UNIT-8: Friendship

Interpersonal conflict – Developing Assertive skills. Perspectives on close relationships – Initial attraction and relationship development – Friendship –

UNIT-9: Love

Romantic love: Myths about love- Sexual orientation and love – Theories of love – Gender differences in love – The course of romantic love.

UNIT-10: Loneliness

Loneliness- Definition- Meaning-Nature – Prevalence – Roots – Correlates – Conquering loneliness.

UNIT- 11: Marital Adjustment

Moving Toward Marriage:The motivation to marry – Selecting a mate – Predictors of marital success.

UNIT-12: Family Life

Marital Adjustment Across the Family Life Cycle - Vulnerable areas in Marital Adjustment: Gaps in role expectations –

UNIT-13: Work and Career Issues

Work and career issues – Financial difficulties – Inadequate communication.

UNIT-14: Divorse

Divorce: Nature- Causes – Adjusting to divorce – Remarriage.

References

1. Wayne Weiten, Margret A. Lloyd. (2004). Psychology Applied to modern Life: (7th Edition). Singapore: Thompson Wadsworth.
2. Crisp, R.J. and Turner, R.N. (2007). Essential Social Psychology. New Delhi: Sage Publications.